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South Africa

The South African Medical Research Council

recognises the catastrophic and persisting consequences of colonialism and apartheid, including land dispossession and the intentional imposition of educational and health inequities.

Acknowledging the SAMRC's historical role and silence during apartheid, we commit our capacities and resources to the continued promotion of justice and dignity in health research in South Africa.

STEAM-3D

Building African Institutional Capacity for Evidence-Informed Modelling, Biostatistics and Decision-Making

WP: Knowledge translation (evidence synthesis) curriculum · Workshop 1, 15 October 2026

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WELCOME & OVERVIEW

Evidence synthesis matters for modelling

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Teleportation


THE AGE
OF TIME
TRAVEL
IS NOW

PATTERN ONE

WHEN PARAMETERS WERE WELL-GROUNDED, PREDICTIONS HELD

H1N1 — GLEAM

Parameters (R0, generation interval) were drawn from decades of prior influenza pandemic and seasonal literature already synthesised by 2009.

SARS — Lipsitch et al.

Multiple independent teams estimated R0 for the same outbreak in parallel; the numbers converged quickly, effectively self-synthesising across studies.

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PATTERN ONE · STUDY 1 OF 3

H1N1 2009: THE GLEAM GLOBAL FORECAST

Balcan et al. (2009) forecast the 2009 pandemic wave in real time; Tizzoni et al. (2012) later tested it against surveillance data from 48 countries.

Where the inputs came from

- **Transmissibility was fitted, not borrowed.** R0 = 1.75 (95% CI 1.64–1.88) and generation time 3.6 days (2.2–5.1) came from a Monte Carlo maximum-likelihood fit to the dates the virus arrived in 12 countries seeded from Mexico.
- **Secondary inputs came from earlier work.** Latent period (1.1 days) followed early estimates compatible with seasonal influenza, then was varied from 1.1 to 2.5 days; the generation-time range was checked against previous pandemic and seasonal influenza studies (mostly 2–4 days).
- **Mobility was measured:** airline and commuting data for 3,362 subpopulations in 220 countries.

What the papers found

- **Forecast (Sept 2009, before the northern peaks):** a worst-case, no-mitigation epidemic peaking in October–November in the northern hemisphere.
- **Peak timing held up.** Observed peaks fell inside the prediction interval in 87% of countries; the other 13% were off by 2 weeks at most. Median error 0; within 2 weeks in 50% of countries and within 4 weeks in 95%.
- **Size was not forecast reliably.** Attack rate depends on vaccine uptake, pre-existing immunity and the asymptomatic fraction, which became known only after the wave. Model attack rates ran above serological and several surveillance estimates, but close to US estimates.

Lesson for evidence synthesis

- **Verdict: timing held, size did not.**
- **Why it worked:** the transmission signal came from the epidemic itself, the mobility data were high quality, and the authors ran sensitivity analyses and framed the result as a worst case.
- **Where synthesis could still add most:** the inputs behind attack rate (pre-existing immunity, asymptomatic proportion, vaccine uptake) had no synthesised evidence in real time. A rapid, living review of those inputs is the gap.

References (file labels in the STEAM 3D › References folder): 01a Balcan D, Hu H, Goncalves B, et al. Seasonal transmission potential and activity peaks of the new influenza A(H1N1). BMC Med 2009;7:45. 01b Tizzoni M, Bajardi P, Poletto C, et al. Real-time numerical forecast of global epidemic spreading: case study of 2009 A/H1N1pdm. BMC Med 2012;10:165.

PATTERN ONE · STUDY 2 OF 3

SARS 2003: LIPSITCH ET AL.

Lipsitch et al. (Science, June 2003) estimated how infectious SARS was, and what isolation and quarantine could achieve, while the epidemic was still unfolding.

Where the inputs came from

- **Singapore case data:** the first 205 probable cases. Secondary cases per index case fell from a mean of 7 (first week) to 1.6 (second week) to below 1 in most later weeks, as isolation and quarantine began.
- **Serial interval measured directly:** mean 8.4 days (SD 3.8) in Singapore.
- **Early growth in other settings:** control began quickly in Singapore, so there was too little uncontrolled data there. R0 was instead inferred from growth in reported cases (Hong Kong and worldwide counts), using assumed serial intervals of 8–12 days.

What the paper found

- **R0 for uncontrolled spread: about 3.** The Hong Kong growth analysis gave 2.2–3.6 for serial intervals of 8–12 days.
- **Wide uncertainty, stated openly.** The Singapore stochastic model gave a 90% credible interval of 1.5–7.7 (mode about 2.2, mean 3.5), because superspreading was common: 103 of the first 201 probable cases were infected by just five source cases.
- **Under-reporting barely mattered:** the case count enters the estimate through a logarithm, so even large under-reporting moved R0 little.
- **Control looked achievable:** R only has to be held below 1, so isolation and quarantine need not be perfect.

Lesson for evidence synthesis

- **Verdict: reliable because it was honest about the range.**
- **Triangulation within one analysis:** three data sources (Singapore line list, Hong Kong growth, worldwide counts) were used, with explicit sensitivity analyses for serial interval and under-reporting.
- **Decision-makers needed a range, not a point.** A transparent range for R0, rather than a single estimate, is what a living synthesis of parameter estimates formalises.
- **Limit:** too little pre-control data from Singapore, so uncontrolled R0 had to be inferred indirectly.

References (file labels in the STEAM 3D › References folder): 03 Lipsitch M, Cohen T, Cooper B, et al. Transmission dynamics and control of severe acute respiratory syndrome. Science 2003;300(5627):1966–1970. doi:10.1126/science.1086616

PATTERN ONE · STUDY 3 OF 3

HIV WITHIN-HOST DYNAMICS: PERELSON ET AL.

Perelson et al. (Science, 1996) measured how fast HIV is cleared and produced inside patients, by watching viral load fall after a protease inhibitor.

Where the inputs came from

- **Measured in patients, not taken from the literature.** Five HIV-infected people received ritonavir (600 mg twice daily) and plasma HIV-1 RNA was tracked closely: 15 data points in the first 7 days.
- **Denser sampling than before.** The same group's earlier study sampled every 3 days and gave only minimum estimates.
- **Model fitted to each patient's decay curve** (non-linear least squares), separating clearance of free virions from loss of infected cells, after estimating and subtracting the drug's pharmacokinetic delay.

What the paper found

- **Free virions:** half-life 0.24 days (about 6 hours); clearance rate 2.06–3.81 per day across patients.
- **Infected cells:** mean life-span 2.2 days (half-life 1.6 days).
- **Production:** at least 10.3×10^9 virions per day, about 15 times the group's earlier minimum estimate.
- **Cycle timing:** minimum life cycle 1.2 days; generation time 2.6 ± 0.8 days.
- **Caveat from the authors:** values are minimum estimates, assuming the drug was fully effective and target cells did not recover.

Lesson for evidence synthesis

- **Verdict: the clean end of Pattern One.** The parameters were measured in the system being modelled, so no synthesis was needed to feed the model.
- **Independent replication:** other groups (Wei et al., Nowak et al.) had found essentially identical decay kinetics.
- **Study design matters:** denser sampling revised production upward by about 15-fold. When pooling evidence, check how each estimate was measured.
- **Scope:** five patients; the estimates describe within-host biology and would need synthesis before use in population models.

References (file labels in the STEAM 3D › References folder): 06 Perelson AS, Neumann AU, Markowitz M, Leonard JM, Ho DD. HIV-1 dynamics in vivo: virion clearance rate, infected cell life-span, and viral generation time. Science 1996;271(5255):1582–1586.

PATTERN TWO

WHEN EVIDENCE WAS THIN, IDEALISED, OR CONTESTED — PREDICTIONS SLIPPED

Ebola — WHO 2014

The natural-history evidence base was thin (only a handful of small prior outbreaks). The 'no intervention' scenario was one point on a wide plausible range that wasn't clearly communicated as such.

FMD — Ferguson et al.

The transmission-kernel and culling-efficacy evidence drew on a narrow slice of data. Broader international FMD control evidence, including vaccination experience, existed but was not systematically weighed.

HIV TasP — Granich et al.

The model assumed near-ideal testing, linkage and adherence. A substantial literature on real-world care-cascade attrition already existed in Africa by 2009 but was not incorporated.

A seventh case — COVID-19's Imperial Report 9 — doesn't fit either pattern cleanly: its headline number was an explicit non-intervention counterfactual, not a forecast. See the case-by-case verdict table.

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PATTERN TWO · STUDY 1 OF 3

EBOLA 2014: WHO FORWARD PROJECTIONS

WHO Ebola Response Team (NEJM, Oct 2014) described the first 9 months of the West Africa epidemic and projected cases to 2 November, assuming no change in control.

Where the inputs came from

- **Field surveillance data** on 3,343 confirmed and 667 probable cases (Guinea, Liberia, Nigeria, Sierra Leone) as of 14 September 2014.
- **Natural history estimated from those cases:** incubation 11.4 days; serial interval 15.3 ± 9.3 days. The authors report these, and the case fatality rate (70.8%), as within the ranges seen in previous Ebola outbreaks.
- **Growth rates from early exponential growth:** R0 1.71 (Guinea), 1.83 (Liberia), 2.02 (Sierra Leone); doubling times 15.7, 23.6 and 30.2 days.

What the paper found

- **Conditional projection:** assuming no change in control measures, cumulative cases by 2 November would be 5,740 (Guinea), 9,890 (Liberia) and 5,000 (Sierra Leone): more than 20,000 in total.
- **Limits the authors flagged:** the data understate the true size; reliable estimates of the current reproduction number reached only early September because of reporting delays.
- **Their conclusion:** the epidemic was large because of the affected populations and insufficient control, not unusual virus biology. In the week of 8–14 Sept about 995 patients needed care versus roughly 610 beds.

Lesson for evidence synthesis

- **Verdict: a scenario, not a forecast.** The >20,000 figure assumed nothing changed, so it can't be scored as a prediction against what followed.
- **The weak link was not the natural-history parameters.** Earlier outbreaks supplied a check on incubation, fatality and R0. The weaker inputs were incomplete, delayed surveillance data and the assumption that control stayed fixed.
- **What synthesis could not supply:** how behaviour and the response would change. Stating that conditionality clearly is part of reporting the evidence.

References (file labels in the STEAM 3D › References folder): 02 WHO Ebola Response Team. Ebola virus disease in West Africa – the first 9 months of the epidemic and forward projections. N Engl J Med 2014;371(16):1481–1495. doi:10.1056/NEJMoa1411100

PATTERN TWO · STUDY 2 OF 3

FOOT-AND-MOUTH 2001: FERGUSON ET AL.

Ferguson et al. (2001) modelled culling and vaccination in the UK epidemic; Kitching et al. (2006) criticised the inputs; Tildesley et al. (2008) tested the model.

Where the inputs came from

- **Farm-level data:** disease, culling and census records for every livestock farm in Great Britain (Nature). The Science paper used the epidemic's first two months, with reporting and slaughter delays estimated from surveillance.
- **Contact-tracing data** supplied the spatial kernel and infectivity timing. Kitching et al. note it reached modellers on 13 March, three weeks in, and was incomplete.
- **Infectivity was assumed:** constant from about 3 days after infection until slaughter. Ferguson et al. wrote that data do not exist to estimate a farm's infectiousness over time since infection.

What the papers found

- **Ferguson et al.:** extended culling was essential; faster action could have made the epidemic much smaller. Ring culling and ring vaccination were both modelled, and culling was predicted to be more effective.
- **Kitching et al.:** tracing data were incomplete and biased towards short-range spread; infectivity timing conflicted with biology; parameters were held constant; models were unvalidated. 10,157 premises were culled; 2,026 were infected.
- **Tildesley et al.:** farm-level accuracy 5–15% for infected farms, 20–30% for culls. The low figure matched the spread between independent model runs, so they judged the model a reliable representation.

Lesson for evidence synthesis

- **Verdict: contested. Same evidence, opposite readings.**
- **Vaccination was modelled, not ignored.** The dispute is about input quality: biased tracing data, the timing of infectivity and the absence of validation.
- **The authors named their own gap:** no data on infectiousness over time. A structured appraisal of the contact-tracing data, and a synthesis of what was known about infectivity timing, was the missing step before a culling policy was built on the model.
- **Both critique and test belong in the evidence base.**

References (file labels in the STEAM 3D › References folder): 04a Ferguson NM, Donnelly CA, Anderson RM. Transmission intensity and impact of control policies on the foot and mouth epidemic in Great Britain. Nature 2001;413:542–548. 04b Ferguson NM, Donnelly CA, Anderson RM. The foot-and-mouth epidemic in Great Britain: pattern of spread and impact of interventions. Science 2001;292:1155–1160. 04c Kitching RP, Thrusfield MV, Taylor NM. Use and abuse of mathematical models. Rev Sci Tech 2006;25(1):293–311. 04d Tildesley MJ, Deardon R, Savill NJ, et al. Accuracy of models for the 2001 foot-and-mouth epidemic. Proc R Soc B 2008;275(1641):1459–1468.

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PATTERN TWO · STUDY 3 OF 3

HIV TREATMENT AS PREVENTION: GRANICH ET AL.

Granich et al. (Lancet, 2009) modelled universal annual testing with immediate ART; Eaton et al. (2012) compared 12 models and Hayes et al. (2019) tested the idea in a trial

Where the inputs came from

- **Test case:** South Africa as a hypothetical generalised epidemic, calibrated to R0 = 7, a doubling time of 1.25 years and antenatal prevalence data.
- **Natural history and ART effect:** survival (mean 11 years), 10-fold infectiousness in the 2-month acute phase, and infectiousness cut to 1% on ART, all from sources cited in the paper.
- **Programme assumptions:** annual testing of everyone aged 15+, 100% test accuracy, coverage rising to 90% by 2016. Retention and failure used Malawi programme data (1.5% a year drop-out, 3% a year first-line failure).

What the papers found

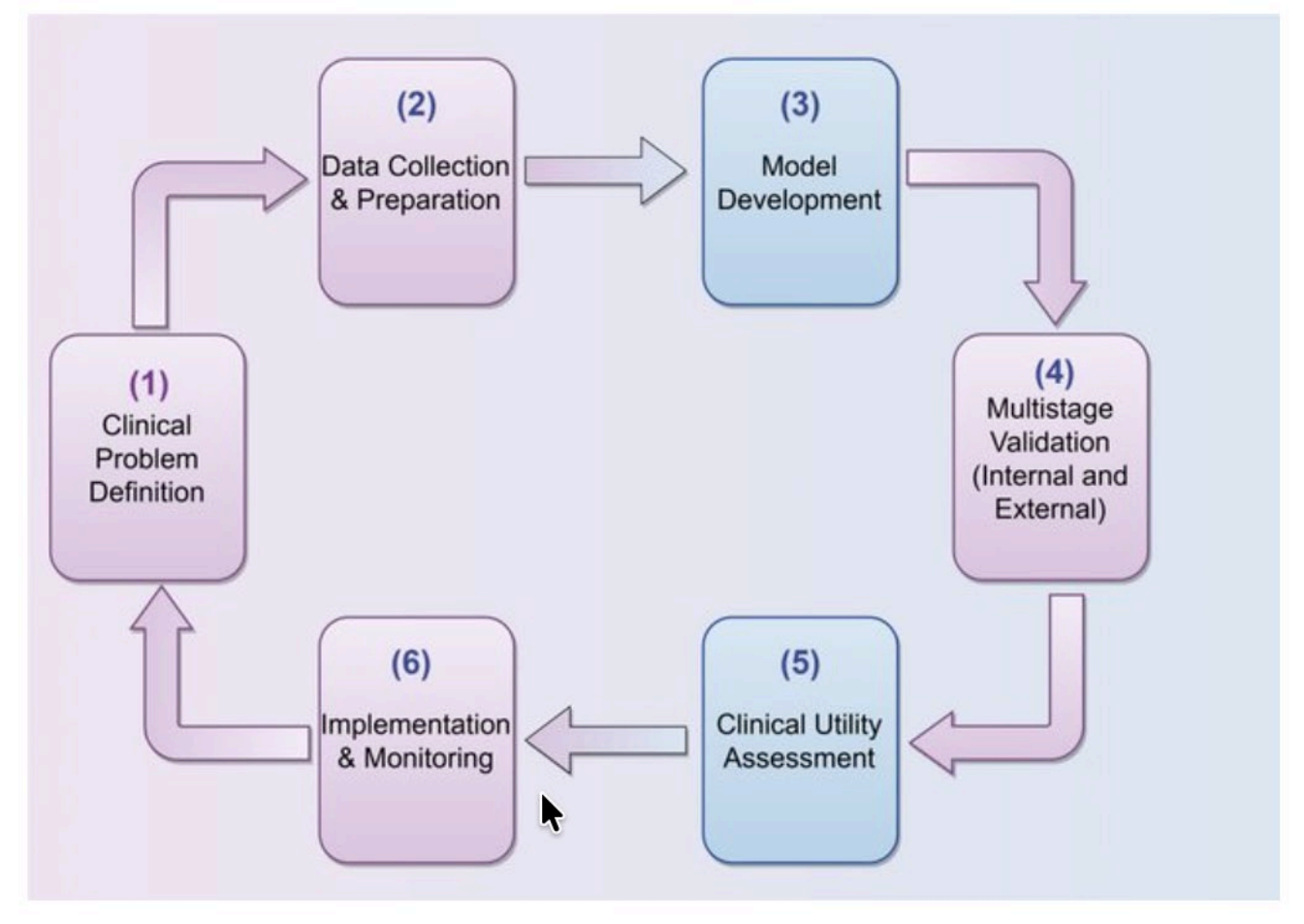
- **Granich:** R0 below 1 needed yearly testing and ART at diagnosis (annual testing with ART at CD4 350 gave R0 about 3). Elimination (< 1 case per 1,000 a year) within about 10 years of full roll-out.
- **Eaton:** 12 models agreed ART at high coverage cuts incidence, but only 3 of 9 reached < 0.1% by 2050 at 95% access and retention. Granich's own model gave 0.13%.
- **Hayes (PopART trial, 21 communities):** incidence rate ratio 0.93 (95% CI 0.74–1.18) with universal ART versus standard care, and 0.70 (0.55–0.88) with ART per local guidelines. The authors called the null result unanticipated.

Lesson for evidence synthesis

- **Verdict: optimistic because the programme was idealised.** The result depended on near-universal testing uptake and coverage.
- **Not ignored, but not stress-tested:** Granich used real Malawi retention data and flagged uptake, adherence and behaviour change as needing better data. The elimination claim still rested on the idealised settings.
- **Synthesis could have mapped the range early:** the same-model re-analysis by Wagner and Blower, cited in Eaton, reached the opposite conclusion using different uptake assumptions.

References (file labels in the STEAM 3D › References folder): 05a Granich RM, Gilks CF, Dye C, De Cock KM, Williams BG. Universal voluntary HIV testing with immediate antiretroviral therapy as a strategy for elimination of HIV transmission: a mathematical model. Lancet 2009;373:48–57. 05b Hayes RJ, Donnell D, Floyd S, et al. Effect of universal testing and treatment on HIV incidence – HPTN 071 (PopART). N Engl J Med 2019;381:207–218. 05c Eaton JW, Johnson LF, Salomon JA, et al. HIV treatment as prevention: systematic comparison of mathematical models. PLoS Med 2012;9(7):e1001245.

MODELLING STUDIES DATA SOURCES



Sources of data?

- Where?
- Types ?
- Format ?
- Reliability ?

MODELLING STUDIES DATA SOURCES

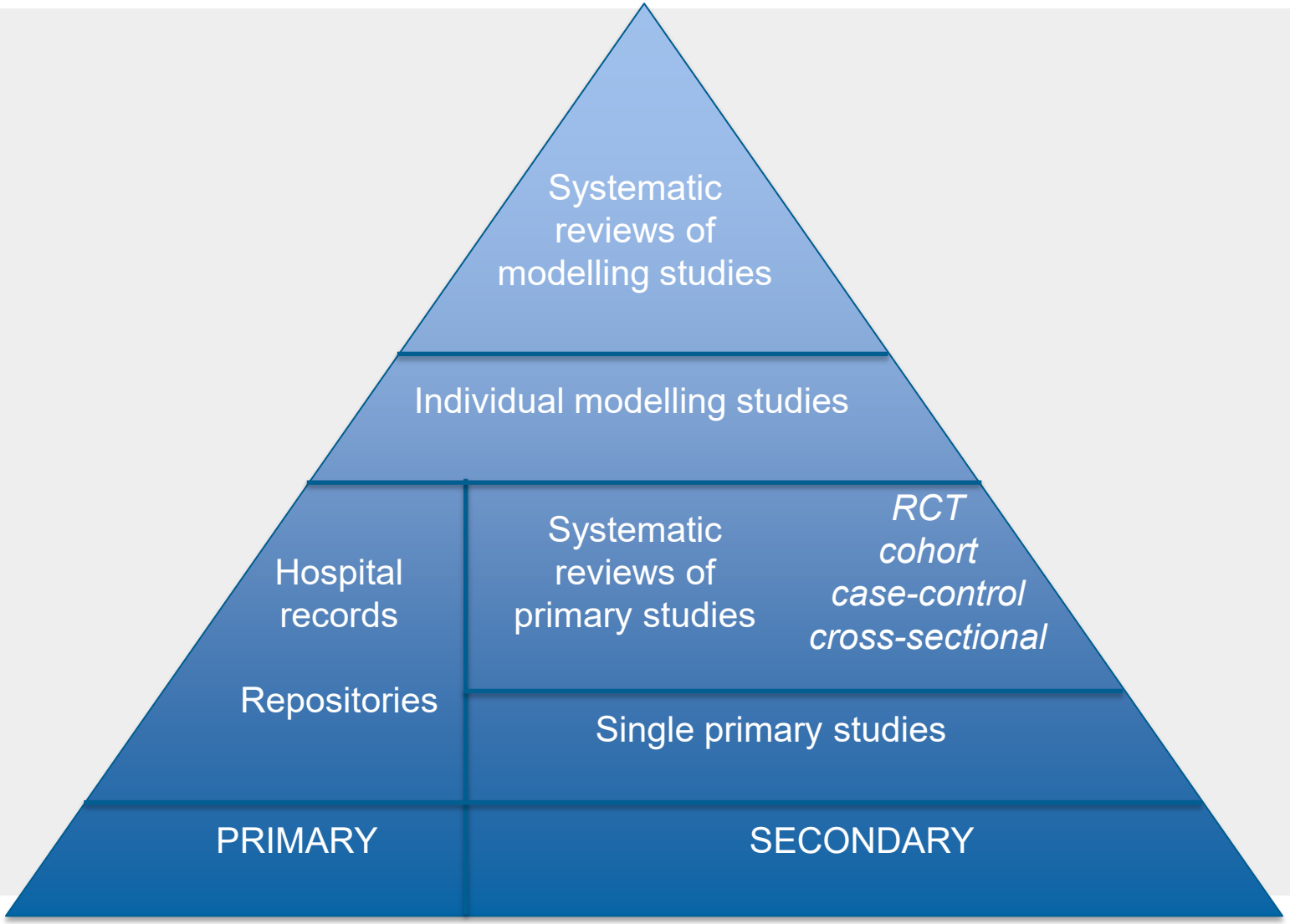


Your national Department of Health requests an update on the risk predictions for the Russian pneumonic "outbreak" coming to your country.

Map out your approach w.r.t. data sources.

Indicate your preference for the types of data your may find.

MODELLING STUDIES DATA SOURCE HIERARCHY



TYPES OF EVIDENCE SYNTHESIS USEFUL FOR MODELLING

Choose the method based on urgency and decision need

Systematic review	Effect estimates and certainty
Rapid review	Urgent policy need
Scoping review	Map available evidence
Evidence map	Identify gaps
Qualitative synthesis	Feasibility and acceptability
Meta-analysis	Pooled parameter estimates
Living review	Fast-changing evidence

MODELLING STUDIES DATA SOURCES

Model Type	Main Data Source	Typical Accuracy
Transmission models using SR-derived epidemiological parameters	Systematic reviews of incubation, serial interval, IFR, vaccine effectiveness	More robust and generalizable for policy analysis and scenario modelling.
Real-time forecasting models	Surveillance data, cases, hospitalizations, mobility data	Better short-term predictive accuracy.
Early-pandemic models using limited case reports	Individual studies and preliminary outbreak data	Often required substantial revision as evidence accumulated.
Hybrid models	Systematic reviews + surveillance + mobility data	Frequently the strongest overall performance because they combine stable and dynamic inputs.

MODELLING STUDIES DATA SOURCES



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HOW DO WE KNOW THE DATA ARE RELIABLE?

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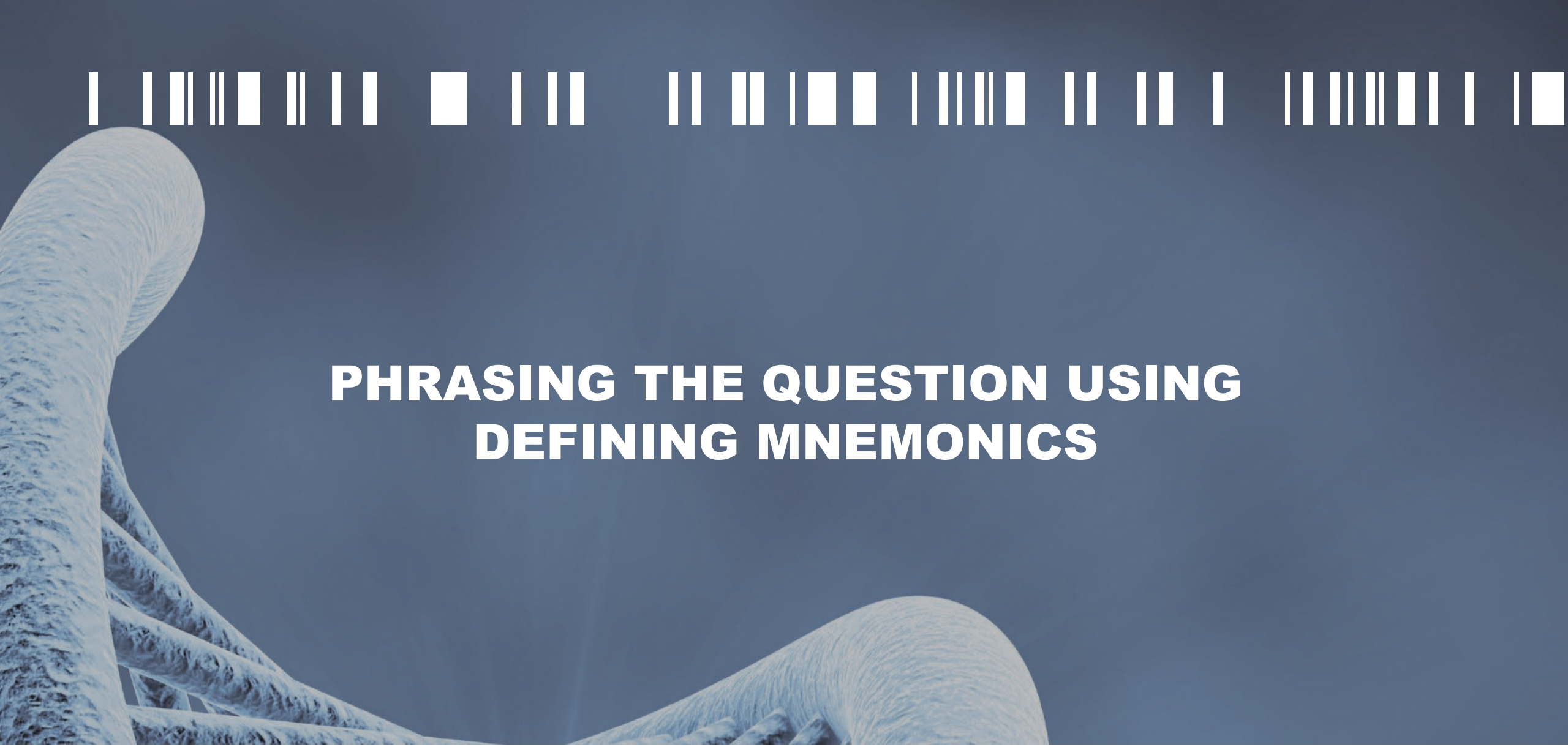
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LEARNING OBJECTIVES

BY THE END OF THE DAY, PARTICIPANTS WILL BE ABLE TO…

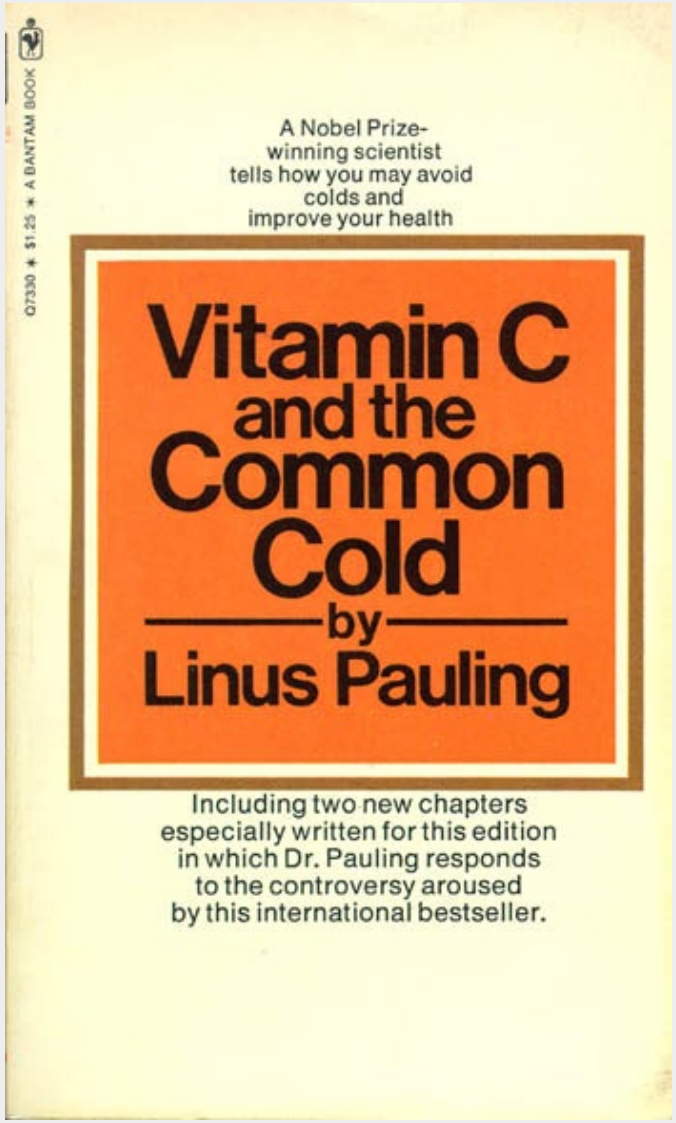
1	**Frame review questions using MECIR standards**	Apply Cochrane's Methodological Expectations for Intervention Reviews as the benchmark for systematic rigour.
2	**Design search strategies**	Mnemonic elements combined with Inclusion Criteria: PubMed overview and tips
3	**Manage results / Data Extraction**	Rayyan overview; Data extraction basics
4	**Meta-Analysis revision/overview**	Forest plot interpretation; heterogeneity
5	**Assess risk of bias SR and studies**	Choose and apply the right tool for each design (RoB 2, ROBINS-I, QUADAS-2) to trial and observational evidence.
6	**Critically appraise modelling studies**	Judge a model's credibility and decision-relevance — the standard their own outputs will be held to.



PHRASING THE QUESTION USING DEFINING MNEMONICS

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Vitamin C for preventing and treating the common cold

Authors' conclusions:

The failure of vitamin C supplementation to reduce the incidence of colds in the general population indicates that routine vitamin C supplementation is not justified, yet vitamin C may be useful for people exposed to brief periods of severe physical exercise. Regular supplementation trials have shown that vitamin C reduces the duration of colds, but this was not replicated in the few therapeutic trials that have been carried out. Nevertheless, given the consistent effect of vitamin C on the duration and severity of colds in the regular

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A Systematic Review

"A review in which bias has been reduced by the:

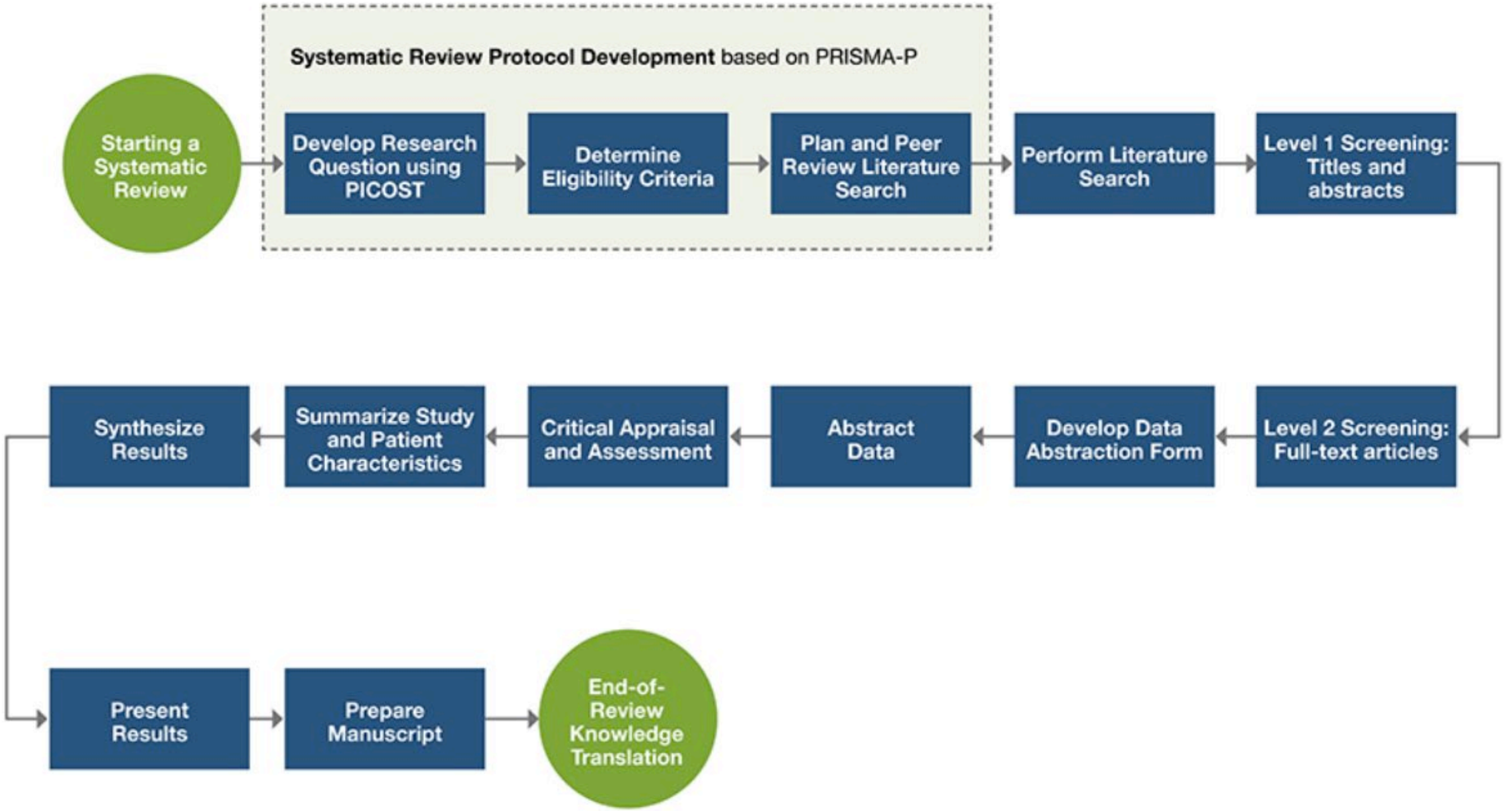
- systematic identification,
- appraisal,
- synthesis, and
- statistical aggregation (*if relevant)*

of *all* relevant studies on a specific topic according to a predetermined and explicit method."

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Moher et al. Lancet; 354: 1896-900

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Systematic Review Protocol Development based on PRISMA-P
Starting a Systematic Review
Develop Research Question using PICOST
Determine Eligibility Criteria
Plan and Peer Review Literature Search
Perform Literature Search
Level 1 Screening: Titles and abstracts
Level 2 Screening: Full-text articles
Develop Data Abstraction Form
Abstract Data
Critical Appraisal and Assessment
Summarize Study and Patient Characteristics
Synthesize Results
Present Results
Prepare Manuscript
End-of-Review Knowledge Translation



Scenario / Possible questions?

Worldwide efforts have been made by some countries to offer HPV vaccination since its introduction in 2006.

You wish to review available evidence on the effect of HPV vaccine among young women to prevent HPV infection.



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Phrasing questions using a mnemonic as a tool

- **P** = **P**atient / **p**erson / **p**opulation
- **I/E** = **I**ntervention (action) / **e**xposure (observation) / **i**ndex test
- **C** = **C**omparator for the I / E
- **O** = Desired **o**utcome
- **(T)** = **T**ime frame
- **(S)** = **S**tudy design



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Review Question: PECO approach

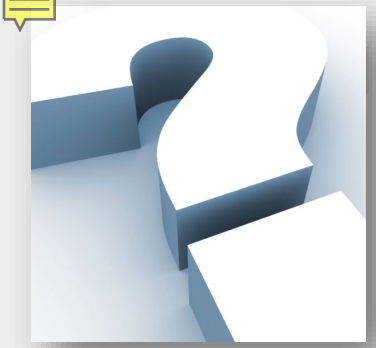


Participants:	Women aged 12 to 34 years
Intervention/Exposure:	HPV Vaccination
Comparison:	Non-vaccinated women
Outcome:	Prevalence of HPV infection

What is the prevalence of HPV following HPV vaccination in women when compared with non-vaccinated women?

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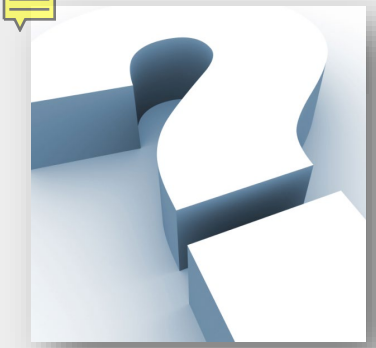




Consider this Scenario

A patient with a severe fungal infection mentions that he does not like mouthwash or oral medication. He asks you how effective is lazer treatment are in comparison. You tell him that you will read up on this and get back to him later

Participants | Intervention | Comparison | Outcome





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Participants | Intervention | Comparison | Outcome

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INTERVENTION QUESTION

Intervention: PICO

Participants:
Intervention:
Comparison:
Outcome:



The review question:

Is photodynamic therapy compared with standard anti-fungal treatment modalities, effective for the treatment of oral fungal infections in humans?

INTERVENTION QUESTION

Intervention: PICO

Participants:	Patients with oral fungal infections
Intervention:	Photodynamic therapy
Comparison:	Standard treatment
Outcome:	Effective treatment



The review question:

Is photodynamic therapy compared with standard anti-fungal treatment modalities, effective for the treatment of oral fungal infections in humans?



TBI PROGNOSIS



Treatment













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Category	Typical question
Diagnosis & Screening	How accurate is this test?
Frequency/Prevalence/Incidence	How often does this disease occur?
Risk/Aetiology/Cause	What factors increase the risk of this disease?
Harm	Is this intervention safe?
Prognosis	What are the consequences of this disease?
Treatment/Intervention	Does treatment change the course of this disease?
Prevention/Intervention	Does early intervention keep this disease from occurring, or slows its development?
Cost-effectiveness	Are the benefits of the treatment worth the costs incurred?
Experiences & Meaning	Why patients behave in a certain way and how do they experience the disease, test or treatment?

Best study design according to question

- For each type of question, there is a study design that is most appropriate for the specific type of question.
- So when you search for research for a specific type of question, you need to search for the study design that is best for that question.

What are Study Designs?

- Structured approaches to address specific research questions
- Overall research strategy that will be undertaken/implemented
- Provide general guidelines for thinking about specific aspects of study conduct:
  - sampling populations
  - systematically collecting measurements
  - analysing data
- Strengths & limitations of specific designs are well-established
- Some questions can be answered using more than one study design—other considerations

Question type guides study design

Question Type	Best study design ?
Therapy or prevention →	
Aetiology & risk factors →	
Diagnosis →	
Screening →	
Experiences & meaning →	
Prognosis & incidence →	

Question type guides study design

Question Type	Best study design
Therapy or prevention	Randomised controlled trial (RCT)
Aetiology & risk factors	Cohort or case-control study
Diagnosis	Cross-sectional study
Screening	Randomised controlled trial (RCT)
Experiences & meaning	Qualitative study
Prognosis & incidence	Cohort study

Aetiology and Risk

PEO

- Population
- Exposure / Association
- Outcomes

Aetiology and Risk

> Are adults exposed to radon at risk of lung cancer?

Participants | Association | Outcome

Prevalence or Incidence

CoCoPop

- Condition
- Context
- Population

Prevalence or Incidence

> What is the prevalence of **peri-natal depression** among **women** in **Australia**?



Qualitative Reviews

> What is the **experience of recovering from burns injuries** in **adult (over 18 years of age) patients** during their stay in a **hospital burns care unit**?

Population | **Phenomena of Interest** | **Context**

Scoping Reviews

PCC

- Population
- Concept
- Context



HOWEVER not all elements necessarily apply to every scoping review, and use of mnemonics should be considered a guide

Scoping Reviews

> What types of **neurological reactions to the Human Papilloma (HPV) Virus vaccination** have been reported in **people who have received the HPV vaccine**?



Population | Concept | Context

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Review Type	Question Format	Question Example
Costs/Economic Evaluation	Population, Intervention, Comparator/s, Outcomes, Context **(PICOC)**	What is the cost effectiveness of self-monitoring of blood glucose in type 2 diabetes mellitus in high income countries?
Diagnostic Test Accuracy	Population, Index Test, Reference Test, Diagnosis of Interest **(PIRD)**	What is the diagnostic test accuracy of nutritional tools (such as the Malnutrition Screening Tool) compared to the Patient Generated Subjective Global Assessment amongst patients with colorectal cancer to identify undernutrition?
Expert opinion/policy	Population, Intervention or Phenomena of Interest, Context **(PICo)**	What are the policy strategies to reduce maternal mortality in pregnant and birthing women in Cambodia, Thailand, Malaysia and Sri Lanka?
Psychometric	Construct of interest or the name of the measurement instrument(s), Population, Type of measurement instrument, Measurement properties	What is the reliability, validity, responsiveness and interpretability of methods (manual muscle testing, isokinetic dynamometry, hand held dynamometry) to assess muscle strength in adults?
Prognostic	Population, Prognostic Factors (or models of interest), Outcome **(PFO)**	In adults with low back pain, what is the association between individual recovery expectations and disability outcomes?

PHRASING QUESTIONS USING PICO/PECO MNEMONIC



- Structuring qualitative questions? Try **PIC** or **SPIDER:**

  Population, Phenomena of **I**nterest, **C**ontext

  Sample, **P**henomenon of **I**nterest, **D**esign, **E**valuation, **R**esearch type

- Question about aetiology or risk? Try **PEO**

  Population, **E**xposure, **O**utcomes



- Evaluating an intervention, policy or service? Try **SPICE**:

  Setting , **P**opulation or **P**erspective, **I**ntervention, **C**omparison, **E**valuation

PHRASING QUESTIONS USING PICO/PECO MNEMONIC

- Investigating the outcome of a service or policy? Try **ECLIPSE:**

  Expectation, **C**lient group, **L**ocation, **I**mpact, **P**rofessionals, **SE**rvice

- Working out prevalence or incidence? Try **CoCoPop:**

  Condition, **Co**ntext, **Pop**ulation



- Determining prognosis? Try **PFO**:

  Population, Prognostic **F**actors, **O**utcome

- Conducting an economic evaluation? Try **PICOC**:

  Population, **I**ntervention, **C**omparator/s, **O**utomes, **C**ontext

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Training Working Group

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List of Research Question Types with Framework Mnemonic

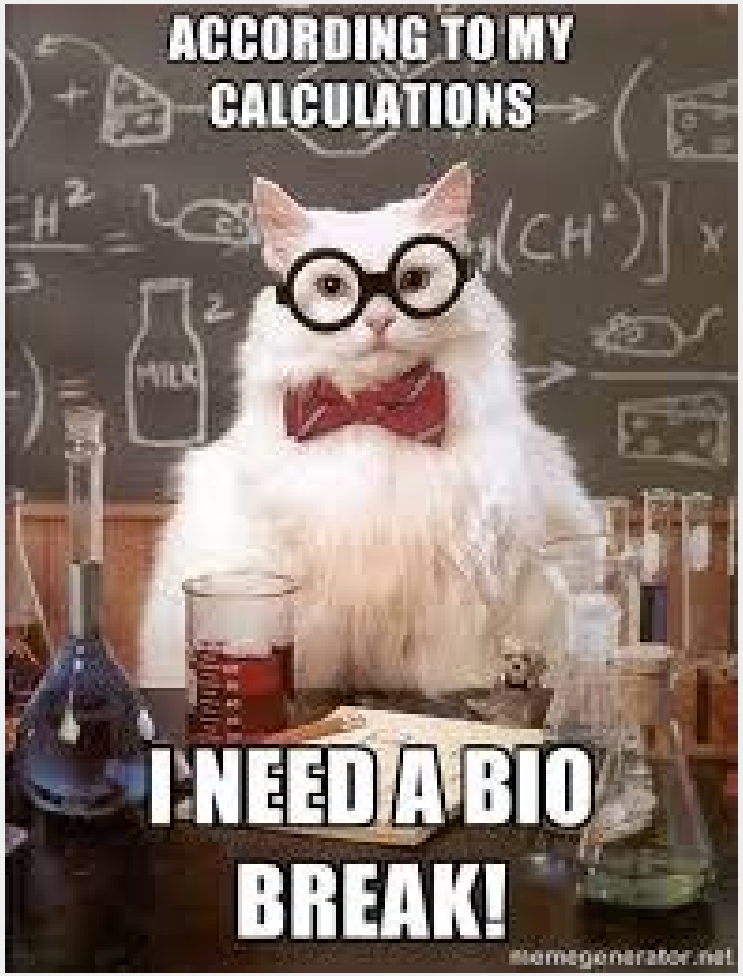
Systematic Review	**PICO**
Patient, **P**roblem, **P**opulation **I**ntervention **C**omparison **O**utcome	
Comparative effectiveness	**PICO**
Patient, **Problem**, **P**opulation **I**ntervention **C**omparison **O**utcome	
Diagnostic Test Accuracy	**PIRD**

Diagnostic Test Accuracy	**PIRD**
Population **I**ndex test **R**eference test **D**iagnosis of interest	
Prognostic Review	**PFO**
Population Prognostic **F**actors (for models of interest) **O**utcome	
Etiology/Risk/Benefit	**PEO**
Population **E**xposure(s) **O**utcomes	
Systematic Review of Epidemiology Studies	**CoCoPop**
Condition **Co**ntext **Pop**ulation	

Umbrella Review	**PICO**
Patient, **P**roblem, **P**opulation **I**ntervention **C**omparison **O**utcome	
Overview (of Reviews)	**PICO**
Patient, **P**roblem, **P**opulation **I**ntervention **C**omparison **O**utcome	
Scoping review	**PCC**
Population **C**oncept **C**ontext	
Qualitative Review	**SPICE**
Setting **P**erspective **I**nterest **P**henomenon of comparison **E**valuation	

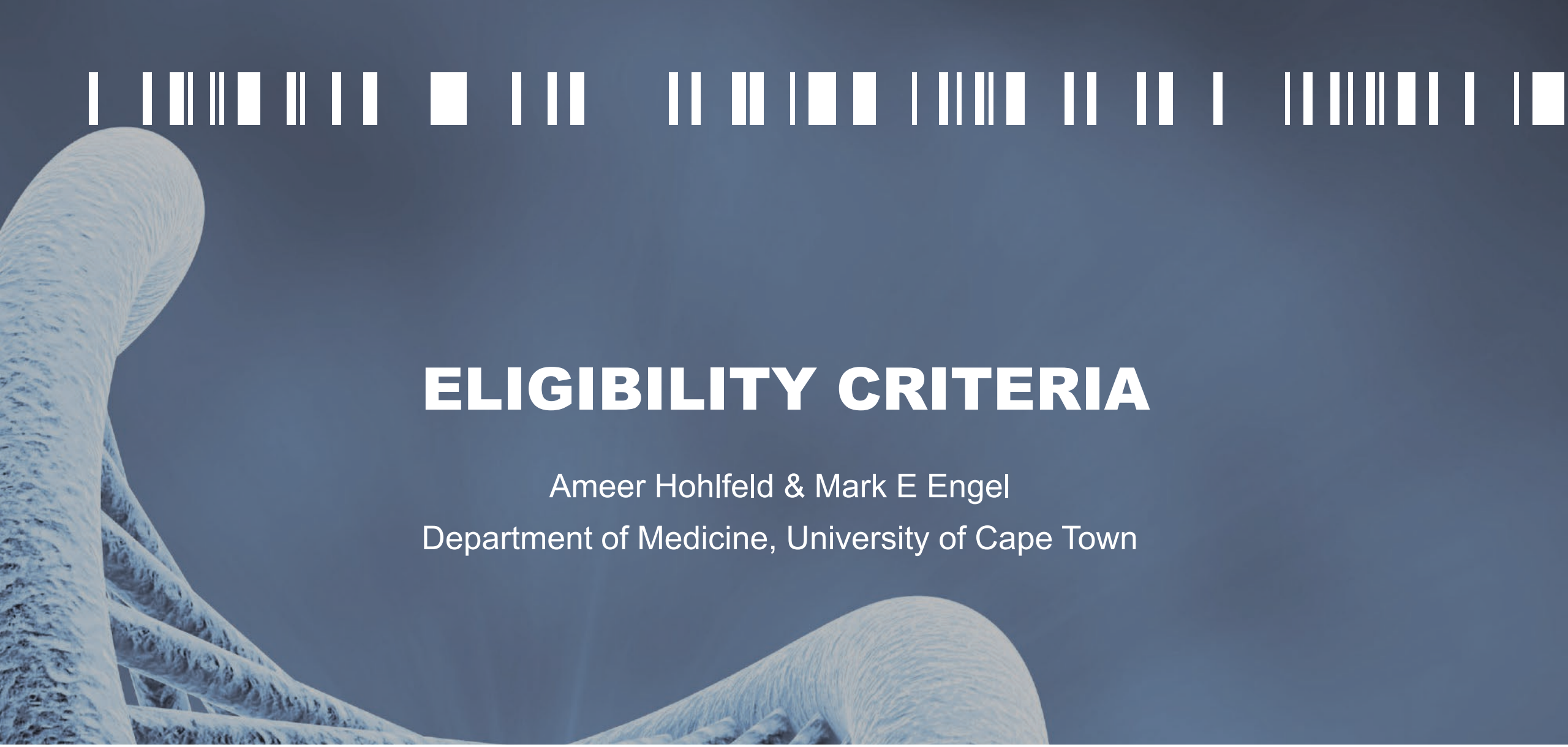






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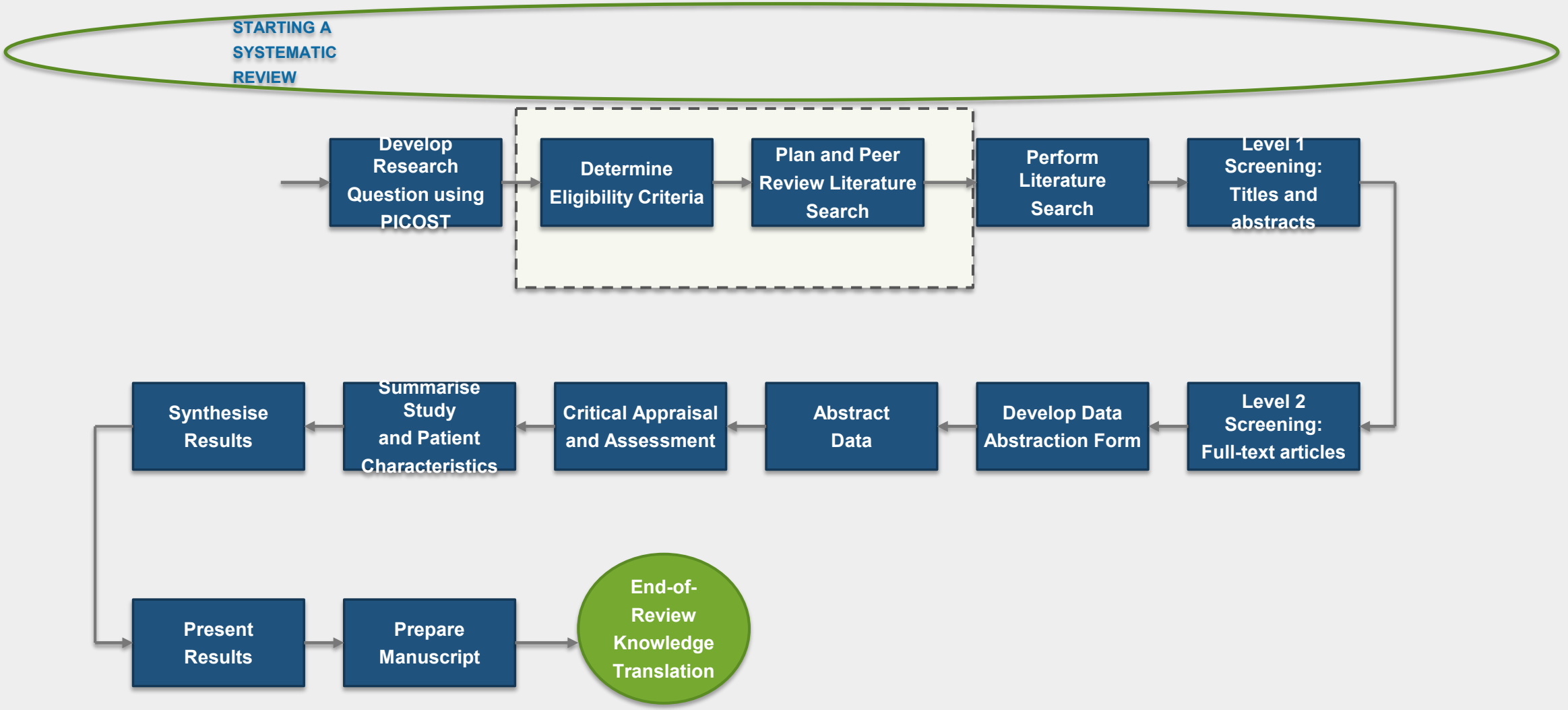
ELIGIBILITY CRITERIA

Ameer Hohlfeld & Mark E Engel

Department of Medicine, University of Cape Town

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Cochrane South Africa


STARTING A SYSTEMATIC REVIEW
Develop Research Question using PICOST
Determine Eligibility Criteria
Plan and Peer Review Literature Search
Perform Literature Search
Level 1 Screening: Titles and abstracts
Level 2 Screening: Full-text articles
Develop Data Abstraction Form
Abstract Data
Critical Appraisal and Assessment
Summarise Study and Patient Characteristics
Synthesise Results
Present Results
Prepare Manuscript
End-of-Review Knowledge Translation

THE "SCIENCE" OF A SYSTEMATIC REVIEW

1. Formulate review question (using a mnemonic)
2. *Define selection criteria for studies (mnemonic, study designs)*
3. Determine search strategy (sources, keywords)
4. Conduct data extraction using pre-designed form
5. Critically appraise study methodologies (bias, confounding)
6. Summarise outcomes data (RR, OR, means etc.)
7. Write a *coherent and logical* summary

Eligibility (Inclusion / Exclusion) criteria

- Key feature of EBHC
- Pre-specifies criteria for identifying studies (to be included in the review)
- Comprises aspects of the clinical question plus specification of study design
- Mnemonic
- Outcomes usually are not part of the criteria
- Typically seek all studies (but could restrict to e.g. randomized trials) of a particular comparison of interventions in a particular population of participants

Defining types of participants: which people and populations?

Usually, the “P” component of a mnemonic

1. How is the disease/condition **defined**?
2. The most **important characteristics** that describe these people (participants)?
3. Relevant **demographic factors** (e.g. age, sex, ethnicity)?
4. The **setting** (e.g. hospital, community, etc)?
5. Who should make the **diagnosis**?
6. **Other types of people** who should be excluded from the review
7. How will studies involving only a subset of relevant participants be handled?

Defining interventions/Control and how they will be grouped

1. What are the **experimental and control** (comparator) interventions of interest?
2. Does the intervention have **variations** (e.g. dosage/intensity, mode of delivery)?
3. Will studies including **only part of the intervention** be included?
4. Will studies including the intervention of interest **combined with another intervention** (co-intervention) be included?
5. Have the **different meanings of phrases** such as ‘control’, ‘placebo’, ‘no intervention’ or ‘usual care’ been considered?

Selecting and prioritizing review outcomes

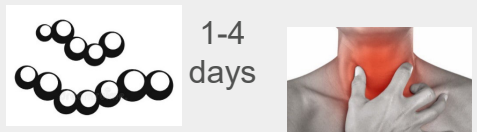
- Consider outcomes **relevant to all potential decision makers**.
  - ❑ **Not** trivial or meaningless
- Additional **outcomes important** to decision makers may also be included in the review.
- Any **outcomes not considered important** to decision makers should be excluded from the review.
- Ensure that outcomes cover potential as well as actual adverse effects.

Determining which study designs to include

- Define in advance the eligibility criteria for study designs in a clear and unambiguous way, **with a focus on what is the best study's design for your question.**

- Justifying choice of study designs
  - ❑ Authors should justify **why they have chosen** either to **restrict the review** to randomized trials or to include non-randomized studies.

UNDERSTANDING THE SPECTRUM OF DISEASE



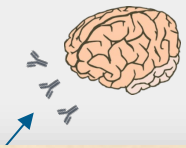
Strep A

1-4 days

Strep throat

2-6 weeks



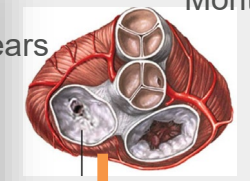


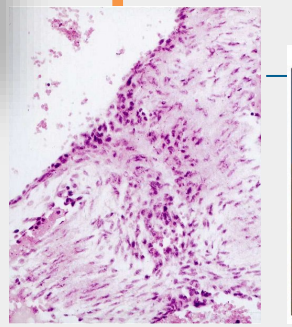


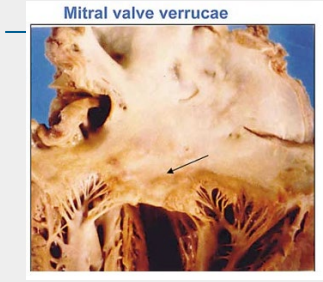


Months to ye

years







Outcomes

- Heart failure
- Atrial fib
- Stroke
- IE

Slide: Ms T Agenson

SCENARIO

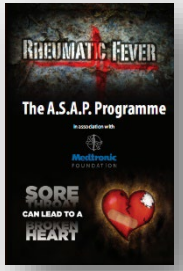
A public health research team in a low-income rural district of KwaZulu-Natal, South Africa, is concerned about the high burden of rheumatic heart disease among young adults. Local clinicians suspect that untreated streptococcal throat infections in childhood may be contributing to the problem. To address this, the team proposes a cluster randomised controlled trial in local primary care clinics and schools. Children aged 5–15 years presenting with a sore throat will be assessed clinically and enrolled if they meet the criteria for pharyngitis. Clinics are randomised to either: Intervention group: administer a single dose of intramuscular benzathine penicillin or a 10-day course of oral penicillin, based on local availability. Control group: provide symptomatic treatment only (e.g. paracetamol), with no antibiotics unless complications develop. The children will be followed for 12 months to assess the incidence of acute rheumatic fever, confirmed by clinical diagnosis and echocardiography, using the revised Jones criteria.

THE REVIEW QUESTION

In people with sore throat, are antibiotics given by any route, compared with no antibiotics, effective in reducing the incidence of rheumatic fever?

BMC Cardiovascular Disorders 2005, 5:11
doi:10.1186/1471-2261-5-11

ANTIBIOTICS FOR REDUCING THE INCIDENCE OF ACUTE RHEUMATIC FEVER FOLLOWING GAS PHARYNGITIS



- Participants
  - People who present with sore throat
- Interventions or Exposures
  - Antibiotics given by any route.
- Comparisons
  - No antibiotics
- Outcome measures
  - Rheumatic Fever



BMC Cardiovascular Disorders

BioMed Central

Research article

Open Access

Antibiotics for the primary prevention of acute rheumatic fever: a meta-analysis

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This article is available from: http://www.biomedcentral.com/1471-2261/5/11

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BMC Cardiovascular Disorders

BioMed Central

Justification

Treatment of streptococcal pharyngitis with antibiotics is currently standard practice in most of the developed world. To note some exceptions, there are a few countries that now recommend no investigation or treatment of sore throat based on recent fears of increasing antibiotic resistance compounded with the findings of a recent study highlighting the negligible benefits gained from antibiotic treatment [14]. This new rationale for treating sore throat with antibiotics may not be applicable to a developing country setting where there remains a real threat of rheumatic fever. Antibiotic treatment of streptococcal pharyngitis appears to be an effective health intervention that is simple to administer; however, its benefits have not been realised in much of the developing world [1].

Controversy exists over the priority that the primary prevention of rheumatic fever deserves in the competition for scarce resources for healthcare in developing countries [1]. In this context it has been argued that secondary rather than primary prevention should be the mainstay of community-based approaches to the control of rheumatic fever and rheumatic heart disease [1]. Unfortunately, these discussions take place in an information vacuum, since the benefits of primary prevention of rheumatic fever, to the best of our knowledge, have not been quantified in a systematic fashion. This review aims to summarise the evidence on the effectiveness of antibiotics for the prevention of acute rheumatic fever, and seeks to determine to what extent the introduction of primary prevention programmes for rheumatic fever is supported by existing research.

Robertson 2005

Study design, case definition, et al

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tify which studies were trials. There were no language restrictions.

Inclusion criteria

We included only randomised and quasi-randomised controlled trials on a patient population presenting with a sore throat (pharyngitis) with or without confirmation of GAS infection by a throat culture and/or a rapid test, and no history of rheumatic fever. Trials that did not specify the method of randomisation and for which the authors were not available for follow-up were excluded.

Robertson 2005

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Case definition

Trials were required to use the Jones Criteria for the diagnosis of rheumatic fever. A positive diagnosis required the presence of 2 major criteria [carditis, migrating polyarthritis of the big joints, chorea, erythema marginatum, subcutaneous nodules] or 1 major and 2 minor criteria [fever, arthralgia without arthritis, previous history of RF or RHD, prolonged P-R interval, elevated erythrocyte sedimentation rate, positive C-reactive protein, leucocytosis] [15].

Interventions et al

Acceptable interventions included any antibiotic versus placebo or no treatment. Acceptable trials included the incidence of a first attack of acute rheumatic fever following throat infection as an outcome. Secondary outcomes that were noted but not required included adverse events to antibiotic use, adherence to antibiotic therapy and mortality associated with the first attack of rheumatic fever.

Robertson 2005

Searching in PubMed

Title of research question:

_____

	Concept	Search terms
P- participants		
I - intervention		
C - control / comparison		
O - outcome		
S - study design		

Methods

Study search

We systematically searched the Cochrane Central Register of Controlled Trials (Cochrane Library Issue 4, 2003), MEDLINE (1966–2003), EMBASE (1966–2003), and the reference lists of identified studies. Search terms included: sore throat, pharyngitis, rheumatic, streptococcal pharyngitis, strep throat, antibiotics, and tonsillitis. We reviewed the selected titles and abstracts (when available) to identify which studies were trials. There were no language restrictions.

Robertson 2005

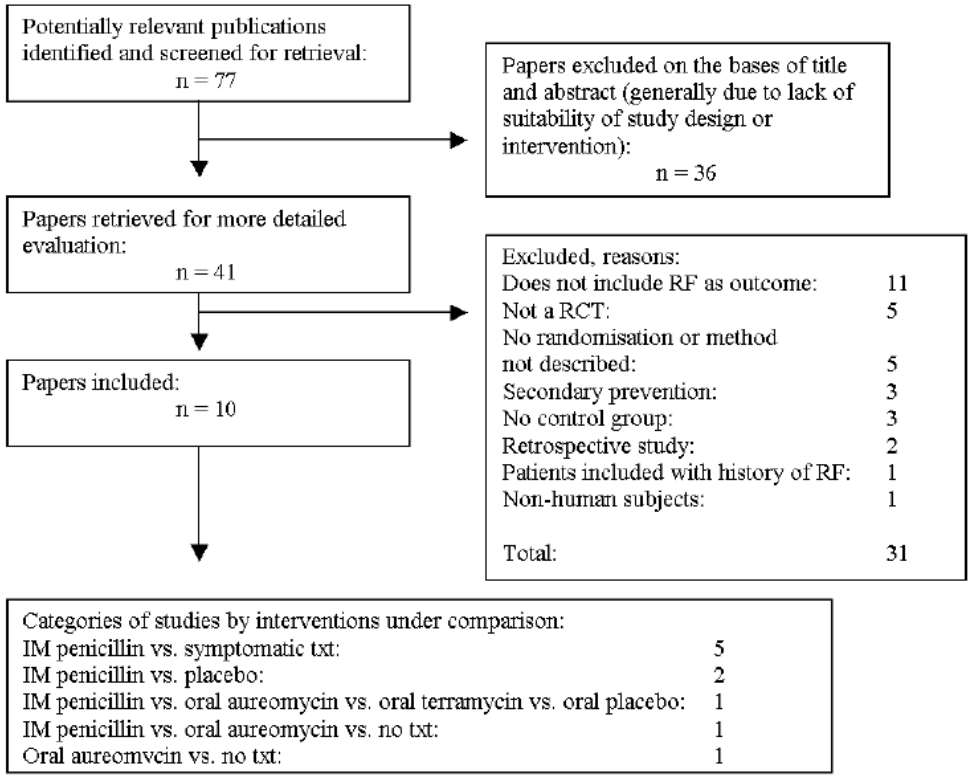


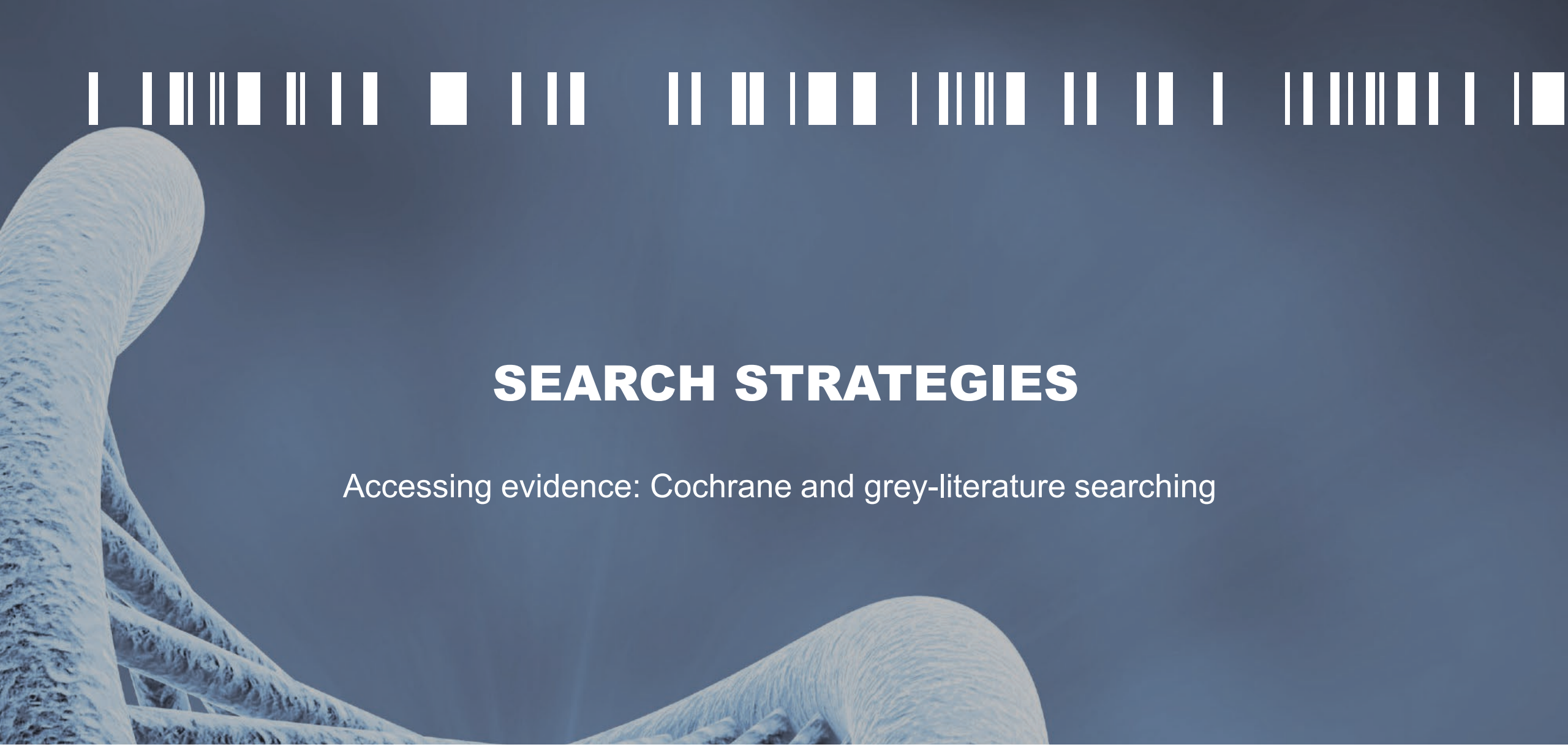
Figure 1
The QUORUM statement.

Robertson 2005

Table 1: Characteristics of included trials

Study ID	Randomisation and concealment of treatment allocation	Participants	Intervention
Bennike, 1951	Quasi-randomised; Inadequate concealment of treatment	349 admitted to hospital with ordinary acute tonsillitis, plegmonous tonsillitis or ulcerative tonsillitis	1. Penicillin: IM, 300,000 units/day for 6 days (adults) 2. Control – symptomatic treatment
Brink, 1951	Quasi-randomised; No concealment of treatment	475 males, aged 17–21, admitted to U.S. military hospital with respiratory symptoms or fever with exudate on tonsils or pharyngeal mucosa	1. Procaine penicillin G: IM 300,000 units/day for 4 days 2. Aureomycin: avg. 2 g/day orally for 4 days 3. Control – no treatment
Brock, 1953	Randomised; No concealment of treatment	349 males admitted to U.S. military hospital with exudative pharyngitis and laboratory-confirmed GAS infection	1. Procaine penicillin G: IM 600,000 units/day for 3 days 2. Control: IM saline placebo, day 1 and day 5
Brumfitt, 1957	Quasi-randomised; No concealment of	121 males, aged 18–21, admitted to U.S. military	1. Combination of procaine penicillin G: IM

Robertson 2005

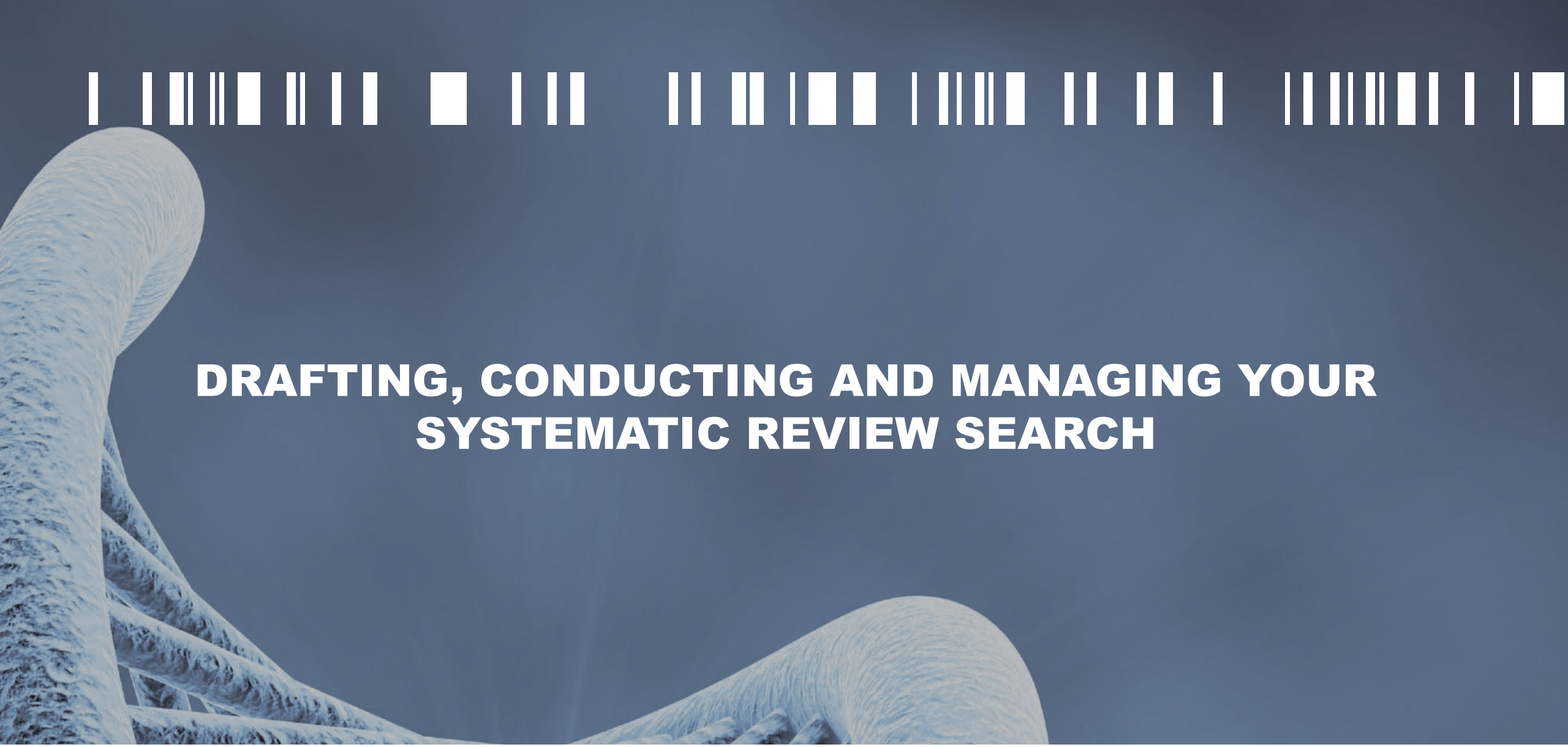


SEARCH STRATEGIES

Accessing evidence: Cochrane and grey-literature searching

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DRAFTING, CONDUCTING AND MANAGING YOUR SYSTEMATIC REVIEW SEARCH

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“WHAT DID TRAVEL MEASURES DO TO COVID-19 SPREAD IN AFRICA?”

OBJECTIVES

- Understand the need for a comprehensive search
- Know what databases and types of literature to search
- Identify key concepts for a search
- Identify search terms
- Understand Boolean operators and MeSH terms
- Develop a search strategy
- Conduct a search in PubMed

PLANNING YOUR SEARCH IS IMPORTANT…

PLANNING YOUR SEARCH IS IMPORTANT…

- Helps you locate relevant information
- Conduct an effective and comprehensive search
- Able to replicate, verify and report your search (important for systematic reviews)

WHERE SHOULD YOU SEARCH?

WHERE SHOULD YOU SEARCH?

- Your systematic review should be comprehensive in its coverage of bibliographic databases
- Search a range of general, specialist, grey literature databases and trial registries
- Choice of where to search is determined by the topic area

ELECTRONIC DATABASES

General	• PubMed • Cochrane Library • Embase • Web of Science • Scopus
Specialist	• African Index Medicus (African health) • ERIC (education) • PsycINFO (psychology, psychiatry and behavioral sciences) • CINAHL (nursing and allied health)

OTHER SOURCES TO SEARCH

- Identify grey literature and new or missed studies
- Reference lists of included studies
- E-table of contents of key journals
- Contact experts
  - Individuals working in the field
  - Authors of included studies
  - Research networks, e.g. mailing lists

GREY LITERATURE

Websites/ Databases	• WHO Library http://www.who.int/library/databases/wholis_tutorial/en/ • OpenGrey (European focus)
Trial registries for ongoing studies. (completed and those in progress)	• Pan African Clinical Trials Registry (www.pactr.org) • WHO International Clinical Trials Registry (http://apps.who.int/trialsearch/)
Dissertations/ theses	• www.netd.ac.za (SA theses and dissertations)
Government reports/research	• SADC http://www.sadc.int/documents-publications/themes/health/
Conference abstracts	• Web of Science • Societies and organizations e.g. www.iasociety.org

BEFORE WE CAN SEARCH, WE NEED TO?

IDENTIFYING KEY CONCEPTS

- Determine the PICO (Mnemonic) from your research question
- Create search terms from the PICO (Mnemonic)
  - Start with the 2-3 most important concepts
- Focus on terms most likely to be found in the title and abstract

SCENARIO: THE MINISTER'S QUESTION

The question WHO put to Cochrane in 2020, asked again by your government. Four steps from question to appraised model

Brief from the National Department of Health

SARS-CoV-2 is spreading internationally and imported cases are arriving.
Are international travel-related control measures (border closures, travel bans, screening and quarantine of travellers) effective in containing the epidemic?

We need an evidence-based answer in two weeks.

.

Burns J, Movsisyan A, Stratil JM, et al. International travel-related control measures to contain the COVID-19 pandemic: a rapid review. Cochrane Database Syst Rev 2021;3:CD013717.

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PICO

CONCEPTS

Participants	Intervention	Comparison	Outcome

CONCEPTS

Participants	Intervention	Comparison	Outcome
Humans of any age susceptible to SARS-CoV-2. Models had to use SARS-CoV-2-specific transmission parameters	Measures on travel across national borders: border closure; travel restrictions or bans; screening at borders; quarantine or isolation of travellers; any combination	No measure; relaxing the measure; later implementation; a less stringent or alternative measure	Primary: cases avoided; cases detected; shift in epidemic development (delay, probability of an outbreak, Rt)

Study design: Experimental and quasi-experimental; observational (cohort, case-control); modelling (compartmental SEIR, Bayesian hierarchical, spatial, time series)

Participants	Intervention	Comparison	Outcome
People susceptible to COVID-19	International travel-related control measures	No measure, or a less strict or later measure	COVID-19 cases avoided, cases detected, epidemic delayed

STEP 1: THE BURNS 2021 REVIEW QUESTION AND PICO

An intervention question, so PICO, with S for study design: Burns included trials, observational studies and models

P	**Population**	Humans of any age susceptible to SARS-CoV-2. Models had to use SARS-CoV-2-specific transmission parameters
I	**Intervention**	Measures on travel across national borders: border closure; travel restrictions or bans; screening at borders; quarantine or isolation of travellers; any combination
C	**Comparator**	No measure; relaxing the measure; later implementation; a less stringent or alternative measure
O	**Outcomes**	Primary: cases avoided; cases detected; shift in epidemic development (delay, probability of an outbreak, Rt)
S	**Study design**	Experimental and quasi-experimental; observational (cohort, case-control); modelling (compartmental SEIR, Bayesian hierarchical, spatial, time series)

The Burns 2021 review question
"To assess the effectiveness of international travel-related control measures during the COVID-19 pandemic on infectious disease transmission and screening-related outcomes."

Why PICO and not another mnemonic?

- **PECO:** for exposures, not deliberate interventions
- **PCC:** for mapping what exists, not effectiveness
- **CHARMS:** for prediction models, not transmission models

Source: Burns et al. 2021, Objectives and 'Criteria for considering studies'. Secondary outcomes (not shown): other transmission outcomes, healthcare use, resources, adverse effects, acceptability.

IDENTIFYING YOUR SEARCH TERMS

- Express each concept of PICO (Mnemonic) in as many ways as possible
- Use text words (used by authors in title and abstract)
  - Include synonyms, alternative spellings, and opposites
- Use truncation to identify
  - Variations of words
  - Example: protect* will retrieve:
    - protect, protected, protection, protective
  - Be careful when using truncation
    - Car* will retrieve cars, carcinoma, cardiovascular, care, etc.

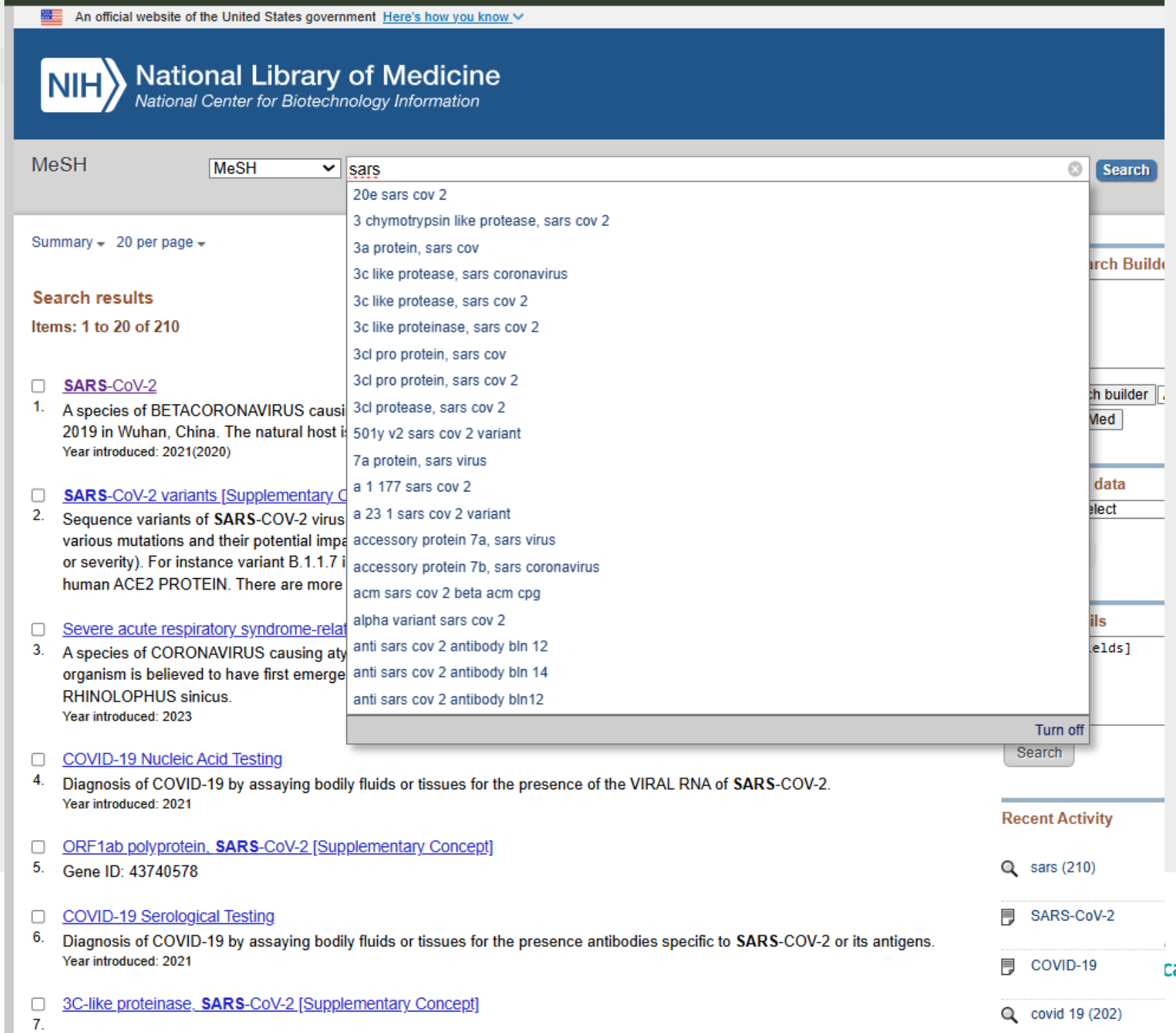
CONSTRUCTING YOUR SEARCH TERMS

- Include indexed terms/subject headings/ thesaurus
  - Terms assigned by indexers, so may not be perfect
  - Ensure the definition of the subject heading/index term is what you need
  - Explode or focus

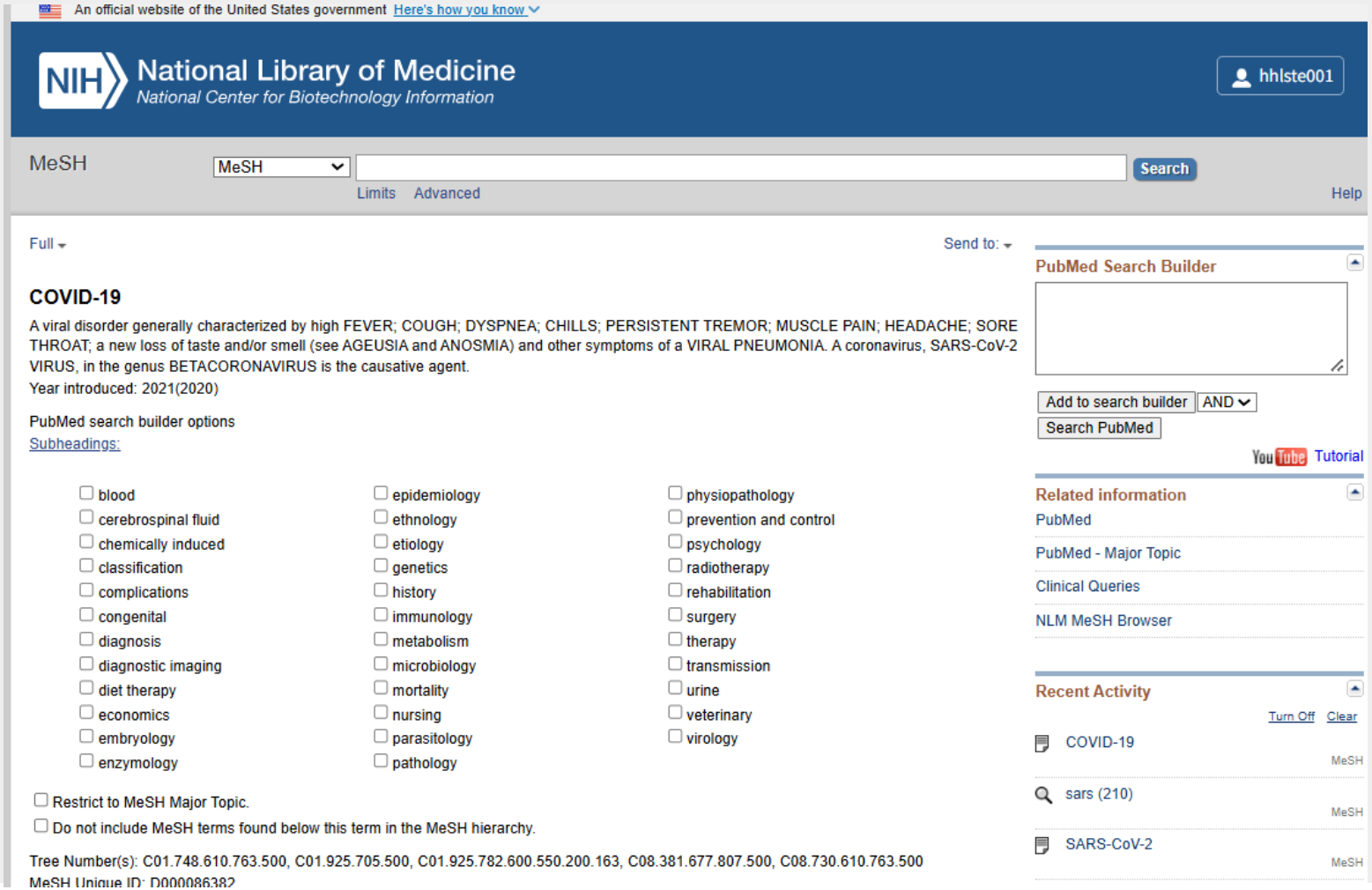
MESH TERMS (YOUR FIRST POINT OF CALL!!)

- National Library of Medicine's controlled and hierarchically-organised vocabulary thesaurus.
- It is used for indexing, cataloguing, and searching of biomedical and health-related information.
- Sets of descriptors in a hierarchical structure that permits searching at various levels of specificity.
- Goes from general (broad headings such as “Mental Disorders”) to specific (e.g., Conduct Disorder)
- 30,000 descriptors in 2022

EXAMPLE OF A MESH TREE



EXAMPLE OF A MESH TREE



- COVID-19 Virus Infection
- COVID 19 Virus Infection
- COVID-19 Virus Infections
- Infection, COVID-19 Virus
- Virus Infection, COVID-19
- COVID-19 Pandemic
- COVID 19 Pandemic
- Pandemic, COVID-19
- COVID-19 Pandemics

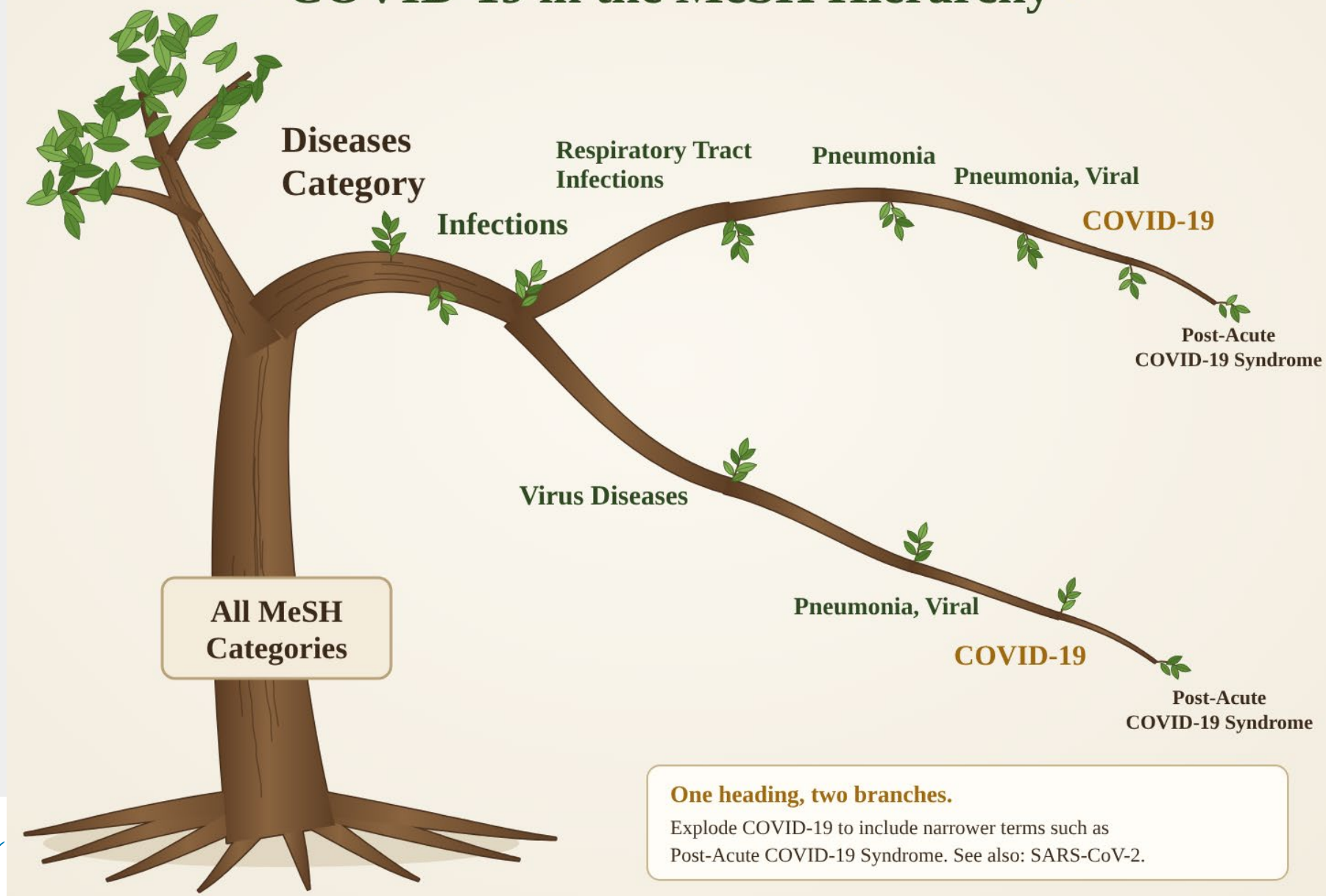
See Also:

- SARS-CoV-2

All MeSH Categories
Diseases Category
Infections
Respiratory Tract Infections
Pneumonia
Pneumonia, Viral
COVID-19
Po cute COVID-19 Syndrome

All MeSH Categories
Diseases Category
Infections
Virus Diseases
Pneumonia, Viral
COVID-19
Post-Acute COVID-19 Syndrome

COVID-19 in the MeSH Hierarchy



One heading, two branches.

Explode COVID-19 to include narrower terms such as Post-Acute COVID-19 Syndrome. See also: SARS-CoV-2.

Concepts

TURNING CONCEPTS INTO SEARCH TERMS

	Participants	Intervention	Comparison	Outcome	Study design
Concept	People susceptible to COVID-19	International travel-related control measures	No measure, or a less strict or later measure	Cases avoided, cases detected, epidemic delayed	
Search terms	? ? ?	? ? ? ?	-	-	

TURNING CONCEPTS INTO SEARCH TERMS

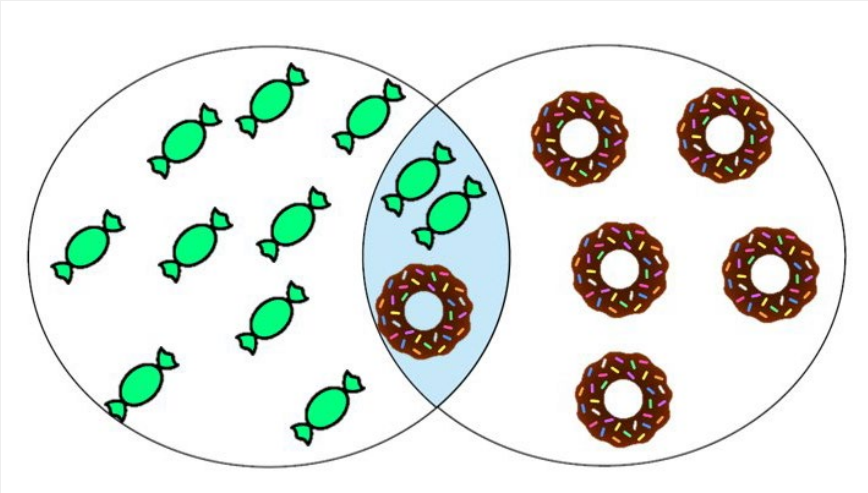
Participants	Intervention	Comparison	Outcome
People susceptible to COVID-19	International travel-related control measures	No measure, or a less strict or later measure	Cases avoided, cases detected, epidemic delayed
COVID-19 SARS-CoV-2 Coronavirus	Travel Border closure Quarantine		

COMBINING YOUR SEARCH TERMS

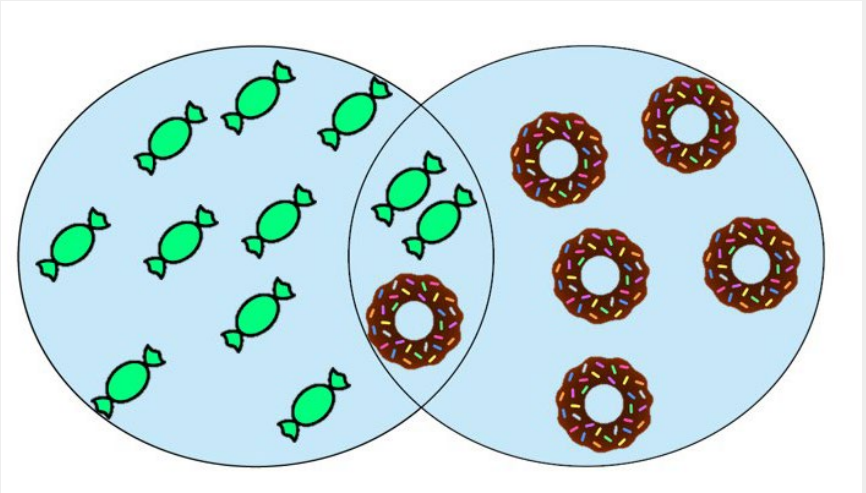
- Search terms are combined using BOOLEAN operators
- The most commonly used are:
  - **OR**
    - Used to join synonyms or related words (within one concept)
    - Instructs the search tool to retrieve any record that contains either (or both) of the terms, thus broadening your search results
  - **AND**
    - Used to join words or phrases when both (or all) the terms must appear in the search results

Boolean Operators

BOOLEAN OPERATORS

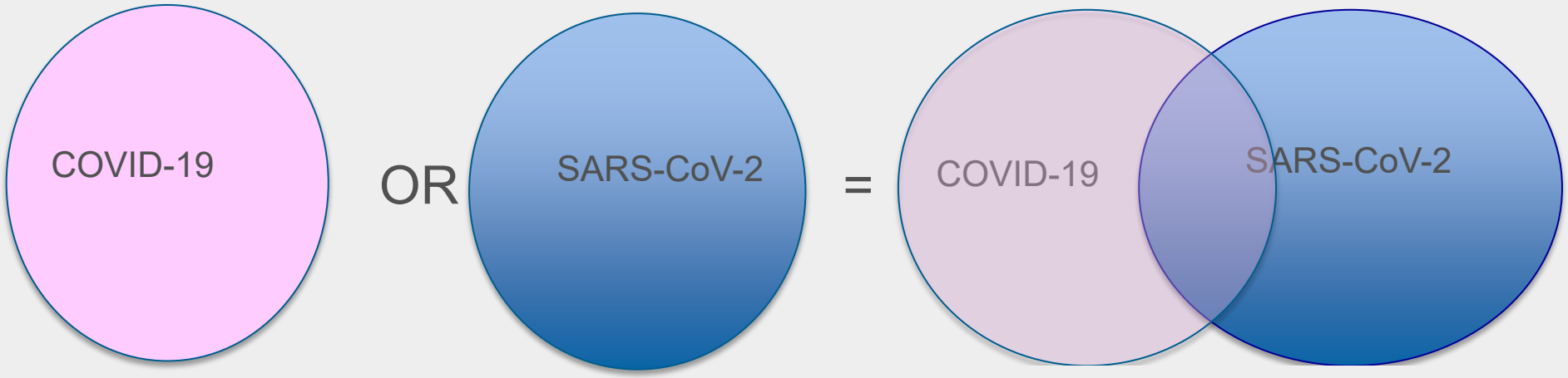


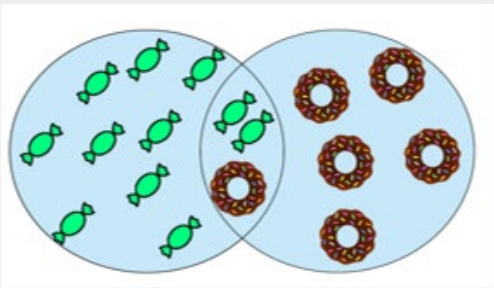
AND - to narrow the search



OR - to expand the search

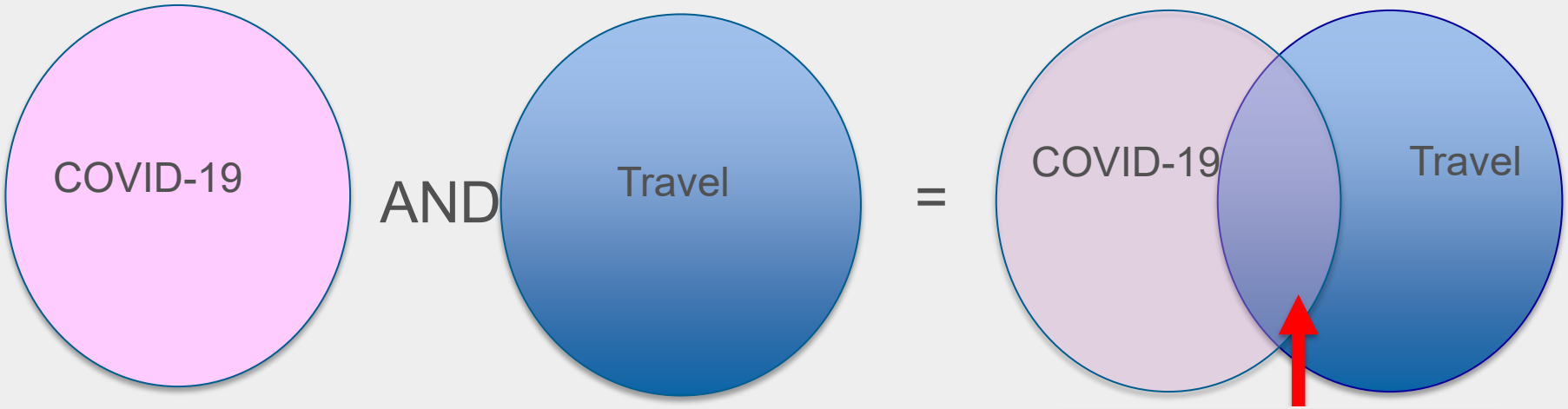
BOOLEAN “OR”

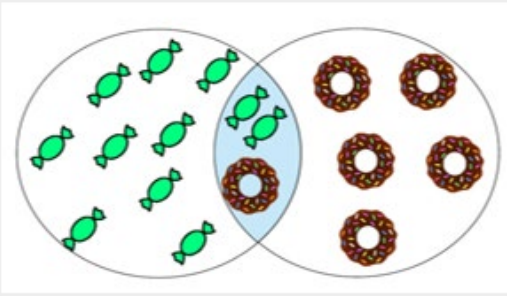




OR - to expand the search

THE BOOLEAN "AND" HYPOTHETICALLY





AND - to narrow the search

BRINGING ALL TOGETHER

Participants
COVID-19
OR
SARS-CoV-2
OR
Coronavirus

AND

Intervention
Travel
OR
Border closure
OR
Quarantine

STEP 2: START FROM A SEARCH

Two concept sets combined with AND. No study-design filter, so trials, observational studies and models are all found

Sources Burns searched

- MEDLINE and Embase (Ovid)
- Cochrane COVID-19 Study Register
- WHO COVID-19 global research database
- Preprints included; all languages
- Searches verified by a content expert and peer reviewed

COCHRANE STRATEGY 1/3: CORONAVIRUSES

Line	Query	Items found
1	exp Coronavirus/	41,876
2	Coronavirus Infections/	40,179
3	COVID-19.rs.	35,243
4	severe acute respiratory syndrome coronavirus 2.os.	29,910
5	(2019 nCoV or 2019nCoV or 2019-novel CoV).ti,ab,kf.	1,285
6	(Coronavir* or corona virus* or Middle East Respiratory Syndrome* or MERS or Severe Acute Respiratory Syndrome* or SARS*).ti,ab,kf.	60,294
7	COVID 19.mp.	68,909
8	(COVID19 or COVID 2019).ti,ab,kf.	1,036
9	(nCov 2019 or nCov 19).ti,ab,kf.	94

Source: Burns et al. 2021, Appendix 3. Ovid MEDLINE, searched 13 November 2020. exp = explode the heading · / = subject heading · .ti,ab,kf = title, abstract, keyword · .rs / .os = COVID-19 registry and supplementary concept · .mp = multi-purpose field · * = truncation

COCHRANE STRATEGY 2/3: TRAVEL MEASURES

Line	Query	Items found
11	Air Travel/	396
12	Travel/	25,483
13	(border? adj3 (clos* or restrict* or control* or measure?)).ab,kf.	1,139
14	((isolat* or quarantin*) adj6 (exposed or suspected or travel* or airport? or border?)).ti,ab,kf.	8,173
15	((mobility or movement*) adj2 (reduc* or restrict*)).ti,ab,kf.	10,041
16	((questionnaire* or RT-PCR or screen* or surveil* or test* or telethermographic* or temperature or thermal imag* or thermal scan* or thermomet* or thermograph*) adj4 (traveller? or entr* or exit or border? or airport?)).ti,ab,kf.	5,328
17	(travel* or border?).ti.	26,695
18	(travel adj4 (measure? or intervention? or NPI?)).ab,kf.	463
Ovid proximity: adj3 = within 3 words of each other · ? = zero or one extra character (border? finds border and borders) · .ti = title only. PubMed has no direct equivalent of adj; use phrases or AND instead.		
20	visa?.ti,ab,kf.	2,175
21	**or/11-20 [Set 2: Travel measures]**	**69,713**

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COCHRANE STRATEGY 3/3: COMBINE AND LIMIT

Line	Query	Items found
10	**Set 1: Coronaviruses**	**97,490**
21	**Set 2: Travel measures**	**69,713**
22	and/10,21 [Sets 1 & 2]	1,700
23	limit 22 to "humans only (removes records about animals)"	1,675

What else Cochrane did

Embase: the same two sets, adapted to Emtree headings; animal studies and conference abstracts removed (2,204 records).

Cochrane COVID-19 Study Register and WHO COVID-19 database: Set 2 travel terms only, because every record is already about COVID-19 (1,064 and 2,287 records).

No study-design filter: trials, observational studies and modelling studies all needed to be found.

STEP 2: START FROM A SEARCH

Two concept sets combined with AND. No study-design filter, so trials, observational studies and models are all found

Ovid MEDLINE, searched 13 November 2020 (Burns 2021, Appendix 3)

Set 1: Coronaviruses (lines 1–10)

```
exp Coronavirus/ OR Coronavirus Infections/ OR COVID-19.rs. OR (Coronavir* or
corona virus* or SARS* or MERS).ti,ab,kf. OR (COVID19 or COVID 2019).ti,ab,kf. …
```

Set 2: Travel measures (lines 11–21)

```
Air Travel/ OR Travel/ OR (border? adj3 (clos* or restrict* or control* or
measure?)).ab,kf. OR ((isolat* or quarantin*) adj6 (exposed or suspected or
travel* or airport? or border?)).ti,ab,kf. OR (travel* adj3 (restrict* or reduc*
or control* or limit* or lockdown? or ban*)).ab,kf. OR visa?.ti,ab,kf. …
```

Line 22: Set 1 AND Set 2 · Line 23: limit to humans

Our cascade reuses Burns' two sets and adds a filter or a model block for each step

- **A** **Reviews of primary studies:** Set 1 AND Set 2 AND systematic review filter
- **B** **Primary studies:** Set 1 AND Set 2 (screen for trials and observational designs)
- **C** **Reviews of modelling studies:** Set 1 AND Set 2 AND model block AND systematic review filter
- **D** **Modelling studies:** Set 1 AND Set 2 AND model block

STEP 3: WORK DOWN THE EVIDENCE CASCADE

Stop at the highest level that answers the question. In this scenario the first three searches come back empty

A

Reviews of primary studies

None found

→

B

Primary studies

None found

→

C

Reviews of modelling studies

None found

→

D

Modelling studies

Found: Adekunle 2020, Anderson 2020, Banholzer 2020

Decision: build our own model, using these models for structure and comparison, not for parameter values. First, appraise them. Which tool?

Teaching scenario. In reality the cascade would stop at a review: Burns 2021 itself. It found 62 studies: 49 modelling and 13 observational, no trials. So the evidence on this question really is mostly models.

WHAT ARE WE SEARCHING FOR?

Building your own model starts with finding what already exists: three searches, one workflow

1 Existing systematic reviews

Search 1

Reviews of models or of primary studies. The fastest route to model structures, assumptions and primary evidence. Check first: has this question already been reviewed?

2 Modelling studies

Search 2

Earlier models of the same disease and intervention. Take their structure and assumptions, and compare your results with theirs. Do not copy their parameter values.

3 Data for model inputs

Search 3

Parameter values come from primary studies, or from reviews of them. One targeted search per input, built as the model develops.

By the end of this block you can: frame a model question as search concepts, build and run a PubMed search for modelling studies, find existing reviews, search for parameters, test and report your search, and re-run it for a rapid update.

FRAME THE MODEL QUESTION FIRST

A PICO-style frame adapted for modelling: disease, scenario, model type and outcome

Worked example question How well have models estimated the effect of travel measures (screening, quarantine, travel restrictions) on the international spread of Ebola virus disease?

Disease	Scenario	Model type	Outcome
Concept 1	Concept 2	Concept 3	Rarely searched
Ebola virus disease	Entry or exit screening, quarantine, travel restrictions	Transmission, simulation or forecasting models	International spread, importations, cases averted

Rule of thumb: search the two or three most distinctive concepts. Outcomes are worded in too many ways to search well. Use your own topic if you have one.

A TERM BANK FOR MODELLING STUDIES

Combine indexed headings with the words authors actually use. Always check the heading in the MeSH browser

Indexed terms (MeSH)

- Models, Theoretical
- Computer Simulation
- Markov Chains
- Monte Carlo Method
- Forecasting

Free-text words (title and abstract)

`model*`	covers model, models, modelling, modelling	`"agent-based"`	agent-based models
`simulat*`	simulation, simulated, simulate	`SEIR`	compartmental model family
`forecast*`	forecast, forecasting	`compartmental`	model structure
`projection*`	projection, projections	`"reproduction number"`	R0 and Rt

Watch out: model* also finds animal models and statistical models. The disease and scenario blocks do the narrowing.

Illustrative terms. Few filters for non-trial designs are widely validated (Kaltenthaler et al., NICE DSU TSD 13, 2011), so test your own.

ASSEMBLE THE SEARCH: OR INSIDE, AND BETWEEN

Each concept becomes a block of alternatives. The blocks are then joined, and every AND narrows the result

#1 COVID-19

```
COVID-19 [mh]
OR
covid* [tiab]
OR
SARS-CoV-2 [tiab]
```

AND

#2 Model

```
Models, Theoretical [mh]
OR
model* [tiab]
OR
simulat* [tiab]
OR
forecast* [tiab]
```

AND

#3 Travel measures

```
Travel [mh]
OR
Quarantine [mh]
OR
border* [tiab]
OR
airport* [tiab]
```

Too many results? Add a block, or limit a word to title and abstract.

Too few? Drop the model block (as Burns 2021 did), add synonyms, or check your MeSH choices.

SEARCH 1: FIND EXISTING SYSTEMATIC REVIEWS

Look for reviews first, before you build a model. #1 and #3 are the COVID-19 and travel blocks from the previous slide

In PubMed

```
(#1 AND #3) AND systematic[sb]
```

The [sb] filter combines a search strategy with the Systematic Review publication type.
systematic review[pt] alone is narrower and can miss records not yet indexed.
Add the model block (#1 AND #2 AND #3 AND systematic[sb]) to find reviews of modelling studies only.
Follow the references of the best review you find, forwards and backwards.

Free sources beyond PubMed

Epistemonikos	reviews linked to their primary studies
Cochrane Library	Cochrane reviews and protocols
PROSPERO	reviews planned, ongoing or completed
Reference lists	of reviews you already trust

Before you use a review: appraise it with ROBIS, as in Exercise 1, and check whether a better or newer one is registered.

Source: PubMed User Guide, systematic reviews subset.

SEARCH 2: COVID-19 TRAVEL-MEASURE MODELS

Worked example. The full search in PubMed Advanced search. Colours match the blocks in "Assemble the search"

#1	("COVID-19"[mh] OR "SARS-CoV-2"[mh] OR covid*[tiab] OR "SARS-CoV-2"[tiab] OR "2019-nCoV"[tiab] OR "novel coronavirus"[tiab])
#2	("Models, Theoretical"[mh] OR "Computer Simulation"[mh] OR model*[tiab] OR simulat*[tiab] OR forecast*[tiab] OR projection*[tiab] OR "agent-based"[tiab] OR SEIR[tiab])
#3	(Travel[mh] OR "Air Travel"[mh] OR Quarantine[mh] OR "Mass Screening"[mh] OR travel*[tiab] OR border*[tiab] OR airport*[tiab] OR quarantin*[tiab] OR visa*[tiab])
#4	**#1 AND #2 AND #3**

Check. Do known papers appear? Look for Adekunle 2020. Anderson and Banholzer were preprints: search medRxiv too.

Read the translation. Open Search details in PubMed to see how it interpreted each line.

Save. Copy the final line and the hit count into your search log.

Illustrative strategy for teaching. Check each MeSH term and run it before class; hit counts change over time.

WHAT TO TAKE FROM AN EXISTING MODEL

Earlier models are a guide to structure. Parameter values should come from primary evidence

Take from existing models

- Model structure and the questions it answers
- Assumptions, and which are contested
- Scenarios and interventions already modelled
- Outputs to compare with your own
- Gaps that justify a new

Take from primary evidence

- Parameter values: incubation period, serial interval, fatality, contact rates
- From primary studies, or reviews of them
- With their uncertainty, not a single number
- Searched one input at a time (Search 3)

Be careful copying values from a model

- Values may be calibrated or assumed, not measured
- The same study can enter your model twice
- The original uncertainty is easily lost
- Their assumptions come with their numbers

Record the source of every input: a primary study, a review of primary studies, or another model. Say which, and treat borrowed model values as the weakest option.

Sources: ISPOR-SMDM Task Force-2, Conceptualizing a Model (2012); Egger et al. F1000Research 2017;6:1584. The points on calibrated values and double counting are methodological reasoning.

SEARCH 3: FIND DATA FOR MODEL INPUTS

Parameter searches are targeted and iterative. They follow the model, not the other way round

A reusable pattern: disease AND parameter words

```
(covid*[tiab] OR "COVID-19"[mh]) AND ("incubation period"[tiab] OR "serial
interval"[tiab] OR "infectious period"[tiab] OR "case fatality"[tiab] OR
"reproduction number"[tiab])
```

1 Search as you build

Information needs emerge as the model develops, so expect several rounds of searching.

2 Favour precision

You rarely need every study. You need good, defensible estimates for each input.

3 Start with high-yield sources

Reviews of primary studies can answer many needs at once. Check where each value came from.

4 Say what you did

Shortcuts raise the risk of missing studies, so justify and document them.

Mind the dates: early COVID-19 inputs come from January to March 2020, often as preprints. A peer-reviewed-only limit would drop them.

Source: Kaltenthaler et al. NICE DSU Technical Support Document 13, 2011 (principles paraphrased).

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BEYOND PUBMED: FREE PLACES TO LOOK

Search at least one more source and say which. Add Embase or Scopus if you have access

Preprints and fast outputs

- Europe PMC (includes preprints)
- medRxiv
- WHO and ministry situation reports

Reviews and registers

- Epistemonikos
- Cochrane Library
- PROSPERO

Regional and citation sources

- African Index Medicus
- WHO Global Index Medicus
- Google Scholar

Why bother: models of a live outbreak often appear first as preprints or reports, and regional sources hold local data PubMed misses.

Fuller lists, with links, are in the resource tables earlier in this section.

LIMITS AND FILTERS: USE THEM WITH CARE

Every limit is a decision to leave something out. Make it deliberately, and report it

Date

Risk. A recent date limit can drop the 1995 and 2000 outbreak studies that supply key parameters.

Safer. Limit by date only for the model and review searches, not for parameter searches.

Language

Risk. English-only searches can miss French-language reports from affected countries.

Safer. Keep all languages at search stage and limit at screening, if you must.

Humans only

Risk. The usual animal-exclusion filter would also drop zoonotic spillover models.

Safer. Skip it, or check what it removes before using it.

Design filters

Risk. No validated filter exists for modelling studies, and the RCT filter does not apply.

Safer. Use your model block, test it on known papers, and report it as untested.

Rule: restrict as late as you can, and write down every limit and why.

TEST AND PEER REVIEW YOUR SEARCH

One attempt rarely gives the final strategy. Check it, then have someone else check it

1 Test it

Build a benchmark set of known relevant papers.
Run the search: are they all retrieved?
If not, find out why: a missing synonym, a wrong heading or a logic error.
Fix it and run again.

2 Peer review it

Ask a second person or an information specialist.
Use the PRESS 2015 checklist:
question translation, operators, subject headings, text words, spelling and syntax, limits and filters.
Involve a content expert to check the jargon.

3 Reuse before you reinvent

Look for a published strategy on a similar topic.
Cochrane Library, PubMed, PROSPERO and Epistemonikos are good places.
Adapt it and credit it. That is good practice, not plagiarism.
Still test it against your own benchmark set.

Source: McGowan et al. PRESS 2015 guideline statement. J Clin Epidemiol 2016;75:40-46.

REPORT AND MANAGE YOUR SEARCH

If someone else cannot repeat your search, it is not a systematic search

Keep a search log

- Source and interface (e.g. PubMed)
- Date the search was run
- Full strategy exactly as run
- Limits applied
- Number of records found
- Who searched, and any peer review
- Later re-runs and changes

Report it: PRISMA-S has 16 items

Items 1 to 7	Where you searched: databases, registers, websites, citation searching, contacts
Items 8 to 12	How: full strategies, limits, filters, strategies reused, updates
Items 13 to 14	When and who checked: search dates, peer review
Items 15 to 16	What you got: record totals and deduplication

Source: Rethlefsen et al. PRISMA-S. Syst Rev 2021;10:39. Use a reference manager (Zotero, EndNote, Mendeley) to hold and deduplicate results.

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RAPID UPDATES: RE-RUN, DON'T RESTART

A saved, logged strategy turns an update into a short job

1 Save
Store the final line in your log. In PubMed, use My NCBI to create an alert

2 Re-run
On a schedule, or when a new outbreak report or model appears.

3 New only
Restrict by entry date: add a date range on [edat].

4 Screen new
Deduplicate against earlier results and screen only the new records.

5 Log
Record the date, hits and any change to the strategy.

`(#1 AND #2 AND #3) AND YYYY/MM/DD:YYYY/MM/DD[edat]` dates are placeholders: use the day after your last search

Rapid-review trade-offs. Fewer sources, date or language limits and a single screener are common shortcuts. Each raises the risk of missing studies, so report them as limitations.

See Garritty et al. Cochrane Rapid Reviews Methods Group guidance. J Clin Epidemiol 2021;130:13-22. PubMed Help for alerts.

SCENARIO EXERCISE: ONE BRIEF

Groups of 3 or 4 · everyone works on the same question · the question WHO put to Cochrane in 2020

Brief from the Department of Health

Travel measures and COVID-19

Imported COVID-19 cases are arriving. Cabinet asks: are international travel-related control measures, such as border closures, travel bans, screening and quarantine of arriving travellers, effective in containing the epidemic?

Before commissioning a new model, your team must find the evidence that already exists.

Work down the evidence cascade

- Search 1: reviews of primary studies
- Primary studies
- Reviews of modelling studies
- Search 2: modelling studies

Tools: the PICO, term bank and COVID-19 MeSH tree slides

Your job: find what already exists before building a model. The real answer is a Cochrane review (Burns et al. 2021) and the models it included. Can your search find them?

SCENARIO EXERCISE: YOUR TASK

Groups of 3 or 4 · 40 minutes · PubMed Advanced search · one brief: the Burns 2021 question

Steps

1	Read the brief. It asks about the effect of an intervention: write it as a PICO question.	3 min
2	Pick the elements worth searching (COVID-19, travel measures). List MeSH headings and free-text words for each.	7 min
3	Build one block per element in Advanced search. Combine with AND, then add systematic[sb] to look for reviews first (Search 1).	8 min
4	Add the model block to find modelling studies (Search 2). Record the hit count for every line.	10 min
5	Test it: are Burns et al. 2021 and Adekunle et al. 2020 retrieved? Fix the search and run it again.	5 min
6	Report back: search lines, hit counts and one change you made. Then compare with Cochrane's strategy.	7 min

Hints

- Use the PICO, term bank and COVID-19 MeSH tree slides.
- The model block is the same as in Search 2.
- Early papers used many names: 2019-nCoV, novel coronavirus, SARS-CoV-2.
- Check Search details to see how PubMed read your line.

Bring to the report-back

Your worksheet, search lines, hit counts, and whether Burns 2021 and Adekunle 2020 were retrieved.

SCENARIO EXERCISE: WORKSHEET

Copy this onto paper or a shared document. One row for each element of your framework

Framework chosen	Our question

Element	Our wording	Search it?	MeSH headings	Free-text words

Combined search line	Hit count	Benchmark papers retrieved

SCENARIO KEY: THE BURNS 2021 BRIEF

Facilitator key (hidden slide). COVID-19, travel measures, model block (as in Search 2). Run every line before class

```
#1  ("COVID-19"[mh] OR "SARS-CoV-2"[mh] OR covid*[tiab] OR "SARS-CoV-2"[tiab] OR "2019-nCoV"[tiab] OR
    "novel coronavirus"[tiab])
#2  (Travel[mh] OR "Air Travel"[mh] OR Quarantine[mh] OR "Mass Screening"[mh] OR travel*[tiab] OR
    border*[tiab] OR airport*[tiab] OR quarantin*[tiab] OR visa*[tiab])
#3  ("Models, Theoretical"[mh] OR "Computer Simulation"[mh] OR model*[tiab] OR
    simulat*[tiab] OR forecast*[tiab] OR projection*[tiab] OR "agent-based"[tiab]
    OR SEIR[tiab])
#4  ("2019/12/01"[edat] : "2020/11/13"[edat])
```

```
#5  #1 AND #2 AND systematic[sb]   Search 1: reviews (should find Burns 2021)
#6  #1 AND #2 AND #3               Search 2: modelling studies (should find Adekunle 2020)
#7  #6 AND #4                      Search 2 within Burns' search window, to compare
```

Benchmarks. #5 should retrieve Burns 2021 (or Hohlfeld 2022); #6 should retrieve Adekunle 2020.

Discuss #4. The date window reproduces Burns' search (to 13 Nov 2020). Use it only to compare: it would miss later studies.

Noise. Anderson and Banholzer were preprints in 2020: PubMed alone may miss them. Burns also searched medRxiv via the WHO database.

Illustrative and untested. Confirm every MeSH heading in the Ovid Embase browser before class.

THE REALITY



All on “Article Management Using Rayyan software” Not for testing

ARTICLE MANAGEMENT USING RAYYAN SOFTWARE

- **User-friendly and intuitive in use**
- **Its free, subscription for additional features**
- **Invite collaborators**
- **Features allow for QUALITY and UNBIASED data extraction**
  - ❑ Predict inclusion and exclusion criteria
  - ❑ Highlights important keywords for identification e.g., RCT, placebo, case study etc.
  - ❑ Add reasons for excluding or including articles e.g., wrong population/study design

ARTICLE MANAGEMENT USING RAYYAN

Three phases take a COVID-19 search result from import to a list of included studies

(1) Planning & organising

- **Creating a profile**
- **Inclusion & exclusion criteria**
- **Filters set up**
- **Deduplication**

(2) Execution

- At least two reviewers
- Blinding
- Screening (title & abstract)

(3) Finalising

- Resolving conflicts
- Screening: full text
- Export included results

Where we start: 4,286 records from PubMed, Embase and Scopus for the COVID-19 travel-measures example. Phase 1 gets them into Rayyan cleanly, before anyone screens.

Illustrative mock-up of the Rayyan screen. Every record, name and number is fictional and made up for teaching.

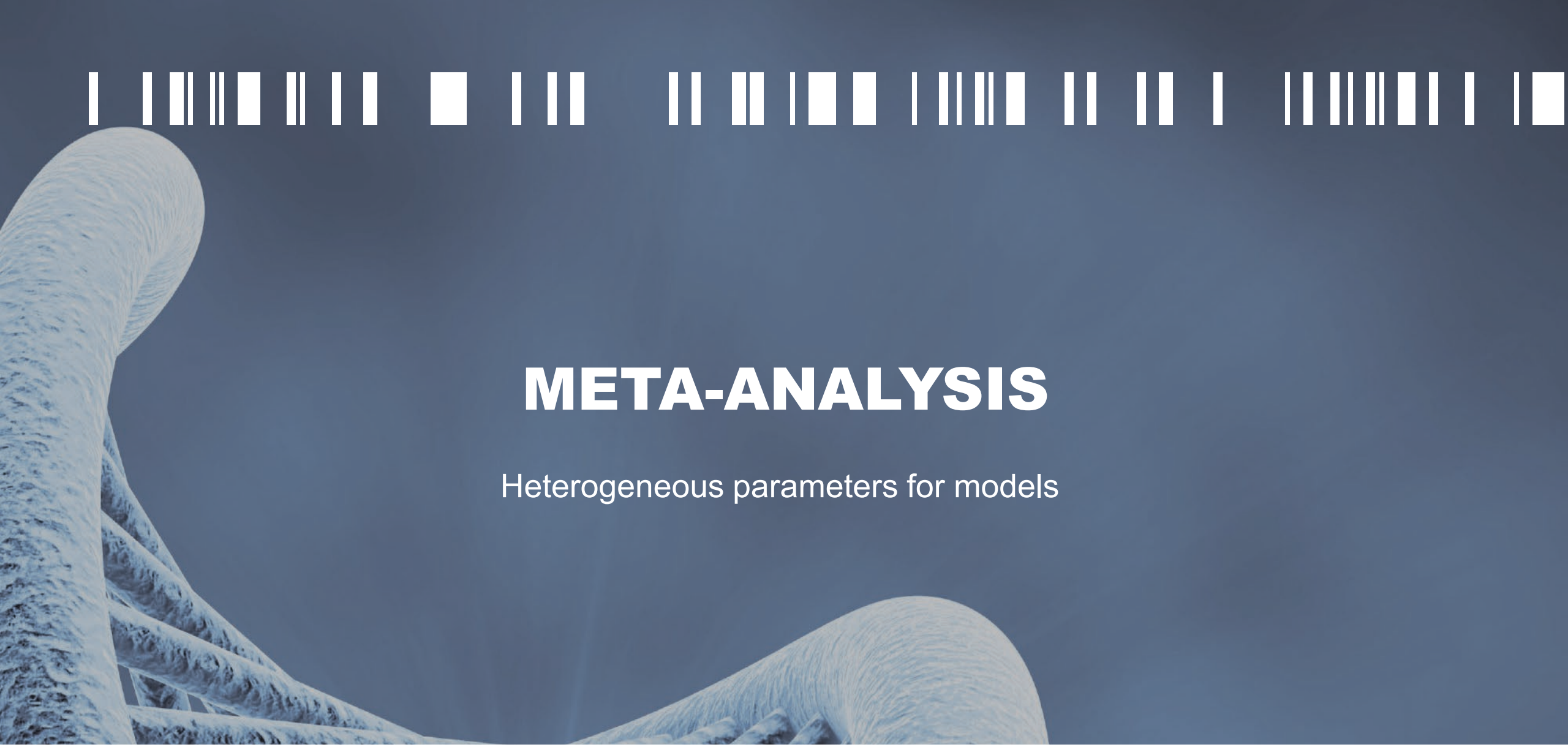
PHASE 2: EXECUTION

Two reviewers screen the same 3,674 records, independently and blinded

(1) Planning & organising	(2) Execution	(3) Finalising
Creating a profile	**At least two reviewers**	Resolving conflicts
Inclusion & exclusion criteria	**Blinding**	Screening: full text
Filters set up	**Screening (title & abstract)**	Export included results
Deduplication		

Why two: A single reviewer misses relevant studies. Independent screening, with blinding on, is the safeguard that makes the included list defensible.

Illustrative mock-up of the Rayyan screen. Every record, name and number is fictional and made up for teaching.

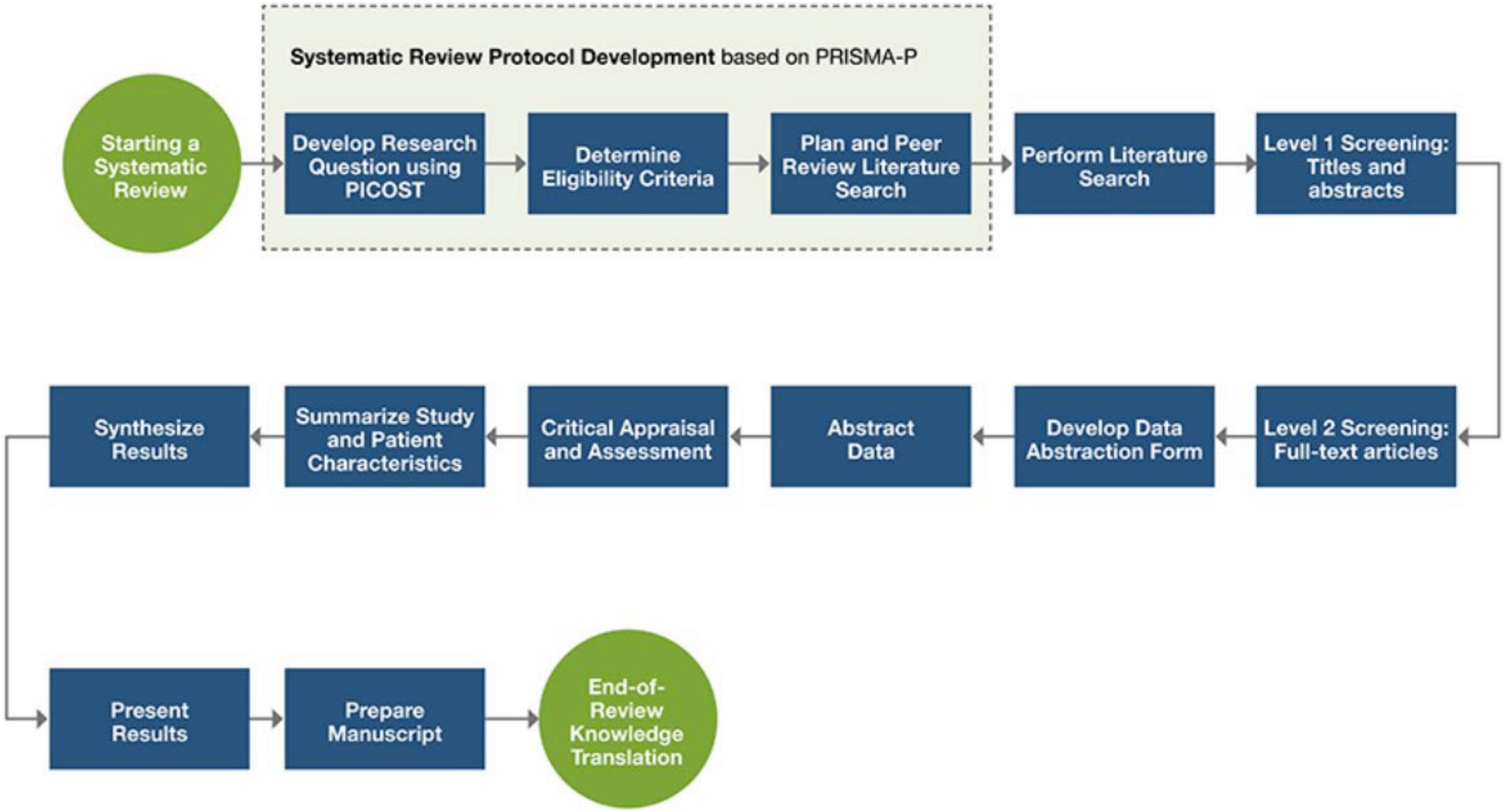


META-ANALYSIS

Heterogeneous parameters for models

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Systematic Review Protocol Development based on PRISMA-P
Starting a Systematic Review
Develop Research Question using PICOST
Determine Eligibility Criteria
Plan and Peer Review Literature Search
Perform Literature Search
Level 1 Screening: Titles and abstracts
Level 2 Screening: Full-text articles
Develop Data Abstraction Form
Abstract Data
Critical Appraisal and Assessment
Summarize Study and Patient Characteristics
Synthesize Results
Present Results
Prepare Manuscript
End-of-Review Knowledge Translation

Data Extraction

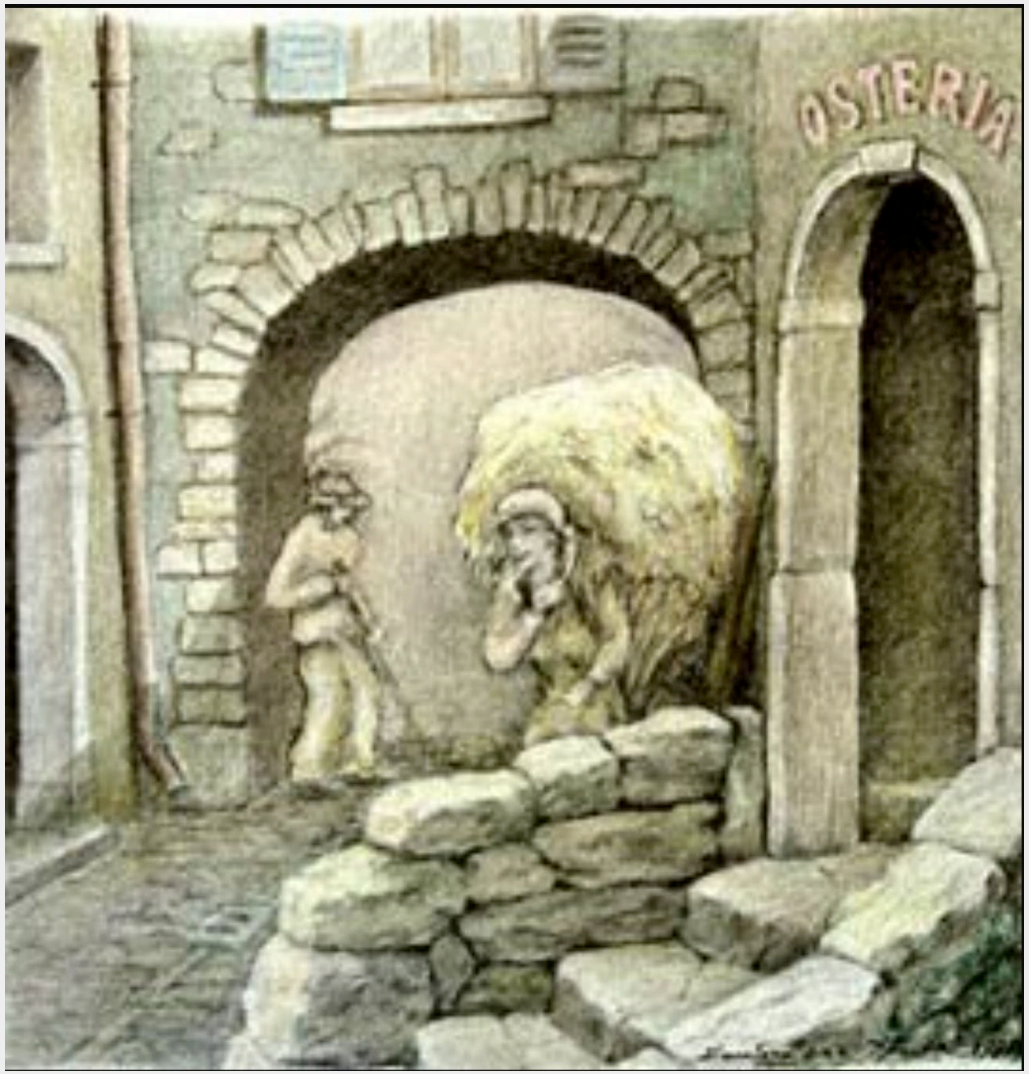
Data extraction serves to focus your preparation for write-up

It can be difficult and time consuming

Often tend to extract too much data – protocol NB

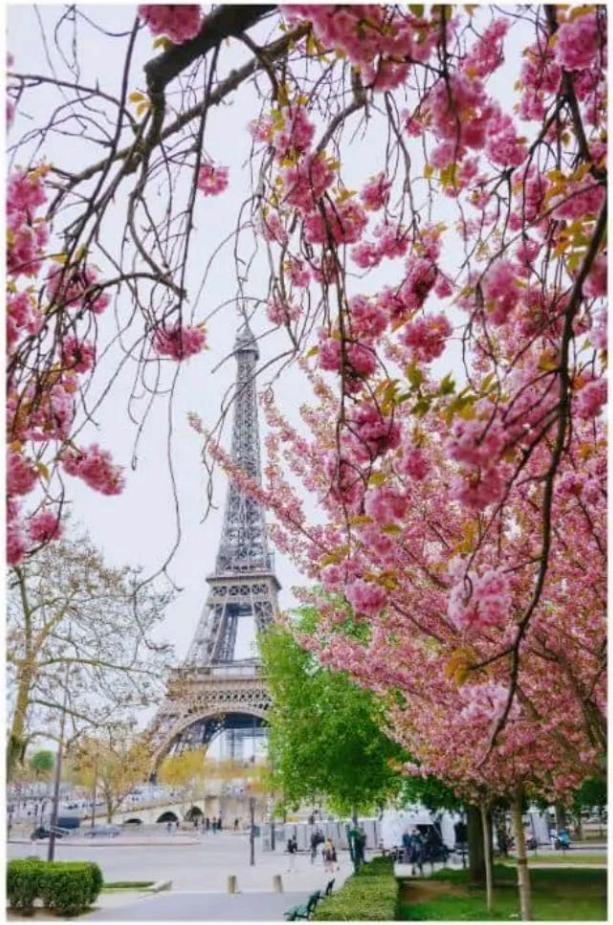
Often limited by reporting




OSTERIA

CHERRY BLOSSOM IN PARIS





PARIS
IN THE
THE SPRING

Overview of process

Design and pilot data extraction forms

Data extracted independently by 2 authors

Compare data extracted and resolve discrepancies

Extracted data transferred into Stats Programme and analysed

What is a Meta-analysis?

- A statistical procedure that pools the results of several independent studies considered sufficiently similar (homogeneous)
- Provides a quantitative summary of the overall treatment effect
- Meta-analysis forms part of a systematic review when appropriate



Wasserman *et al. BMC Infectious Diseases* (2016) 16:482
DOI 10.1186/s12879-016-1809-3

BMC Infectious Diseases

RESEARCH ARTICLE Open Access

Burden of pneumocystis pneumonia in HIV-infected adults in sub-Saharan Africa: a systematic review and meta-analysis

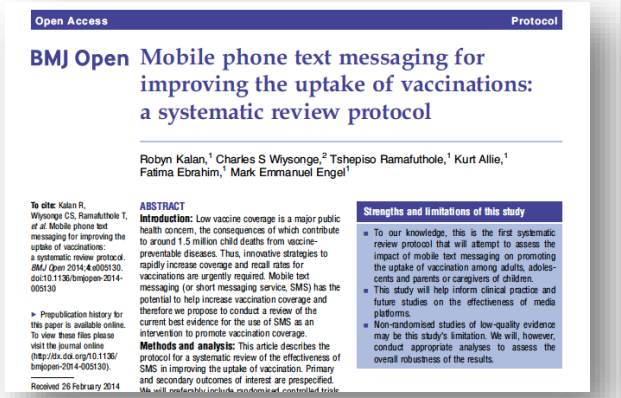
Sean Wasserman[1*], Mark E. Engel[2], Rulan Griesel[3] and Marc Mendelson[1]

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Outcome measures in MAs

- Dichotomous
  - risk ratio, odds ratio, prevalence ratio or risk difference
- Continuous
  - mean difference, standardised mean difference or prevalence (*proportion between 0...1*)



Open Access Protocol

BMJ Open

Mobile phone text messaging for improving the uptake of vaccinations: a systematic review protocol

Robyn Kalan,[1] Charles S Wiysonge,[2] Tshepiso Ramafuthole,[1] Kurt Allie,[1] Fatima Ebrahim,[1] Mark Emmanuel Engel[1]

To cite: Kalan R, Wiysonge CS, Ramafuthole T, *et al*. Mobile phone text messaging for improving the uptake of vaccinations: a systematic review protocol. *BMJ Open* 2014;**4**:e005130. doi:10.1136/bmjopen-2014-005130

► Prepublication history for this paper is available online. To view these files please visit the journal online (http://dx.doi.org/10.1136/bmjopen-2014-005130).

Received 26 February 2014

ABSTRACT

Introduction: Low vaccine coverage is a major public health concern, the consequences of which contribute to around 1.5 million child deaths from vaccine-preventable diseases. Thus, innovative strategies to rapidly increase coverage and recall rates for vaccinations are urgently required. Mobile text messaging (or short messaging service, SMS) has the potential to help increase vaccination coverage and therefore we propose to conduct a review of the current best evidence for the use of SMS as an intervention to promote vaccination coverage.

Methods and analysis: This article describes the protocol for a systematic review of the effectiveness of SMS in improving the uptake of vaccination. Primary and secondary outcomes of interest are prespecified.

Strengths and limitations of this study

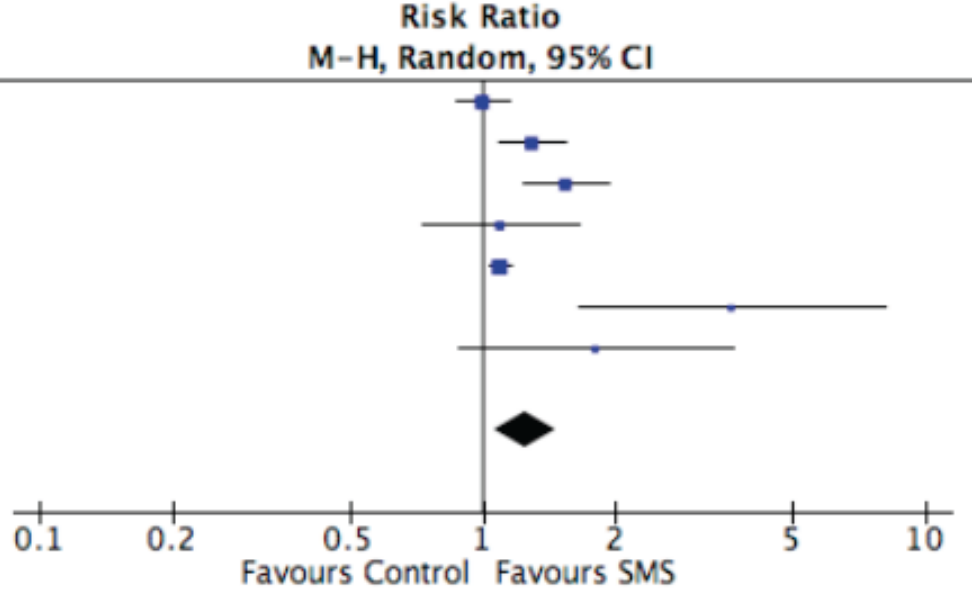
- To our knowledge, this is the first systematic review protocol that will attempt to assess the impact of mobile text messaging on promoting the uptake of vaccination among adults, adolescents and parents or caregivers of children.
- This study will help inform clinical practice and future studies on the effectiveness of media platforms.
- Non-randomised studies of low-quality evidence may be this study's limitation. We will, however, conduct appropriate analyses to assess the overall robustness of the results.

Figure 1: SMS and Vaccination: All trials

Study or Subgroup	SMS Events	SMS Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Ahlers-Schmidt 2012	45	50	36	40	21.5%	1.00 [0.87, 1.15]
Chai 2013	234	999	181	999	19.6%	1.29 [1.09, 1.54]
Kharbanda 2011	67	124	108	308	17.0%	1.54 [1.23, 1.92]
Moniz 2013	34	108	31	108	9.5%	1.10 [0.73, 1.65]
Stockwell 2012a	1653	4607	1509	4606	24.9%	1.10 [1.03, 1.16]
Stockwell 2012b (Adolesc)	30	195	7	166	3.4%	3.65 [1.65, 8.09]
Stockwell 2012b (Paed)	18	87	10	87	4.1%	1.80 [0.88, 3.67]
Total (95% CI)		**6170**		**6314**	**100.0%**	**1.25 [1.07, 1.46]**
Total events	2081		1882			

Heterogeneity: $Tau^2 = 0.03$; $Chi^2 = 24.34$, df = 6 (P = 0.0005); $I^2 = 75\%$

Test for overall effect: Z = 2.78 (P = 0.005)



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Heterogeneity

"variation in study outcomes between studies…"

1- Due to chance
2- Due to scale used to measure the effect
3- Due to treatment characteristics
4- Due to patient level covariates
5- Unexplainable
6- Characteristics of the design and conduct of the studies.

When not to do MA?

Clinical diversity / heterogeneity

- variability in the participants, interventions and outcomes studied

Methodological diversity / heterogeneity

- variability in trial design and quality

Poor quality studies

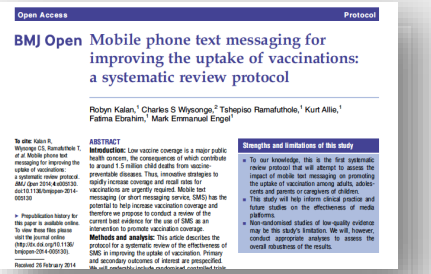
- variability in risk of bias

Cochrane's Q test for heterogeneity

- Null hypothesis – all studies measuring same effect
- Q = Sum of squared deviations of study estimates from the overall estimate (weighted)
- Compared to chi-square distribution with k-1 df (k = number of trials included)
- Low power if k small
- $p < 0.10$ taken as evidence that inter-study differences were not solely due to chance

Exploring heterogeneity by means of subgroup analyses

- Used if different effect anticipated in particular category of participants
- E.g. subgroup by gender, HIV status, class of drug, etc.
- Guidelines
  - Consider only if indirect evidence in support of a difference
  - Specify in review protocol
  - Keep to a minimum


Open Access Protocol

BMJ Open

Mobile phone text messaging for improving the uptake of vaccinations: a systematic review protocol

Robyn Kalan,[1] Charles S Wiysonge,[2] Tshepiso Ramafuthole,[1] Kurt Allie,[1] Fatima Ebrahim,[1] Mark Emmanuel Engel[1]

ABSTRACT

Strengths and limitations of this study

- To our knowledge, this is the first systematic review protocol that will attempt to assess the impact of mobile text messaging on promoting the uptake of vaccination among adults, adolescents and parents or caregivers of children.
- This study will help inform clinical practice and future studies on the effectiveness of media platforms.
- Non-randomised studies of low-quality evidence may be this study's limitation. We will, however, conduct appropriate analyses to assess the overall robustness of the results.

Study or Subgroup	SMS Events	SMS Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.9.1 blinding reported						
Chai 2013	234	999	181	999	27.9%	1.29 [1.09, 1.54]
Moniz 2013	34	108	31	108	7.2%	1.10 [0.73, 1.65]
Stockwell 2012a	1653	4607	1509	4606	64.9%	1.10 [1.03, 1.16]
Subtotal (95% CI)		**5714**		**5713**	**100.0%**	**1.15 [1.02, 1.29]**
Total events	1921		1721			
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 3.21$, df = 2 (P = 0.20); $I^2 = 38\%$						
Test for overall effect: Z = 2.35 (P = 0.02)						
1.9.2 Blinding not reported						
Ahlers-Schmidt 2012	45	50	36	40	30.4%	1.00 [0.87, 1.15]
Kharbanda 2011	67	124	108	308	29.5%	1.54 [1.23, 1.92]
Stockwell 2012b (Adolesc)	30	195	7	166	19.3%	3.65 [1.65, 8.09]
Stockwell 2012b (Paed)	18	87	10	87	20.8%	1.80 [0.88, 3.67]
Subtotal (95% CI)		**456**		**601**	**100.0%**	**1.65 [0.93, 2.92]**
Total events	160		161			
Heterogeneity: $Tau^2 = 0.27$; $Chi^2 = 39.43$, df = 3 (P < 0.00001); $I^2 = 92\%$						
Test for overall effect: Z = 1.72 (P = 0.09)						
Test for subgroup differences: $Chi^2 = 1.49$, df = 1 (P = 0.22), $I^2 = 32.9\%$						

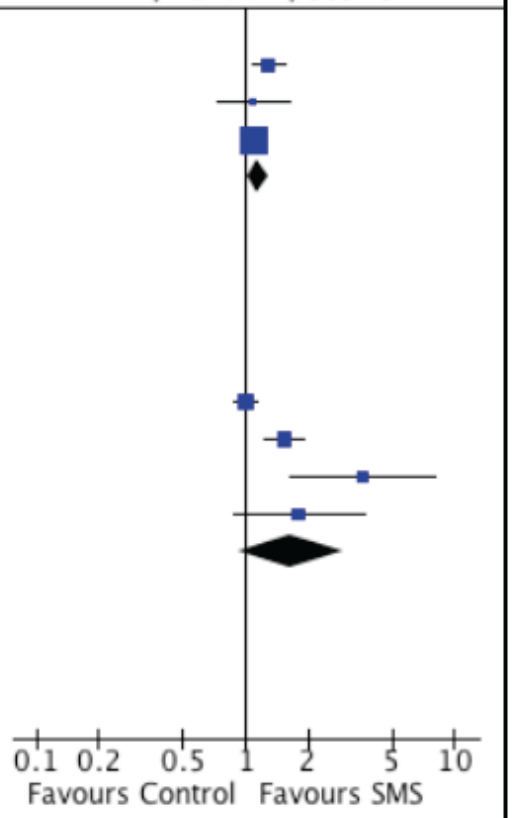


Figure 8: Blinding

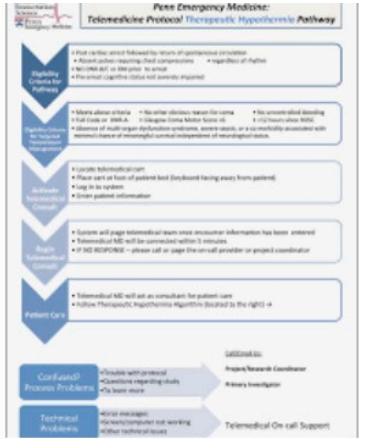
British Journal of Anaesthesia **110** (3): 357–67 (2013)
Advance Access publication 25 January 2013 · doi:10.1093/bja/aes500

BJA

Role of therapeutic hypothermia in improving outcome after traumatic brain injury: a systematic review

A. P. Georgiou[1]* and A. R. Manara[2]





ic review was undertaken to review the evidence now available and to re-evaluate the risks and benefits of PTH when used in the management of TBI. It aims to answer the following specific questions:

(i) Does PTH improve survival in patients with TBI?
(ii) Does PTH improve subsequent neurological function in patients who survive TBI?
(iii) Is PTH safe when used in the context of TBI?

PTH may be defined as the deliberate lowering of core body temperature initiated on presentation of the patient, in order to achieve a beneficial outcome. This should be differentiated

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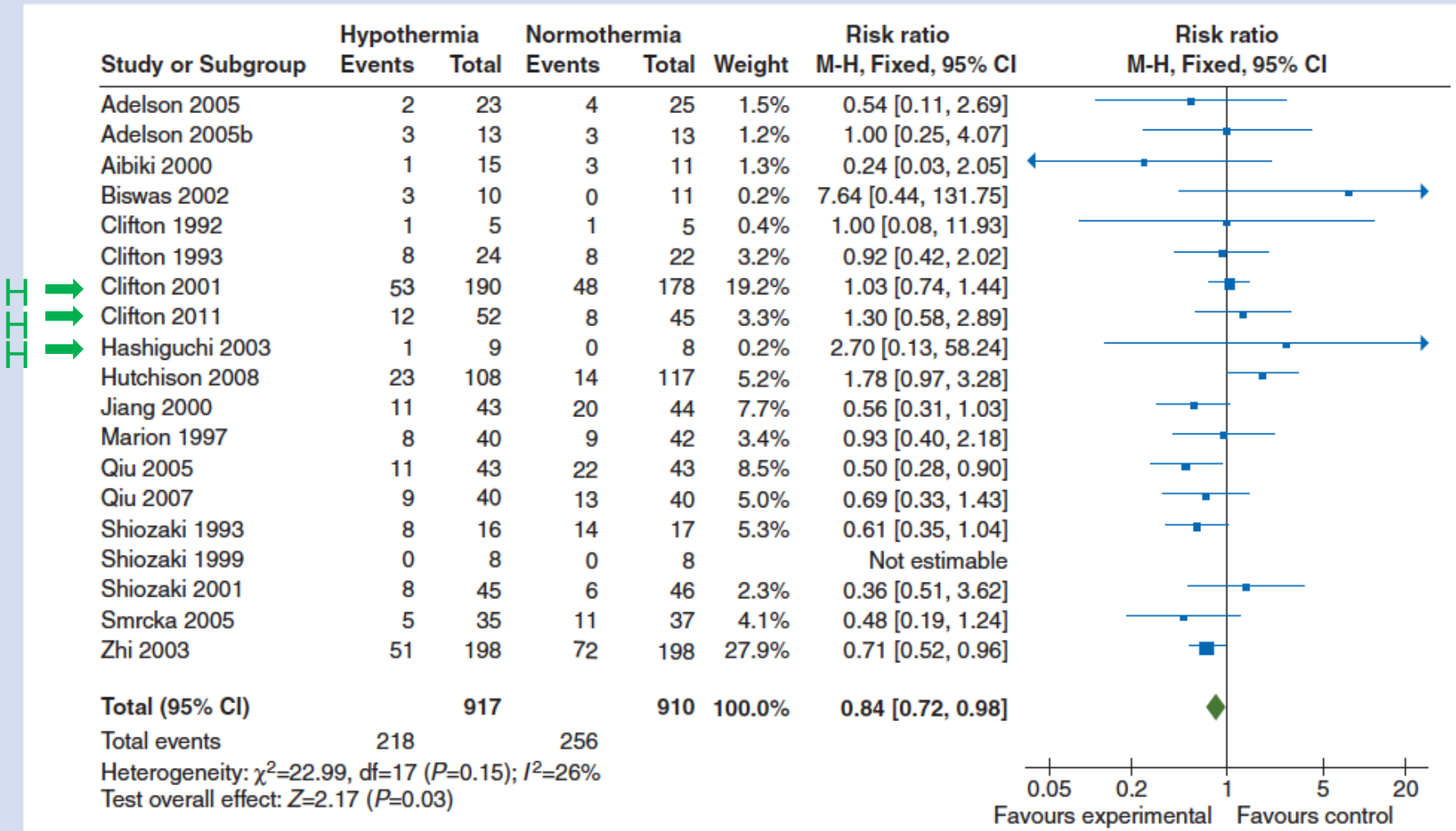


Fig 2 Forest plot of all included trials examining the effect of prophylactic therapeutic hypothermia *vs* normothermia on mortality in TBI. The estimate of heterogeneity places a numerical value on the variation in the results of individual studies, thereby suggesting the confidence that one can have in the combined estimate. The results from these studies are relatively heterogeneous.

Importance of quality appraisal

Study or Subgroup	Hypothermia Events	Total	Normothermia Events	Total	Weight	Odds ratio M-H, Fixed, 95% CI
Clifton 2001	53	190	48	178	67.5%	1.05 [0.66, 1.66]
Clifton 2011	12	52	8	45	12.5%	1.39 [0.51, 3.77]
Hutchison 2008	23	108	14	117	20.0%	1.99 [0.97, 4.11]
Total (95% CI)		**350**		**340**	**100.0%**	**1.28 [0.89, 1.83]**
Total events	88		70			

Heterogeneity: χ^2=2.19, df=2 (P=0.33); I^2=9%
Test for overall effect: Z=1.34 (P=0.18)

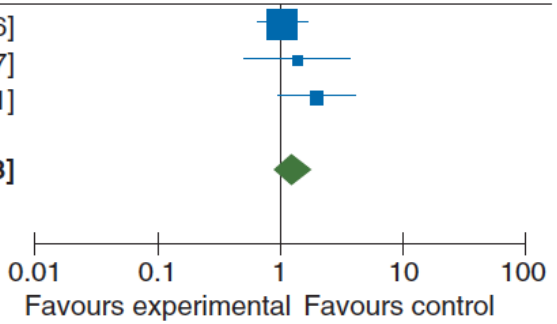
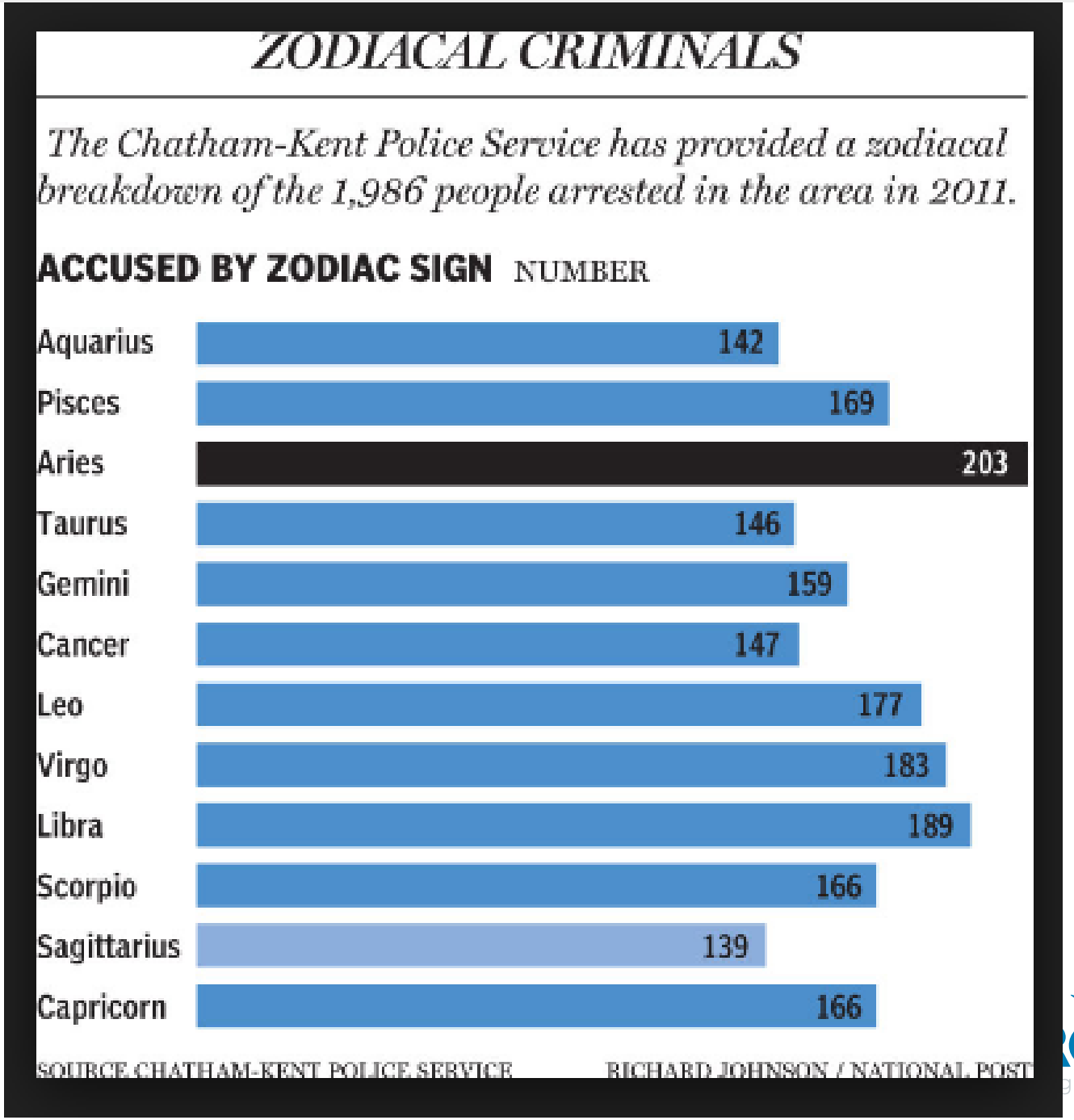


Fig 4 Forest plot of only high-quality trials, examining the effect of prophylactic therapeutic hypothermia *vs* normothermia on mortality in TBI.

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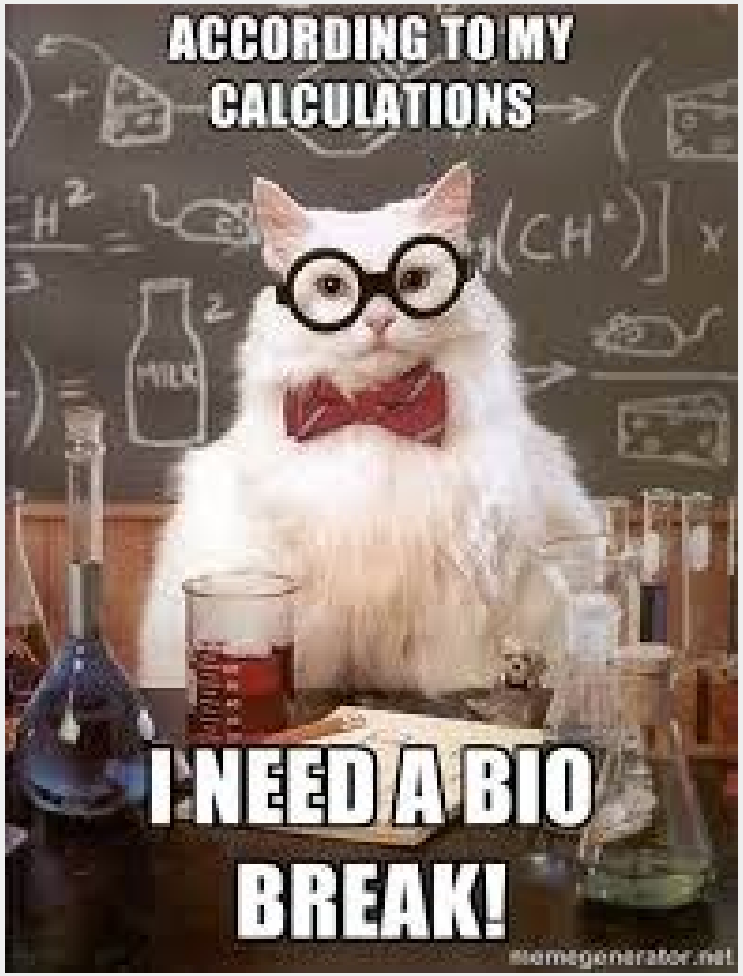
Importance of the protocol – avoids "fishing"



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QUALITY APPRAISAL LAB

Risk of bias in primary studies

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Critically appraise study methodologies (bias, confounding)

Selection bias	Systematic differences between baseline characteristics of the groups that are compared.	*Sequence generation; Allocation concealment.*
Performance bias	Systematic differences between groups in the care that is provided, or in exposure to factors other than the interventions of interest.	*Blinding of participants, personnel and outcome assessors; Other potential threats to validity.*
Attrition bias	Systematic differences between groups in withdrawals from a study.	*Incomplete outcome data; Blinding of participants, personnel and outcome assessors.*
Detection bias	Systematic differences between groups in how outcomes are determined.	*Blinding of participants, personnel and outcome assessors; Other potential threats to validity.*
Reporting bias	Systematic differences between reported and unreported findings.	*Selective outcome reporting*

University of BRISTOL

Announcement of r
bias in randomized
and in non-random

Julian Hig

Cochrane M

Home

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An official website of the United States government Here's how you know

Quality Rating (Good, Fair, or Poor)
Rater #1 initials:
Rater #2 initials:
Additional Comments (If POOR, please state why):

*CD, cannot determine; NA, not applicable; NR, not reported

Guidance for Assessing the Quality of Observational Cohort and Cross-Sectional Studies

The guidance document below is organized by question number from the tool for quality assessment of observational cohort and cross-sectional studies.

Question 1. Research question

Did the authors describe their goal in conducting this research? Is it easy to understand what they were looking to find? This issue is important for any scientific paper of any type. Higher quality scientific research explicitly defines a research question.

Questions 2 and 3. Study population

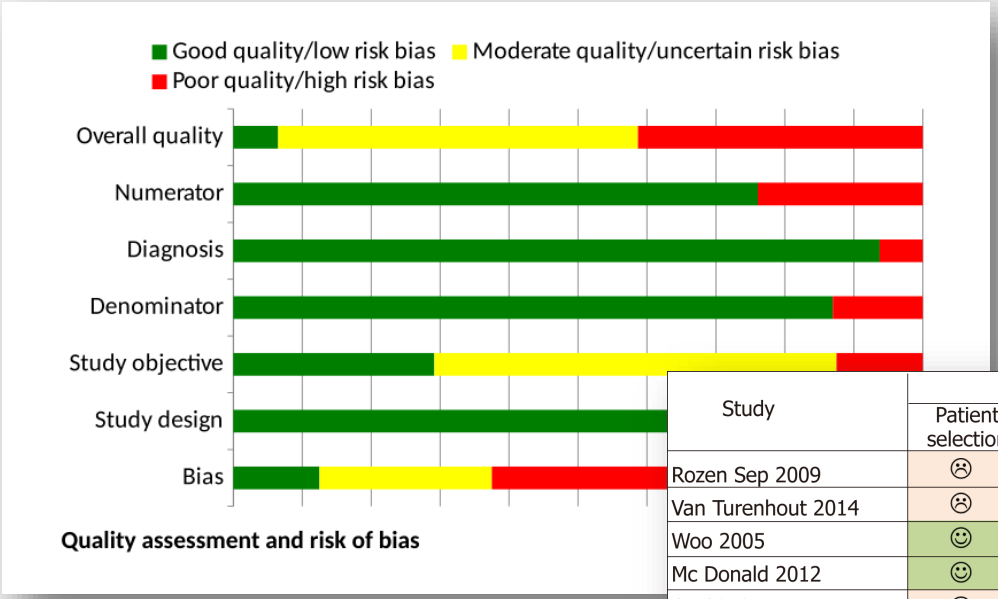
Did the authors describe the group of people from which the study participants were selected or recruited, using demographics, location, and time period? If you were to conduct this study again, would you know who to recruit, from where, and from what time period? Is the cohort population free of the outcomes of interest at the time they were recruited?

An example would be men over 40 years old with type 2 diabetes who began seeking medical care at Phoenix Good Samaritan Hospital between January 1, 1990 and December 31, 1994. In this example, the population is clearly described as: (1) who (men over 40 years old with type 2 diabetes); (2) where (Phoenix Good Samaritan Hospital); and (3) when (between January 1, 1990 and December 31, 1994). Another example is women ages

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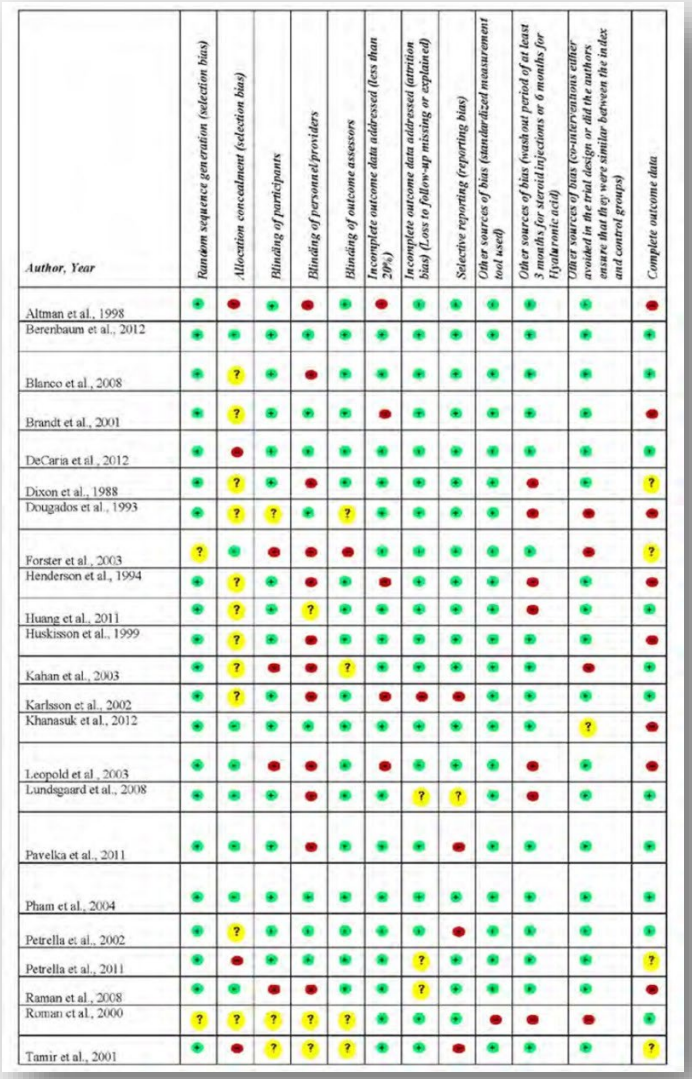
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Quality assessment and risk of bias

Study	Risk of bias				Applicability concerns		
	Patient selection	Index test	Reference standard	Flow and timing	Patient selection	Index test	Reference standard
Rozen Sep 2009	☹	☺	☺	☺	☹	☺	☺
Van Turenhout 2014	☹	☺	?	?	☹	☺	☺
Woo 2005	☺	☺	☺	☺	☹	☺	☺
Mc Donald 2012	☺	☺	?	☺	☹	☺	☺
Ou 2013	☹	☺	☺	☺	☹	☺	☺
Auge 2015	☹	☺	☺	☺	☹	☺	☺
Auge 2018	?	☺	☺	☺	☹	☺	☺
Symonds 2015	?	☺	☺	☺	☹	☺	☺
Parente 2012	☹	☺	☺	☺	☹	☺	☺
Cubiella 2012	☺	☺	☺	☺	☹	☺	☺
Cubiella 2016	☺	☺	☺	☺	☹	☺	☺
Rodríguez-alonso 2015	☹	☺	☺	☺	☹	☺	☺
Mowat 2015	?	☺	☺	☺	☹	☺	☺
Godber 2015	☺	☺	?	☺	☹	☺	☺
Widlack 2016	☺	☺	?	☺	☹	☺	?

☺ Low Risk ☹ High Risk ? Unclear Risk



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Pharyngeal microbiota: Systematic review and Rationale and design of the AFRO*Strep* GAS pharyngitis microbiota study

Kimona Rampersadh, Taariq Salie, Clinton Moodley, Marc Nicol, Kelin K Engel, Liesl J Zuhlke, Mark E Engel

.

	RoB 1	RoB 2	RoB 3	RoB 4	RoB 5	RoB 6	RoB 7	RoB 8	RoB 9
Low	8	8	8	6		8	8	8	6
Unclear	0	0	0	2		0	0	0	2
High	0	0	0	0		0	0	0	0

Coding Low = Unclear= High= N/A=

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RISK OF BIAS TOOLS: MATCH THE TOOL TO THE STUDY DESIGN

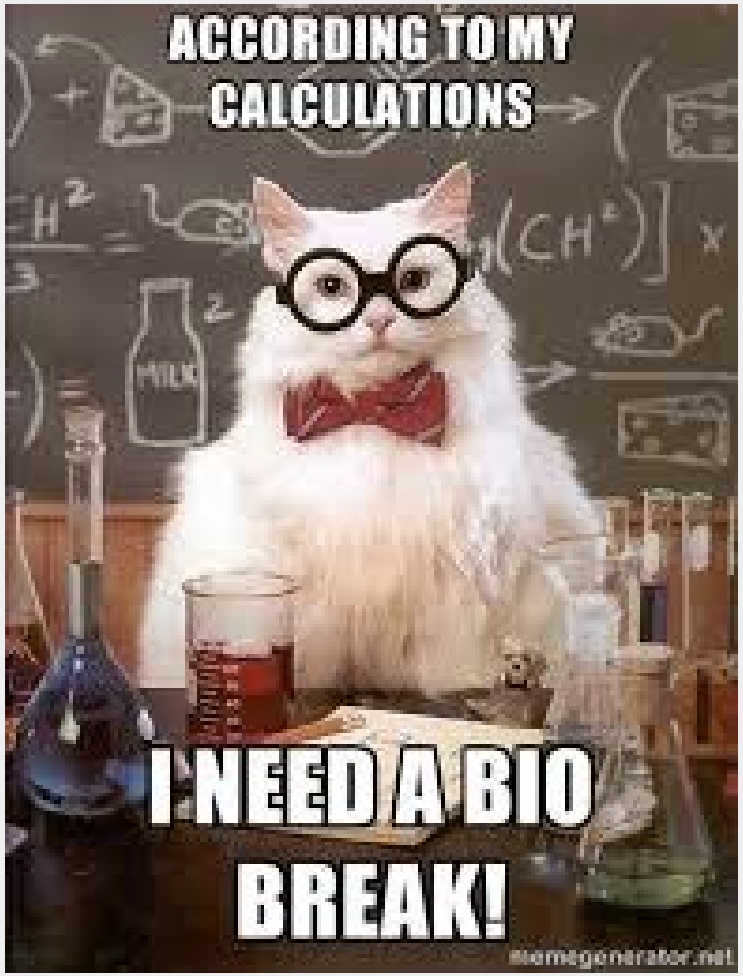
Cochrane-endorsed, domain-based tools — judged per outcome, not per study

Study design	Tool	Judgement scale
Randomised trials incl. cluster & crossover	RoB 2	Low · Some concerns · High
Non-randomised studies of interventions cohort, case-control, controlled before–after	ROBINS-I	Low · Moderate · Serious · Critical
Observational studies of exposures risk factors, environmental exposures	ROBINS-E	Low · Some concerns · High · Very high
Diagnostic & screening test accuracy e.g. entry / exit screening	QUADAS-2	Risk of bias + applicability
Prediction & prognostic models risk scores, prognostic tools	PROBAST	Risk of bias + applicability
Systematic reviews when reviewing reviews	ROBIS / AMSTAR 2	RoB in review / overall confidence
Modelling studies no standard RoB tool	See Section 3	Structured credibility judgement

Also common: Newcastle–Ottawa Scale and JBI checklists — widely used, but item checklists rather than domain-based judgements.

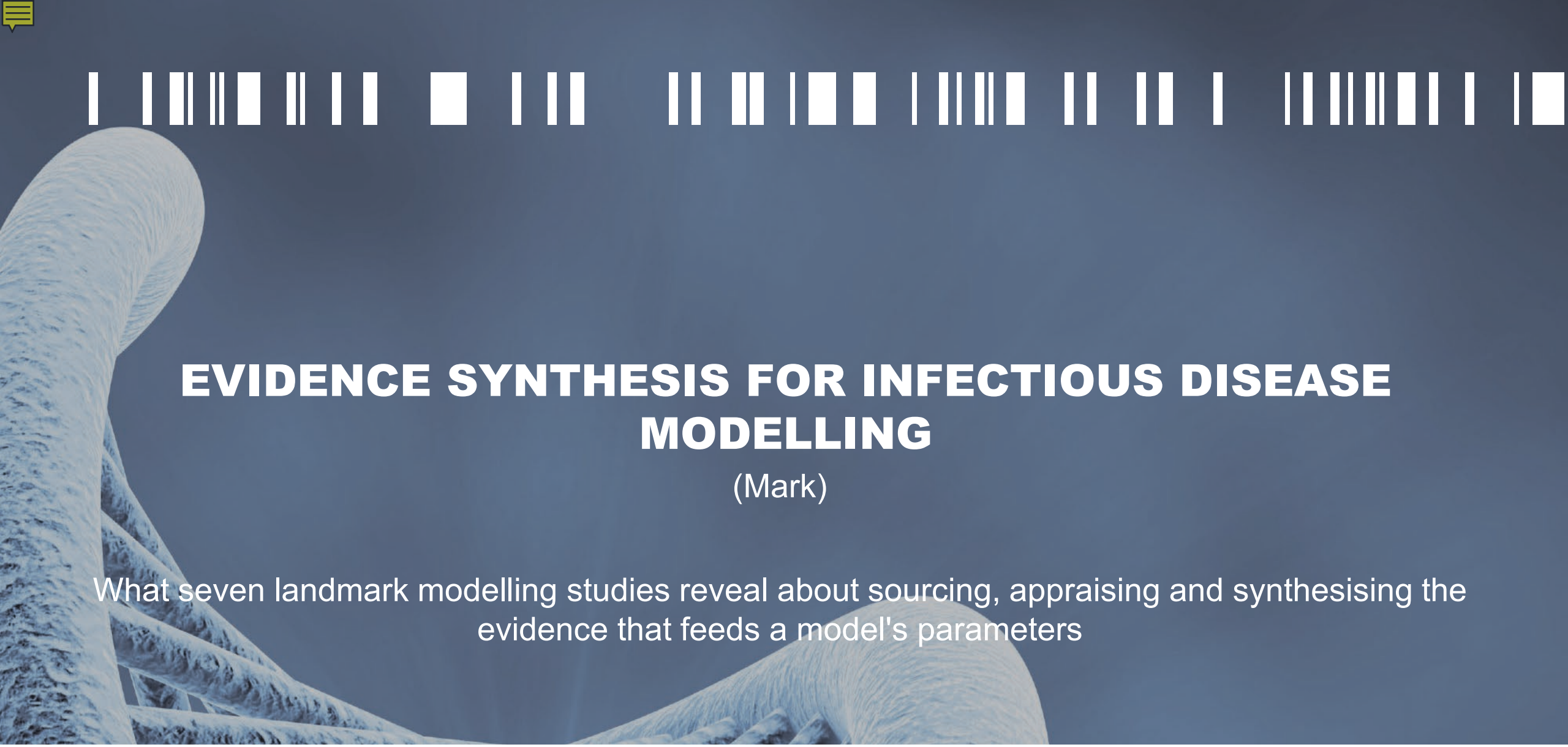






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EVIDENCE SYNTHESIS FOR INFECTIOUS DISEASE MODELLING

(Mark)

What seven landmark modelling studies reveal about sourcing, appraising and synthesising the evidence that feeds a model's parameters

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SYSTEMATIC REVIEWS OF PREDICTION MODELS: WHICH TOOL?

CHARMS and PROBAST help you conduct a review of prediction models; ROBIS or AMSTAR 2 appraise a finished review

1 Frame the question and extract data

CHARMS

7 items to frame the review question and 11 domains to extract from every included study.

2 Assess each included study

PROBAST / ROBIS

Risk of bias (4 domains) and applicability to your review question, for each model in each study.

3 Interpret and report

PRISMA

Carry the PROBAST judgements into how you interpret the results; report the review with PRISMA.

Appraising a published review instead?

CHARMS is not a quality-appraisal tool for a finished review. To judge a review you want to use (for example as a parameter source), use ROBIS for risk of bias in the review, or AMSTAR 2 for overall confidence.

ROBIS / AMSTAR 2

Source: Moons et al. Ann Intern Med 2019;170:W1-W33 (PROBAST explanation and elaboration), Table 1; Moons et al. PLoS Med 2014;11:e1001744

WHY THIS SESSION

EVERY MODEL RUNS ON BORROWED NUMBERS

No model generates its own parameters.

1

R0, generation interval, case-fatality ratio, treatment effect size, testing and retention rates —

every one of these is imported from evidence collected outside the model.

2

“Imported from evidence” should mean synthesised evidence.

A systematically identified, quality-appraised body of estimates — not whichever single paper was easiest to find.

3

Evidence synthesis is the parameter supply chain for modelling.

This session is about that supply chain — not about communicating model results to policymakers (that's the separate Knowledge Translation module).

Seven studies. Seven sets of parameters. One question: was the evidence behind them actually synthesised?

METHODS

THREE CHECKS APPLIED TO EACH STUDY

Before drawing any lesson about parameters, each of the seven studies below was independently verified.

1

Citation accuracy

Does the reference resolve to a real, findable paper — correct authors, journal, volume, year — or is it garbled, merged, or invented?

2

Study authenticity

Is the underlying study genuine and peer-reviewed, and does it actually report what the claim attributes to it?

3

Real-world verdict

Where independent outcome data exist (a later outbreak, a randomised trial, a comparative review), does the model's prediction match what actually happened?

THE VERIFIED RECORD

SEVEN MODELLING STUDIES, CHECKED

Study (Author, Year)	Disease / Model Type	Citation Check	Prediction Verdict
Balcan et al., 2009 (GLEAM)	Pandemic influenza (H1N1) — metapopulation model	Real paper; a widely-quoted accuracy figure belongs to a companion 2012 paper, not this one	CORRECT
WHO Ebola Response Team, 2014	Ebola (West Africa) — SEIR transmission model	Exact	PARTIALLY CORRECT
Lipsitch et al., 2003	SARS-CoV-1 — transmission & control model	Exact	CORRECT
Ferguson, Donnelly & Anderson, 2001	Foot-and-mouth disease — livestock spatial model	Exact (two companion papers, Nature + Science)	CONTESTED
Granich et al., 2009	HIV — Treatment-as-Prevention population model	Exact	INCORRECT
Perelson et al., 1996	HIV — within-host viral dynamics model	Exact	CORRECT
Imperial Report 9 (Ferguson et al., 2020)	COVID-19 — transmission model, NPI scenarios	Exact	NOT DIRECTLY TESTABLE

FOR RESEARCHERS BUILDING THEIR OWN MODELS

WHAT VARIABLES DID EACH MODEL ACTUALLY NEED?

Study	Core Parameters the Model Needed	Best Evidence Source for That Parameter
H1N1 (GLEAM)	R0; generation interval; seasonal transmissibility; real-time international air-travel mobility	Historical pandemic & seasonal-influenza literature, combined with real-time mobility data
Ebola (WHO 2014)	R0; incubation period; serial interval; case-fatality ratio; effectiveness of isolation/contact tracing	Outbreak line-lists plus evidence from prior, smaller Ebola outbreaks (1976–2012)
SARS (Lipsitch)	R0; generation time; effect of case isolation and quarantine on transmission	Contact-tracing and outbreak data across Hong Kong, Singapore and Toronto
FMD (Ferguson et al.)	Farm-to-farm transmission kernel (distance-based); incubation period; culling and vaccination efficacy	Farm-level outbreak data plus international FMD control experience (incl. vaccination)
HIV TasP (Granich)	Testing uptake; linkage to care; ART-initiation threshold; retention & adherence; transmission risk on ART	ART-programme cascade-of-care data already available from African rollout, 2004–2009
HIV within-host (Perelson)	Viral decay rate constant; infected-cell half-life; viral generation time	Direct patient viral-load measurement — primary data, not literature synthesis
COVID-19 (Report 9)	R0; infection-fatality ratio; generation time; incubation period; contact-mixing structure	Early China / Diamond Princess contact-tracing data — revised as serology & contact-survey evidence matured

PATTERN ONE

WHEN PARAMETERS WERE WELL-GROUNDED, PREDICTIONS HELD

H1N1 — GLEAM

Parameters (R0, generation interval) were drawn from decades of prior influenza pandemic and seasonal literature already synthesised by 2009.

SARS — Lipsitch et al.

Multiple independent teams estimated R0 for the same outbreak in parallel; the numbers converged quickly, effectively self-synthesising across studies.

HIV within-host — Perelson et al.

Not literature-dependent at all — parameters were measured directly in patients, sidestepping the need for synthesis entirely.

PATTERN ONE · STUDY 1 OF 3

H1N1 2009: THE GLEAM GLOBAL FORECAST

Balcan et al. (2009) forecast the 2009 pandemic wave in real time; Tizzoni et al. (2012) later tested it against surveillance data from 48 countries.

Where the inputs came from

- **Transmissibility was fitted, not borrowed.** R0 = 1.75 (95% CI 1.64–1.88) and generation time 3.6 days (2.2–5.1) came from a Monte Carlo maximum-likelihood fit to the dates the virus arrived in 12 countries seeded from Mexico.
- **Secondary inputs came from earlier work.** Latent period (1.1 days) followed early estimates compatible with seasonal influenza, then was varied from 1.1 to 2.5 days; the generation-time range was checked against previous pandemic and seasonal influenza studies (mostly 2–4 days).
- **Mobility was measured:** airline and commuting data for 3,362 subpopulations in 220 countries.

What the papers found

- **Forecast (Sept 2009, before the northern peaks):** a worst-case, no-mitigation epidemic peaking in October–November in the northern hemisphere.
- **Peak timing held up.** Observed peaks fell inside the prediction interval in 87% of countries; the other 13% were off by 2 weeks at most. Median error 0; within 2 weeks in 50% of countries and within 4 weeks in 95%.
- **Size was not forecast reliably.** Attack rate depends on vaccine uptake, pre-existing immunity and the asymptomatic fraction, which became known only after the wave. Model attack rates ran above serological and several surveillance estimates, but close to US estimates.

Lesson for evidence synthesis

- **Verdict: timing held, size did not.**
- **Why it worked:** the transmission signal came from the epidemic itself, the mobility data were high quality, and the authors ran sensitivity analyses and framed the result as a worst case.
- **Where synthesis could still add most:** the inputs behind attack rate (pre-existing immunity, asymptomatic proportion, vaccine uptake) had no synthesised evidence in real time. A rapid, living review of those inputs is the gap.

References (file labels in the STEAM 3D › References folder): 01a Balcan D, Hu H, Gonçalves B, et al. Seasonal transmission potential and activity peaks of the new influenza A(H1N1). BMC Med 2009;7:45. 01b Tizzoni M, Bajardi P, Poletto C, et al. Real-time numerical forecast of global epidemic spreading: case study of 2009 A/H1N1pdm. BMC Med 2012;10:165.

PATTERN ONE · STUDY 2 OF 3

SARS 2003: LIPSITCH ET AL.

Lipsitch et al. (Science, June 2003) estimated how infectious SARS was, and what isolation and quarantine could achieve, while the epidemic was still unfolding.

Where the inputs came from

- **Singapore case data:** the first 205 probable cases. Secondary cases per index case fell from a mean of 7 (first week) to 1.6 (second week) to below 1 in most later weeks, as isolation and quarantine began.
- **Serial interval measured directly:** mean 8.4 days (SD 3.8) in Singapore.
- **Early growth in other settings:** control began quickly in Singapore, so there was too little uncontrolled data there. R0 was instead inferred from growth in reported cases (Hong Kong and worldwide counts), using assumed serial intervals of 8–12 days.

What the paper found

- **R0 for uncontrolled spread: about 3.** The Hong Kong growth analysis gave 2.2–3.6 for serial intervals of 8–12 days.
- **Wide uncertainty, stated openly.** The Singapore stochastic model gave a 90% credible interval of 1.5–7.7 (mode about 2.2, mean 3.5), because superspreading was common: 103 of the first 201 probable cases were infected by just five source cases.
- **Under-reporting barely mattered:** the case count enters the estimate through a logarithm, so even large under-reporting moved R0 little.
- **Control looked achievable:** R only has to be held below 1, so isolation and quarantine need not be perfect.

Lesson for evidence synthesis

- **Verdict: reliable because it was honest about the range.**
- **Triangulation within one analysis:** three data sources (Singapore line list, Hong Kong growth, worldwide counts) were used, with explicit sensitivity analyses for serial interval and under-reporting.
- **Decision-makers needed a range, not a point.** A transparent range for R0, rather than a single estimate, is what a living synthesis of parameter estimates formalises.
- **Limit:** too little pre-control data from Singapore, so uncontrolled R0 had to be inferred indirectly.

References (file labels in the STEAM 3D › References folder): 03 Lipsitch M, Cohen T, Cooper B, et al. Transmission dynamics and control of severe acute respiratory syndrome. Science 2003;300(5627):1966–1970. doi:10.1126/science.1086616

PATTERN ONE · STUDY 3 OF 3

HIV WITHIN-HOST DYNAMICS: PERELSON ET AL.

Perelson et al. (Science, 1996) measured how fast HIV is cleared and produced inside patients, by watching viral load fall after a protease inhibitor.

Where the inputs came from

- **Measured in patients, not taken from the literature.** Five HIV-infected people received ritonavir (600 mg twice daily) and plasma HIV-1 RNA was tracked closely: 15 data points in the first 7 days.
- **Denser sampling than before.** The same group's earlier study sampled every 3 days and gave only minimum estimates.
- **Model fitted to each patient's decay curve** (non-linear least squares), separating clearance of free virions from loss of infected cells, after estimating and subtracting the drug's pharmacokinetic delay.

What the paper found

- **Free virions:** half-life 0.24 days (about 6 hours); clearance rate 2.06–3.81 per day across patients.
- **Infected cells:** mean life-span 2.2 days (half-life 1.6 days).
- **Production:** at least 10.3×10^9 virions per day, about 15 times the group's earlier minimum estimate.
- **Cycle timing:** minimum life cycle 1.2 days; generation time 2.6 ± 0.8 days.
- **Caveat from the authors:** values are minimum estimates, assuming the drug was fully effective and target cells did not recover.

Lesson for evidence synthesis

- **Verdict: the clean end of Pattern One.** The parameters were measured in the system being modelled, so no synthesis was needed to feed the model.
- **Independent replication:** other groups (Wei et al., Nowak et al.) had found essentially identical decay kinetics.
- **Study design matters:** denser sampling revised production upward by about 15-fold. When pooling evidence, check how each estimate was measured.
- **Scope:** five patients; the estimates describe within-host biology and would need synthesis before use in population models.

References (file labels in the STEAM 3D › References folder): 06 Perelson AS, Neumann AU, Markowitz M, Leonard JM, Ho DD. HIV-1 dynamics in vivo: virion clearance rate, infected cell life-span, and viral generation time. Science 1996;271(5255):1582–1586.

PATTERN TWO

WHEN EVIDENCE WAS THIN, IDEALISED, OR CONTESTED — PREDICTIONS SLIPPED

Ebola — WHO 2014

The natural-history evidence base was thin (only a handful of small prior outbreaks). The 'no intervention' scenario was one point on a wide plausible range that wasn't clearly communicated as such.

FMD — Ferguson et al.

The transmission-kernel and culling-efficacy evidence drew on a narrow slice of data. Broader international FMD control evidence, including vaccination experience, existed but was not systematically weighed.

HIV TasP — Granich et al.

The model assumed near-ideal testing, linkage and adherence. A substantial literature on real-world care-cascade attrition already existed in Africa by 2009 but was not incorporated.

A seventh case — COVID-19's Imperial Report 9 — doesn't fit either pattern cleanly: its headline number was an explicit non-intervention counterfactual, not a forecast. See the case-by-case verdict table.

PATTERN TWO · STUDY 1 OF 3

EBOLA 2014: WHO FORWARD PROJECTIONS

WHO Ebola Response Team (NEJM, Oct 2014) described the first 9 months of the West Africa epidemic and projected cases to 2 November, assuming no change in control

Where the inputs came from

- **Field surveillance data** on 3,343 confirmed and 667 probable cases (Guinea, Liberia, Nigeria, Sierra Leone) as of 14 September 2014.
- **Natural history estimated from those cases:** incubation 11.4 days; serial interval 15.3 ± 9.3 days. The authors report these, and the case fatality rate (70.8%), as within the ranges seen in previous Ebola outbreaks.
- **Growth rates from early exponential growth:** R0 1.71 (Guinea), 1.83 (Liberia), 2.02 (Sierra Leone); doubling times 15.7, 23.6 and 30.2 days.

What the paper found

- **Conditional projection:** assuming no change in control measures, cumulative cases by 2 November would be 5,740 (Guinea), 9,890 (Liberia) and 5,000 (Sierra Leone): more than 20,000 in total.
- **Limits the authors flagged:** the data understate the true size; reliable estimates of the current reproduction number reached only early September because of reporting delays.
- **Their conclusion:** the epidemic was large because of the affected populations and insufficient control, not unusual virus biology. In the week of 8–14 Sept about 995 patients needed care versus roughly 610 beds.

Lesson for evidence synthesis

- **Verdict: a scenario, not a forecast.** The >20,000 figure assumed nothing changed, so it can't be scored as a prediction against what followed.
- **The weak link was not the natural-history parameters.** Earlier outbreaks supplied a check on incubation, fatality and R0. The weaker inputs were incomplete, delayed surveillance data and the assumption that control stayed fixed.
- **What synthesis could not supply:** how behaviour and the response would change. Stating that conditionality clearly is part of reporting the evidence.

References (file labels in the STEAM 3D › References folder): 02 WHO Ebola Response Team. Ebola virus disease in West Africa – the first 9 months of the epidemic and forward projections. N Engl J Med 2014;371(16):1481–1495. doi:10.1056/NEJMoa1411100

PATTERN TWO · STUDY 2 OF 3

FOOT-AND-MOUTH 2001: FERGUSON ET AL.

Ferguson et al. (2001) modelled culling and vaccination in the UK epidemic; Kitching et al. (2006) criticised the inputs; Tildesley et al. (2008) tested the model.

Where the inputs came from

- **Farm-level data:** disease, culling and census records for every livestock farm in Great Britain (Nature). The Science paper used the epidemic's first two months, with reporting and slaughter delays estimated from surveillance.
- **Contact-tracing data** supplied the spatial kernel and infectivity timing. Kitching et al. note it reached modellers on 13 March, three weeks in, and was incomplete.
- **Infectivity was assumed:** constant from about 3 days after infection until slaughter. Ferguson et al. wrote that data do not exist to estimate a farm's infectiousness over time since infection.

What the papers found

- **Ferguson et al.:** extended culling was essential; faster action could have made the epidemic much smaller. Ring culling and ring vaccination were both modelled, and culling was predicted to be more effective.
- **Kitching et al.:** tracing data were incomplete and biased towards short-range spread; infectivity timing conflicted with biology; parameters were held constant; models were unvalidated. 10,157 premises were culled; 2,026 were infected.
- **Tildesley et al.:** farm-level accuracy 5–15% for infected farms, 20–30% for culls. The low figure matched the spread between independent model runs, so they judged the model a reliable representation.

Lesson for evidence synthesis

- **Verdict: contested. Same evidence, opposite readings.**
- **Vaccination was modelled, not ignored.** The dispute is about input quality: biased tracing data, the timing of infectivity and the absence of validation.
- **The authors named their own gap:** no data on infectiousness over time. A structured appraisal of the contact-tracing data, and a synthesis of what was known about infectivity timing, was the missing step before a culling policy was built on the model.
- **Both critique and test belong in the evidence base.**

References (file labels in the STEAM 3D › References folder): 04a Ferguson NM, Donnelly CA, Anderson RM. Transmission intensity and impact of control policies on the foot and mouth epidemic in Great Britain. Nature 2001;413:542–548. 04b Ferguson NM, Donnelly CA, Anderson RM. The foot-and-mouth epidemic in Great Britain: pattern of spread and impact of interventions. Science 2001;292:1155–1160. 04c Kitching RP, Thrusfield MV, Taylor NM. Use and abuse of mathematical models. Rev Sci Tech 2006;25(1):293–311. 04d Tildesley MJ, Deardon R, Savill NJ, et al. Accuracy of models for the 2001 foot-and-mouth epidemic. Proc R Soc B 2008;275:1459–1468.

PATTERN TWO · STUDY 3 OF 3

HIV TREATMENT AS PREVENTION: GRANICH ET AL.

Granich et al. (Lancet, 2009) modelled universal annual testing with immediate ART; Eaton et al. (2012) compared 12 models and Hayes et al. (2019) tested the idea in a trial

Where the inputs came from

- **Test case:** South Africa as a hypothetical generalised epidemic, calibrated to R0 = 7, a doubling time of 1.25 years and antenatal prevalence data.
- **Natural history and ART effect:** survival (mean 11 years), 10-fold infectiousness in the 2-month acute phase, and infectiousness cut to 1% on ART, all from sources cited in the paper.
- **Programme assumptions:** annual testing of everyone aged 15+, 100% test accuracy, coverage rising to 90% by 2016. Retention and failure used Malawi programme data (1.5% a year drop-out, 3% a year first-line failure).

What the papers found

- **Granich:** R0 below 1 needed yearly testing and ART at diagnosis (annual testing with ART at CD4 350 gave R0 about 3). Elimination (< 1 case per 1,000 a year) within about 10 years of full roll-out.
- **Eaton:** 12 models agreed ART at high coverage cuts incidence, but only 3 of 9 reached < 0.1% by 2050 at 95% access and retention. Granich's own model gave 0.13%.
- **Hayes (PopART trial, 21 communities):** incidence rate ratio 0.93 (95% CI 0.74–1.18) with universal ART versus standard care, and 0.70 (0.55–0.88) with ART per local guidelines. The authors called the null result unanticipated.

Lesson for evidence synthesis

- **Verdict: optimistic because the programme was idealised.** The result depended on near-universal testing uptake and coverage.
- **Not ignored, but not stress-tested:** Granich used real Malawi retention data and flagged uptake, adherence and behaviour change as needing better data. The elimination claim still rested on the idealised settings.
- **Synthesis could have mapped the range early:** the same-model re-analysis by Wagner and Blower, cited in Eaton, reached the opposite conclusion using different uptake assumptions.

References (file labels in the STEAM 3D › References folder): 05a Granich RM, Gilks CF, Dye C, De Cock KM, Williams BG. Universal voluntary HIV testing with immediate antiretroviral therapy as a strategy for elimination of HIV transmission: a mathematical model. Lancet 2009;373:48–57. 05b Hayes RJ, Donnell D, Floyd S, et al. Effect of universal testing and treatment on HIV incidence – HPTN 071 (PopART). N Engl J Med 2019;381:207–218. 05c Eaton JW, Johnson LF, Salomon JA, et al. HIV treatment as prevention: systematic comparison of mathematical models. PLoS Med 2012;9(7):e1001245.

THE CENTRAL QUESTION

WOULD A PRIOR SYSTEMATIC REVIEW HAVE CHANGED THE OUTCOME?

For several of the seven studies — yes, plausibly. And evidence synthesis has two distinct jobs here:

Synthesising the parameter evidence

Systematically pooling existing estimates of R0, generation interval, cascade-of-care metrics, etc. — before they go into the model.

Synthesising the modelling studies themselves

A systematic review or comparison ACROSS multiple independent models addressing the same question — checking whether one model's headline number is an outlier.

ANSWERING THE QUESTION, STUDY BY STUDY

WOULD PRIOR EVIDENCE SYNTHESIS HAVE HELPED?

Study	Why / why not	Verdict
H1N1 (GLEAM)	Already evidence-informed — built on synthesised historical influenza literature.	**NOT NEEDED**
Ebola (WHO 2014)	Historical base was thin; mainly a scenario-communication issue, not a pure data gap.	**PARTIALLY**
SARS (Lipsitch)	Faster cross-study synthesis of concurrent R0 estimates would have sped convergence and fixed the superspreading misattribution.	**YES, MODESTLY**
FMD (Ferguson et al.)	Strongest case — a wider evidence base (incl. vaccination experience) very plausibly tempers the culling policy.	**YES, CLEARLY**
HIV TasP (Granich)	Strongest case — existing real-world cascade-of-care evidence would have produced a far more conservative estimate.	**YES, CLEARLY**
HIV within-host (Perelson)	Primary data, not synthesis-dependent — synthesis is not the relevant mechanism here.	**N/A**
COVID-19 (Report 9)	Predates almost all formal COVID parameter synthesis; even a later rapid review (Gallo et al. 2020) found R0 ranging 0.48–14.8 across the literature.	**MIXED**

SPOTLIGHT

THE MISSING PIECE: A SYSTEMATIC REVIEW OF THE MODELS THEMSELVES

Eaton et al., 2012 — PLoS Medicine

"HIV Treatment as Prevention: Systematic Comparison of Mathematical Models of the Potential Impact of Antiretroviral Therapy on HIV Incidence in South Africa"

- Systematically compared 12 independent mathematical models addressing the same question as Granich et al.
- Found incidence reductions of 35–54% by 2020 under a common moderate-intervention scenario — far more modest than Granich's headline figure.
- Only 3 of 9 models predicted elimination-level reductions, even under optimistic assumptions.

THE LIVESTOCK-DISEASE ANALOGUE

Tildesley et al., 2008 — Proc R Soc B

Retrospectively compared multiple 2001 FMD models' predictions against the observed epidemic — the same comparative logic applied after the fact to livestock-disease models. This is the same paper that grounds the FMD verdict elsewhere in this deck.

A systematic review doesn't just synthesise primary studies — it can synthesise models.

Eaton et al. published in 2012 — three years after Granich (2009). Too late to inform the original model, but exactly the kind of comparative synthesis that should sit alongside any single influential model.

DEEP DIVE — 2001 UK FOOT-AND-MOUTH DISEASE

IMPERIAL COLLEGE TRANSMISSION MODEL

Ferguson NM, Donnelly CA, Anderson RM. Nature. 2001;413:542–548 • Science. 2001;292(5519):1155–1160 • Kitching RP, et al. Rev Sci Tech. 2006;25(1):293–311 •

WHAT FED THE MODEL

- Farm-to-farm spatial transmission kernel (distance-based risk)
- Estimated incubation and infectious period
- Case/contact-tracing data from only the first few weeks of a fast-moving, geographically atypical outbreak

PREDICTED

Epidemic would continue growing steeply without pre-emptive culling of neighbouring, uninfected premises (“contiguous cull”).

ACTUAL

Outbreak was brought under control, but Kitching et al. (2006) argued the kernel overstated local spread risk — driving culling of many uninfected animals — and Tildesley et al. (2008) found the models' predictive accuracy was uneven across the epidemic.

CONTESTED

The clearest case in this deck for prior synthesis mattering — decades of relevant FMD veterinary evidence existed before 2001 but wasn't systematically drawn into the transmission-kernel parameters.

DEEP DIVE — COVID-19, MARCH 2020

IMPERIAL COLLEGE REPORT 9 (FERGUSON ET AL.)

Ferguson NM, Laydon D, Nedjati-Gilani G, Imai N, Ainslie K, et al. Imperial College COVID-19 Response Team, 16 Mar 2020 • Keeling MJ, Eames KT. J R Soc Interface.

WHAT FED THE MODEL

- R0 = 2.4 (range tested 2.0–2.6), from early China / Diamond Princess contact-tracing data
- Infection fatality ratio = 0.9%, age-stratified 0.002%–9.3%
- 6.5-day mean generation time; 5.1-day incubation period

PREDICTED

~510,000 deaths in Great Britain and ~2.2 million in the US under an explicit unmitigated ("do nothing") scenario, 81% eventual attack rate.

ACTUAL

The unmitigated scenario never occurred once suppression measures were adopted — not directly checkable against reality by design.

NOT DIRECTLY TESTABLE

Structural assumptions (homogeneous contact mixing, per Keeling & Eames 2005) and single-point R0/IFR choices drew substantive, documented critique — even later synthesis (Gallo et al. 2020) found huge disagreement in the primary evidence.

PRACTICAL GUIDANCE

BEFORE YOU PARAMETERISE A MODEL: A CHECKLIST

1. Search for existing systematic reviews or meta-analyses of each parameter you need — R0, generation interval, effect sizes, cascade metrics — before adopting a single source study.
2. Search separately for reviews or comparisons of prior MODELS addressing your question, not just of raw primary data.
3. Appraise the quality and applicable context of each source estimate — setting, population, era — before transporting a parameter into a new context.
4. Where synthesised evidence shows a range, run your model across that range (sensitivity analysis) rather than a single point estimate.
5. Treat one landmark paper as a hypothesis to be tested against your synthesis — not as ground truth for a policy claim.

SYNTHESIS FOR RESEARCHERS BUILDING THEIR OWN MODELS

THREE BUCKETS, NOT ONE UNIFORM ANSWER

ALREADY EVIDENCE-INFORMED

GLEAM/H1N1, SARS, HIV within-host

No fix needed: parameters were already grounded in mature synthesised literature, rapid cross-study convergence, or direct primary measurement.

EVIDENCE EXISTED, WASN'T SYNTHESIZED

FMD 2001, HIV TasP

Fix: a systematic review of the existing literature — including of prior MODELS on the same question — before parameterising.

DIDN'T EXIST YET OR STAYED CONTESTED

Ebola (WHO 2014), COVID-19 (Report 9)

Fix: living, iterative synthesis updated as new data accrues, explicit uncertainty ranges carried through, multi-source sensitivity analysis.

The lesson isn't "always do a systematic review first" — it's to first diagnose which bucket you're in.

THE TAKEAWAY

Evidence synthesis is not an add-on to modelling.

It is the parameter pipeline that determines whether a model's predictions can be trusted.

Three of seven landmark models held up because their parameters were already grounded in synthesised or primary evidence. Two were clear cases where prior synthesis would plausibly have helped. Two remain genuinely contested even in hindsight, because the underlying evidence itself was thin or unsettled.

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QUALITY APPRAISAL LAB

(Ameer)

Risk of bias in primary studies

saMRC advancinglife

Cochrane South Africa

CHARMS: FRAME THE REVIEW, THEN EXTRACT THE DATA

Moons et al., PLoS Medicine 2014 — for reviews of studies that develop, validate or update a prediction model

Part 1 · Frame the review question

7 items

1. Prognostic or diagnostic model
2. Intended scope of the review
3. Type of prediction modelling study
4. Target population
5. Outcome to be predicted
6. Time span of the prediction
7. Intended moment of using the model

Part 2 · Extract from each included study

11 domains — each item is tagged for description, applicability and/or risk of bias

1. Source of data
2. Participants
3. Outcome(s) to be predicted
4. Candidate predictors (or index tests)
5. Sample size
6. Missing data
7. Model development
8. Model performance
9. Model evaluation
10. Results
11. Interpretation and discussion

Out of scope: studies of single prognostic factors (use QUIPS) and studies of a prediction model's impact on care. CHARMS extraction feeds the PROBAST assessment on the next slides.

Source: Moons et al. PLoS Med 2014;11(10):e1001744, Tables 1 and 2

RISK OF BIAS TOOLS: MATCH THE TOOL TO THE STUDY DESIGN

Cochrane-endorsed, domain-based tools — judged per outcome, not per study

Study design	Tool	Judgement scale
Randomised trials incl. cluster & crossover	RoB 2	Low · Some concerns · High
Non-randomised studies of interventions cohort, case-control, controlled before–after	ROBINS-I	Low · Moderate · Serious · Critical
Observational studies of exposures risk factors, environmental exposures	ROBINS-E	Low · Some concerns · High · Very high
Diagnostic & screening test accuracy e.g. entry / exit screening	QUADAS-2	Risk of bias + applicability
Prediction & prognostic models risk scores, prognostic tools	PROBAST	Risk of bias + applicability
Systematic reviews when reviewing reviews	ROBIS / AMSTAR 2	RoB in review / overall confidence
Modelling studies no standard RoB tool	See Section 3	Structured credibility judgement

Also common: Newcastle–Ottawa Scale and JBI checklists — widely used, but item checklists rather than domain-based judgements.

OTHER DESIGNS: QUADAS-2, PROBAST AND ROBIS AT A GLANCE

Same logic — domains, signalling questions, then a judgement; two add an applicability rating

QUADAS-2

Diagnostic & screening accuracy

1. **Patient selection**
2. **Index test**
3. **Reference standard**
4. **Flow and timing**

Applicability judged for the first three domains

PROBAST

Prediction & prognostic models

1. **Participants**
2. **Predictors**
3. **Outcome**
4. **Analysis**

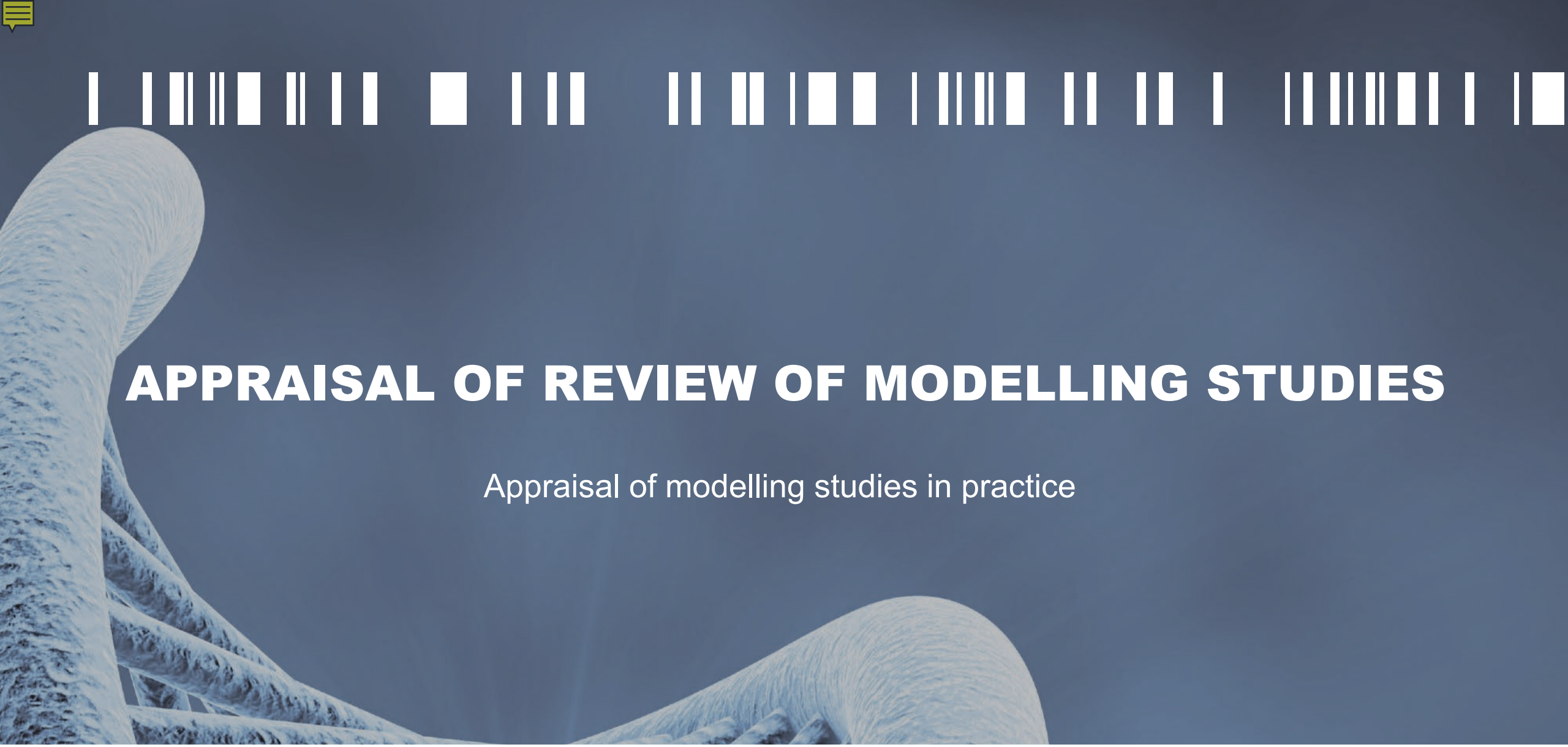
Applicability judged for the first three domains

ROBIS

Systematic reviews

1. **Study eligibility criteria**
2. **Identification & selection of studies**
3. **Data collection & study appraisal**
4. **Synthesis & findings**

Alternative: AMSTAR 2 — 16 items, 7 critical, overall confidence rating



APPRAISAL OF REVIEW OF MODELLING STUDIES

Appraisal of modelling studies in practice

saMRC advancinglife

Cochrane South Africa

HOW DO I APPRAISE A REVIEW OF MODELS?

TOOL: ROBIS

Your task

1. Form four groups. Each group takes one ROBIS domain: 1 eligibility criteria, 2 identification and selection of studies, 3 data collection and study appraisal, 4 synthesis and findings.
2. Read the paper and answer your domain's signalling questions: yes, probably yes, probably no, no or no information. Note where in the paper you found the evidence.
3. Rate your domain's concern as low, high or unclear. Write your reasons on one flipchart row.
4. Report back in two minutes. As a class, answer the three Phase 3 questions and agree an overall rating for the review.

The paper

Wong ZSY, Bui CM, Chughtai AA, MacIntyre CR. A systematic review of early modelling studies of Ebola virus disease in West Africa. Epidemiol Infect 2017;145(6):1069–94.
41 modelling studies from 2014–15.

Time and output

5 min set-up, 15 min group work, 10 min report back.
Output: one flipchart row per group, then a class rating.

Ground rule: judge how the review was done, not whether you like its findings. We found no tool built for reviews of modelling studies; ROBIS is the best general fit.

ROBIS: THE TOOL FOR EXERCISE 1

Whiting et al., Journal of Clinical Epidemiology 2016 — for systematic reviews of interventions, diagnosis, prognosis and aetiology

Phase 1 · Relevance

Does the review match your question? Optional.

Phase 2 · Review process

Four domains. Answer the signalling questions, then rate concern: low, high or unclear

Phase 3 · Overall judgement

Three questions on interpretation. Overall risk of bias: low, high or unclear

1 Eligibility criteria

5 questions

Predefined objectives and criteria?
Criteria fit the review question?
Criteria unambiguous?
Study-based restrictions appropriate?
Source-based restrictions appropriate?

2 Identification and selection

5 questions

Appropriate range of databases?
Methods beyond database searching?
Search terms likely to find all eligible studies?
Date, format and language limits appropriate?
Errors in selection minimised?

3 Data collection and appraisal

5 questions

Errors in data collection minimised?
Enough study detail to interpret results?
All relevant results collected?
Risk of bias formally assessed?
Errors in bias assessment minimised?

4 Synthesis and findings

6 questions

All studies that should be included?
Predefined analyses followed?
Synthesis suits the studies?
Heterogeneity addressed?
Findings shown to be robust?
Biases in studies addressed?

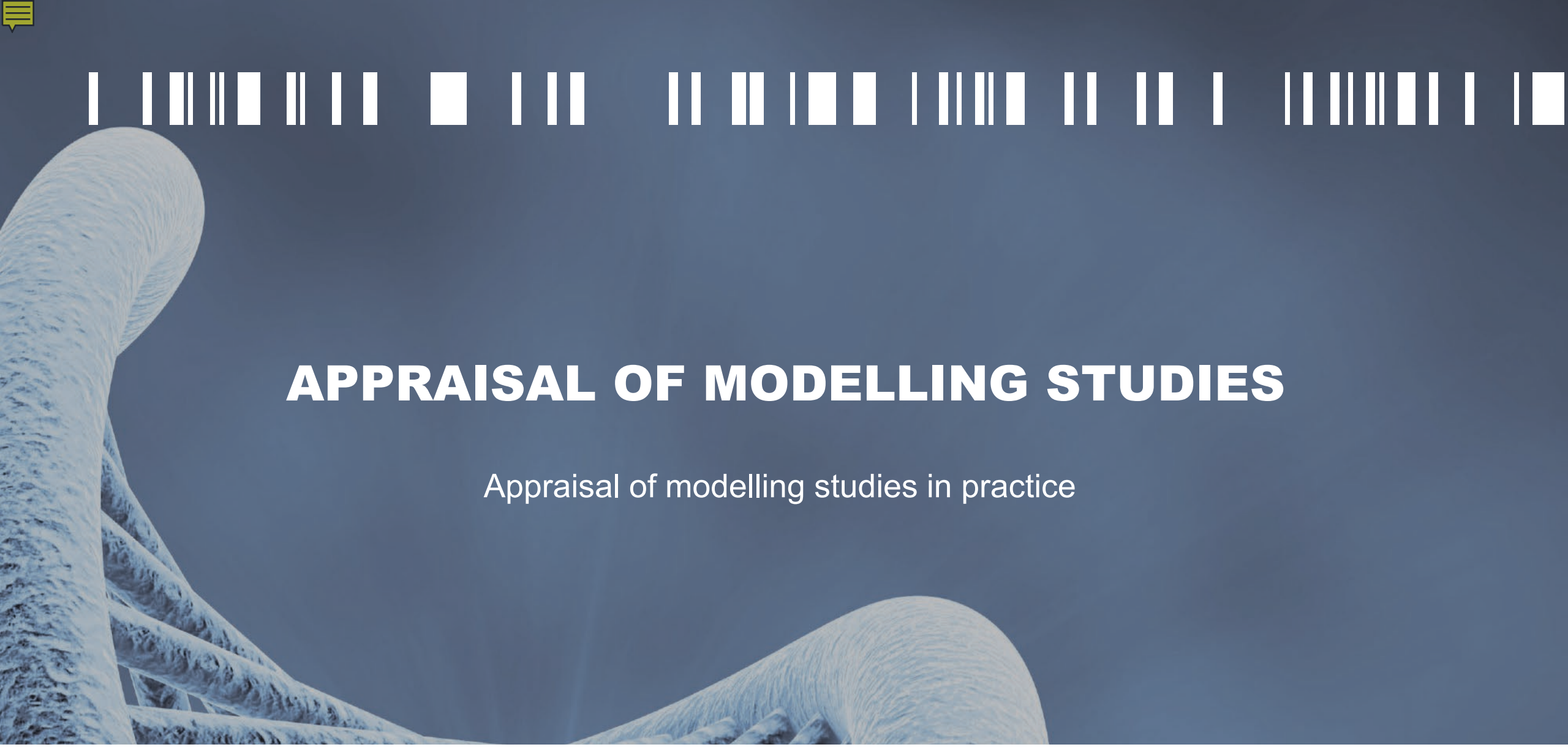
Source: Whiting et al. J Clin Epidemiol 2016;69:225–34; ROBIS guidance, University of Bristol. Questions shortened. Yes always means low concern.

EXERCISE: WONG ET AL. 2017

Facilitator key (hidden slide) — our reading of the paper; check against the full text before class

1 Eligibility criteria	No protocol or registration mentioned. Window January 2014 to December 2015, English only, search “ebola” AND “model*”. Check how “modelling study” is defined in the Methods.	**Unclear**
2 Identification and selection	PubMed, Embase and Scopus only; no grey-literature source. Hand-search of key papers and an earlier scoping review. One narrow term pair may miss “simulation” or “forecast”. Two reviewers screened titles and abstracts independently.	**High or unclear**
3 Data collection and appraisal	Number of independent extractors not stated. No formal risk-of-bias or quality appraisal of the included models (question 3.4: no). It did record whether models corrected for underreporting, which is useful but is not an appraisal.	**High**
4 Synthesis and findings	Descriptive synthesis with medians and non-parametric tests suits varied models. Models with and without an underreporting correction were compared. Check the PRISMA numbers against Figure 1; they did not add up when we checked.	**Unclear**

Phase 3 · suggested overall: high risk of bias. Main reasons: the included models were not appraised and the search was narrow. Discussion point: ROBIS rates the review process, so a review can still be useful descriptively.



APPRAISAL OF MODELLING STUDIES

Appraisal of modelling studies in practice

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EXERCISE: APPRAISE A MODEL UNAIDED

1 What would you look for?

List the five things you would check to decide whether to trust this model's results.

2 Was it well reported?

What is need for someone to follow and to repeat what was done ?

3 How much would you trust it?

Give the model your own risk-of-bias rating: low, moderate or high. Give your two strongest reasons.

COMMON RED FLAGS IN MODELLING STUDIES

Signals that decision-makers should be cautious

Reporting and design red flags

- No clear decision question
- Model structure not justified
- Parameter sources unclear
- No local contextualisation
- No code, equations or supplement

Analytic red flags

- No calibration
- No validation
- Single scenario only
- No uncertainty analysis
- Claims stronger than the model supports

WHY CRITICALLY APPRAISE MODELLING STUDIES?

Complexity does not equal credibility

A useful modelling study is:

- relevant
- credible
- transparent
- validated
- useful for decision-making

Not because it is mathematically sophisticated.



APPRAISAL TOOLS PARTICIPANTS WILL LEARN

Select the tool according to the model type and purpose

Burns 2021 bespoke tool	Feasible rapid-review appraisal of modelling studies
Caro et al. 2014	ISPOR-AMCP-NPC relevance and credibility questionnaire
Philips et al. 2006	Decision-analytic model quality framework
ISPOR-SMDM guidance	Good research practices for modelling
Chaturvedi et al. 2026	Emerging: 7 dimensions for infectious-disease models
EPIFORGE 2020	Reporting checklist for epidemic forecasts (Exercises 2–3)
GRADE for modelled evidence	Certainty of model outputs (Brozek 2021)
AdViSHE	Economic models only: validation status
CHEERS 2022	Economic models only: reporting checklist
PROBAST	Prediction models only: risk of bias and applicability

CORE APPRAISAL DOMAINS

A pragmatic STEAM-3D framework

Applicability	**Model structure**	**Input data**
Parameterisation	**Calibration**	**Validation**
Uncertainty analysis	**Transparency**	**Interpretation**

A BESPOKE TOOL FOR MODELLING STUDIES

Used in the Cochrane rapid review of air-travel control measures (Burns et al. 2021): ten criteria, each reported separately

Model structure

1 Are the structural assumptions transparent and justified?
2 Are they reasonable given the model's objective, perspective and scope?

Input data

3 Are the input parameters transparent and justified?
4 Are the input parameters reasonable?

External validation

5 Has the external validation process been described?
6 Has the model been shown to be externally valid?

Internal validation

7 Has the internal validation process been described?
8 Has the model been shown to be internally valid?

Uncertainty

9 Was there an adequate assessment of the effects of uncertainty?

Transparency

10 Is technical documentation, including code, detailed enough for replication, and is it available?

No summary score. Report each criterion on its own. For the exercise, answer yes, partly, no or unclear and give the evidence.

Source: Burns et al. Cochrane Database Syst Rev 2021, CD013717.pub2, Appendix 6. Items 1–9 as worded; item 10 shortened. The four-point answer scale is added for this workshop.

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STEP 4: APPRAISE THE MODELS WITH THE BURNS 2021 TOOL

No validated risk-of-bias tool exists for modelling studies. Cochrane built this one for the same question

The tool at a glance: 6 domains, 10 questions

Model structure *	Q1–2
Input data *	Q3–4
External validation	Q5–6
Internal validation	Q7–8
Uncertainty	Q9
Transparency	Q10

Ratings: No to minor, Moderate or Major concerns. Q5 and Q7 (was a process described?) are Reported or Not reported.
*** Fatal-flaw indicators.** No summary score: report each criterion.

Worked example: Adekunle 2020
Adekunle A, Meehan M, Rojas-Alvarez D, Trauer J, McBryde E. Delaying the COVID-19 epidemic in Australia: evaluating the effectiveness of international travel bans. Aust N Z J Public Health 2020;44(4):257–9.
A 3-page brief report. A stochastic SEIR meta-population model of global spread, driven by flight data, asks how much Australia's travel bans on China, Iran, South Korea and Italy delayed the epidemic.
Most of the model is in Supplementary Appendix 1. Get it before you start.

Tool: Burns et al. Cochrane Database Syst Rev 2021;CD013717, Appendix 6 (criteria from Philips 2006 and Caro 2014).

ADEKUNLEA, MEEHANM, ROJAS-ALVAREZD, TRAUERJ, MCBRYDEE. DELAYING THE COVID-19 EPIDEMIC IN AUSTRALIA: EVALUATING THE E#ECTIVENESS OF INTERNATIONAL TRAVEL BANS. *AUSTRALIAN AND NEW ZEALAND JOURNAL OF PUBLIC HEALTH* 2020;**44**(4):257-9. [DOI: 10.1111/1753-6405.13016]

ADEKUNLE 2020: MODEL STRUCTURE (Q1–Q2)

Worked example 1 of 3 · * = fatal-flaw indicator

Question	Rating (Burns 2021)	Why	Where to look in the article
Q1 Are the structural assumptions transparent and justified?	**Moderate concerns**	The model type is named (stochastic SEIR meta-population) but the main text defers detail to a reference and the supplement. It does not explain how the travel bans were put into the model, or how 'time to widespread local transmission' was worked out.	Methods p257–8: 'we used a stochastic meta-population model…'; 'we progressively imposed travel restrictions… IATA'. Results p258–9 and Fig 3. Supplementary Appendix 1.
Q2 * Are the assumptions reasonable for the objective, perspective and scope?	**Moderate concerns**	Travel follows 2018 flight volumes until 24 January 2020, and infected people keep travelling: nobody stops because they are ill or in hospital. This could overstate importations without a ban, and so the ban's benefit. Burns recorded this reason under Q1 and left Q2 blank.	Methods p257–8: 'Migration patterns… OAG… March 2018'; 'travel proceeded in accordance with these historic travel data'.

Ratings and reasons: Burns et al. 2021, Appendix 9 (Adekunle 2020). Page numbers refer to Aust N Z J Public Health 2020;44(4):257–9.

ADEKUNLE 2020: INPUT DATA AND EXTERNAL VALIDATION (Q3–Q6)

Worked example 2 of 3 · * = fatal-flaw indicator

Question	Rating (Burns 2021)	Why	Where to look in the article
Q3 Are the input parameters transparent and justified?	No to minor concerns	Each parameter has a source: R of 2.63 falling to 1.73 (Wang 2020), case notifications (LSHTM), OAG flight data.	Methods p257, references 5–6. Parameter table in Supplementary Appendix 1.
Q4 * Are the input parameters reasonable?	Moderate concerns	An infectious period of 4 days looks short and a mortality of 1.8% looks high against evidence at the time. They would shift results, but not enough to call them a fatal	Supplementary Appendix 1 only: not in the main text. Compare each value with reviews or other models of the time.
Q5 Has an external validation process been described?	Reported	Predicted imported cases were compared with observed imported cases in 19 countries at 31 January 2020.	Results p258: 'We first validated our model…'; Fig 1.
Q6 Has the model been shown to be externally valid?	No to minor concerns	Predictions matched observed imports, especially for Australia. The authors say fit was poorer for countries with heavy travel from China.	Results p258; Fig 1 (blue model vs red observed).

Ratings and reasons: Burns et al. 2021, Appendix 9 (Adekunle 2020). Page numbers refer to Aust N Z J Public Health 2020;44(4):257–9.

ADEKUNLE 2020: THE REMAINING DOMAINS (Q7–Q10)

Worked example 3 of 3 · internal validation, uncertainty, transparency

Question	Rating (Burns 2021)	Why	Where to look in the article
Q7 Has an internal validation process been described?	Not reported	No description of code checks, debugging or tests on simulated data.	Methods and Supplementary Appendix 1. The absence is the finding.
Q8 Has the model been shown to be internally valid?	Moderate concerns	Without internal validation the calculations cannot be shown to be correct. Burns rated this moderate, not major.	As Q7.
Q9 Was there an adequate assessment of the effects of uncertainty?	No to minor concerns	1,000 stochastic runs with interquartile ranges, and two R values (2.63, 1.73) as best and worst cases. Burns gave no reason: this one is ours. Debate it: only R	Methods p258: 'simulated… 1,000 times'; Figs 1–2; Results and Discussion p259.
Q10 Was technical documentation available for replication?	Moderate concerns	A technical model description is in the supplement, but the code is not shared.	'Supporting Information' (p259). No code or data availability statement.

Ratings and reasons: Burns et al. 2021, Appendix 9 (Adekunle 2020). Page numbers refer to Aust N Z J Public Health 2020;44(4):257–9.

ADEKUNLE 2020: A PROFILE, NOT A SCORE

Report each criterion, flag the fatal-flaw items, then decide how the model can be used

Structure	Input data	External val.	Internal val.	Uncertainty	Transparency
Q1	Q3	Q5	Q7	Q9	Q10
Moderate	No to minor	Reported	Not reported	No to minor	Moderate
Q2 *	Q4 *	Q6	Q8		
Moderate	Moderate	No to minor	Moderate		

What it tells us

- Both fatal-flaw items (Q2, Q4) raise moderate concerns, not major ones.
- External validation was reported and the fit for Australia was reasonable.
- Internal validation and code are missing.

How we would use it

- As supporting evidence that early bans delayed importation by weeks.
- For structure: meta-population model plus flight data.
- Not for parameter values: infectious period and mortality are doubtful.

Lessons for appraisers

- Get the supplement: the inputs are only there.
- Write a reason and a page for every rating. Burns left Q2 and Q9 blank.
- 'Not reported' is not the same as 'not done'.

** Fatal-flaw indicator. Ratings: Burns et al. 2021, Appendix 9.*

WORKSHEET: BURNS 2021 TOOL

Copy onto paper or a shared document. Rating, reason and where you found it, for every question

Question	Anderson: rating	Reason and where found	Banholzer: rating	Reason and where found
Q1 Structure transparent and justified?				
Q2 * Structure reasonable?				
Q3 Inputs transparent and justified?				
Q4 * Inputs reasonable?				
Q5 External validation described?				
Q6 Externally valid?				
Q7 Internal validation described?				
Q8 Internally valid?				
Q9 Uncertainty assessed adequately?				
Q10 Documentation and code available?				

** Fatal-flaw indicator. Ratings: No to minor, Moderate or Major concerns; Q5 and Q7: Reported or Not reported. No summary score.*

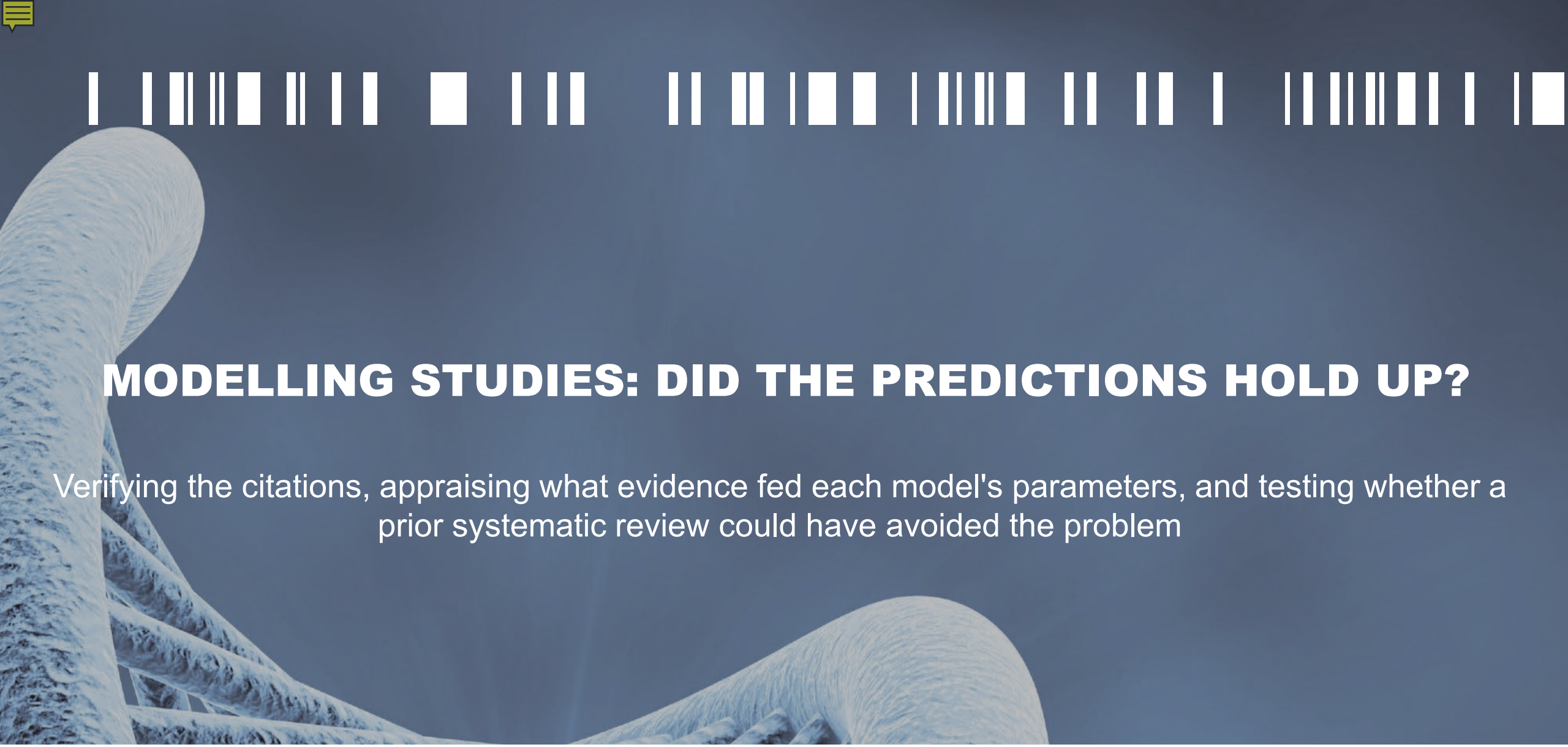
DEBRIEF: THREE MODELS, ONE TOOL

Cochrane's ratings side by side. Compare them with yours

	Q1	Q2 *	Q3	Q4 *	Q5	Q6	Q7	Q8	Q9	Q10
Adekunle 2020	Moderate	Moderate	Minor	Moderate	Reported	Minor	Not rep.	Moderate	Minor	Moderate
Anderson 2020	Minor	Moderate	Minor	Moderate	Reported	Minor	Not rep.	Moderate	Moderate	Minor
Banholzer 2020	Minor	Major	Minor	Minor	Reported	Minor	Not rep.	Moderate	Minor	Minor

1. Which model has a fatal-flaw concern, and what does that mean for using its estimate?
2. Which model would you borrow structure from for your own model, and why?
3. Where did your ratings differ from Cochrane's? Who was right, and how would two reviewers settle it?

** Fatal-flaw indicator. Minor = no-to-minor concerns; Not rep. = not reported. Source: Burns et al. 2021, Appendix 9.*



MODELLING STUDIES: DID THE PREDICTIONS HOLD UP?

Verifying the citations, appraising what evidence fed each model's parameters, and testing whether a prior systematic review could have avoided the problem

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WHY APPRAISE A MODEL LIKE ANY OTHER EVIDENCE

THREE QUESTIONS, APPLIED TO EVERY MODEL

1

Is the citation real?

Every citation checked against PubMed, Crossref and the publishing journal itself — not taken on trust. Of 10 citations originally supplied for this exercise, 2 were fabricated: a real-sounding title stitched to an invented author, year and journal.

2

What evidence fed its parameters?

For each model: where did R0, case-fatality/infection-fatality ratio, transmission kernel or generation time actually come from — and how complete was that underlying data at the time?

3

Could synthesis have helped?

For each model: could a systematic review — of the pathogen's existing natural-history literature, or of prior modelling studies — plausibly have narrowed the error, given what evidence actually existed at the time?

FOUR VERIFIED CASES

FOUR MODELS, FOUR VERDICTS

Model / event	Key parameter(s)	Predicted	Actual	Verdict
CDC Ebola model (Meltzer et al. 2014)	Underreporting ×2.5; fixed doubling time	Up to 1.4M cases by Jan 2015 (unmitigated)	~28,600 cases total, 2014–16	**Overestimated (as a scenario)**
UK FMD model (Ferguson et al. 2001)	Spatial transmission kernel, fit to first weeks	Contiguous culling needed to halt spread	Outbreak controlled; cull scale later disputed	**Partially inaccurate**
H1N1 model (Fraser et al. 2009)	CFR from early Mexico case/death counts	Case fatality ≈0.4% (0.3–1.5%)	True IFR ~2 orders of magnitude lower	**Overestimated (severity)**
Imperial Report 9 (Ferguson et al. 2020)	R0=2.4, IFR=0.9%, generation time 6.5d	~510k UK / ~2.2M US deaths, unmitigated	Scenario never occurred (interventions adopted)	**Not directly testable**

Full detail, evidence base and sourcing for each case follow on the next four slides.

SOURCES

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CLOSING FROM EVIDENCE TO ACTION

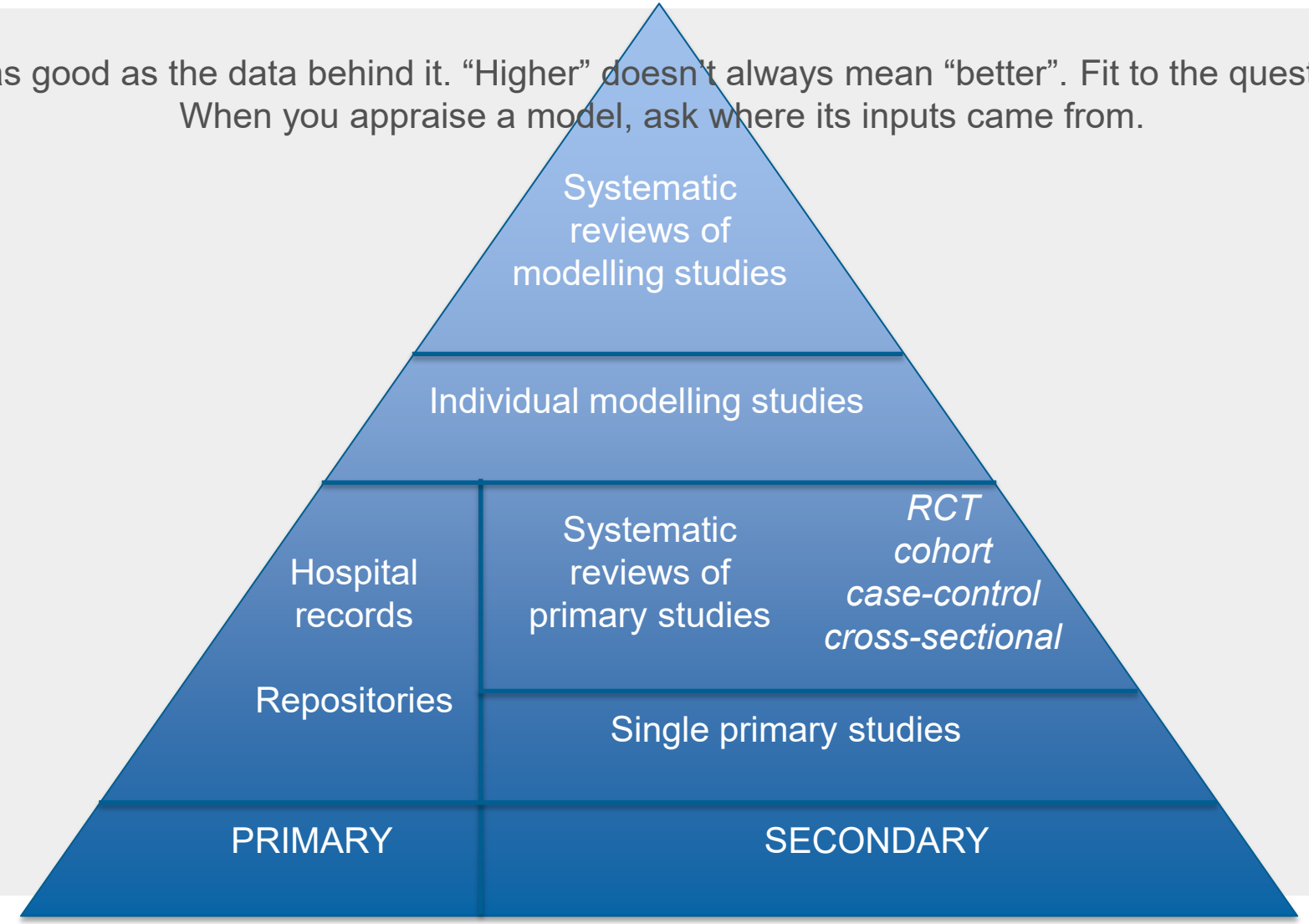
Knowledge Translation primer and next steps

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MODELLING STUDIES DATA SOURCE HIERARCHY

A model is only as good as the data behind it. “Higher” doesn’t always mean “better”. Fit to the question matters most. When you appraise a model, ask where its inputs came from.



KNOWLEDGE TRANSLATION PRIMER

Moving evidence into action

Knowledge translation is the process of moving evidence into action.

- Evidence synthesis
- Dissemination
- Stakeholder engagement
- Contextualisation
- Implementation support
- Evaluation of use and impact



WHY KT MATTERS IN STEAM-3D

KT prevents evidence from stopping at publication

Without KT

- Models may not answer policy questions
- Uncertainty may be misunderstood
- Outputs may not reach decision-makers
- Evidence may not change programmes

With KT

- Questions are co-produced
- Findings are contextualised
- Uncertainty is communicated
- Evidence becomes decision-ready

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